Lecture Notes in Computer Science 6369

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Future Internet – FIS 2010

Third Future Internet Symposium
Berlin, Germany, September 20-22, 2010
Proceedings

Springer

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Arne J. Berre
SINTEF Information and Communication Technology
Forskningsveien 1, 0373 Oslo, Norway
E-mail: arne.j.berre@sintef.no

Asunción Gómez-Pérez
Universidad Politécnica de Madrid, Facultad de Informática
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Kurt Tutschku
University of Vienna, Institute of Distributed and Multimedia Systems
Universitätsstr. 10/T 11, 1090 Vienna, Austria
E-mail: kurt.tutschku@univie.ac.at

Dieter Fensel
University of Innsbruck
Technikerstr. 21a, 6020 Innsbruck, Austria
E-mail: dieter.fensel@sti2.at

Library of Congress Control Number: 2010934213

CR Subject Classification (1998): C.2, H.4, H.3, H.5, J.1, H.2

LNCS Sublibrary: SL 1 – Theoretical Computer Science and General Issues

ISSN 0302-9743
ISBN-10 3-642-15876-5 Springer Berlin Heidelberg New York
ISBN-13 978-3-642-15876-6 Springer Berlin Heidelberg New York

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Printed in Germany

Typesetting: Camera-ready by author, data conversion by Scientific Publishing Services, Chennai, India
Printed on acid-free paper 06/3180

Preface

The current Internet has undergone an essential transformation: it has changed from a network of networks that enables access to remote machines by a consistent protocol suite (TCP/IP), to a network of content, applications, and services. Thus, it has become a modern commodity for everyone.

The Future Internet (FI) is destined to continue this development and to provide improved features and usability for individuals and business. Its applications are expected to originate from areas such as entertainment, health, energy grid, utilities and the environment, transport, mobility, and logistics. Tight economic constraints, however, require the Future Internet to consolidate and converge application-specific networks and support for the Internet of Services (IoS), the Internet of Things (IoT), and the Internet of Content (IoC) in a homogenous and, if possible, a single system.

A simple investigation of network performance requirements of the anticipated FI applications reveals a set of contrary needs that have challenged research on network architectures and protocols for decades. Only a few applications have been successful, e.g., P2P systems, which can adapt easily to heterogeneous environments. Similarly, semantic technology has provided meaningful relationships of content, but has failed when it has come to manageability and performance in universal and heterogeneous network systems.

Thus Internet applications have so far been developed mainly for fitting to specific networks. Now it is time for a change. Networks should be developed for applications. The universe of these networks might be considered as the Future Internet. The future network ecosystem will be supported by a consolidated, preferably single, platform. This platform needs to include support for services, things, and content on both a network and an application level. The FI ecosystem has to be sustainable, meaning that applications are supported efficiently, i.e., consuming a minimal amount of required resources, e.g., capacity, electricity, etc., while providing dedicated security and sufficient performance for the applications throughout their lifetime.

Classical research on network architectures and protocols, semantic technologies, service technologies, content and media, sensors and things is isolated. Thus these disciplines have been unable to meet all requirements. Therefore, an interdisciplinary approach of these research areas is suggested for a sustainable Future Internet.

The aim of the Future Internet Symposium 2010 (FIS 2010) was to bring together scientists and engineers from academia and industry and from various disciplines to exchange and discuss their ideas, views, and research results towards a consolidated, converged, and sustainable Future Internet.

September 2010

Arne J. Berre
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German-Lab Experimental Facility

Dennis Schwerdel[1], Daniel Günther[1], Robert Henjes[2],
Bernd Reuther[1], and Paul Müller[1]

[1] University of Kaiserslautern, Integrated Communication Systems Lab
{schwerdel,guenther,reuther,pmueller}@cs.uni-kl.de
[2] University of Würzburg, Institute of Computer Science
henjes@informatik.uni-wuerzburg.de

Abstract. The G-Lab project aims to investigate concepts and technologies for future networks in a practical manner. Thus G-Lab consists of two major fields of activities: research studies of future network components and the design and setup of experimental facilities. Both is controlled by the same community to ensure, that the experimental facility fits to the demand of researchers. Researchers gain access to virtualized resources or may gain exclusive access to resource if necessary. We present the current setup of the experimental facility, describing the available hardware, management of the platform, the utilization of the Planet-Lab software and the user management.

1 Introduction

Today's Internet has a large economic influence but is based on legacy mechanisms and algorithms from the 70ies and 80ies. The rapid evolution of applications and transport technologies demands for changes even of core technologies of the Internet. Thus several research efforts worldwide currently investigate concepts and technologies for future networks. The goal of the G-Lab project is to foster experimentally driven research to exploit future network technologies.

The G-Lab project [1] has started in 2008 as a distributed joint research and experimentation project for Future Internet studies and development. Initially this BMBF[1] funded project was distributed across six universities in Germany: Wrzburg, Kaiserslautern, Berlin, Mnchen, Karlsruhe, and Darmstadt. G-Lab can be divided in two major work areas, the Future Internet research and the experimental platform.

Multiple research groups focus on theoretical and practical studies from architectural questions to routing, mobility and security. The goal of the G-Lab project is not limited to explore theoretical possibilities and novel ideas but also to use experimental approaches to verify the derived results while using the experimental facility. To investigate the functional aspects of novel Internet architecture approaches (e.g. routing, addressing, control, monitoring & management aspects) and their interaction with each other is such an intricate task which could not be validated only by analytical research and methods.

[1] German Federal Ministry of Education and Research, "Bundesministerium fr Bildung und Forschung".

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 1–10, 2010.
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The project is composed of 8 working groups that are dedicated to different aspects of future Internet research: project coordination, architecture, routing, wireless and mobility, monitoring, QoS and security, service composition and the experimental facility. In the working group 7, a distributed experimental facility consisting of wired and wireless hardware with over 170 nodes, which are fully controllable by the G-Lab partners, is built up and managed. This platform provides a facility to G-Lab working groups (e.g., 1-6) to test their proposed approaches and ideas for the future Internet architecture. The whole network of the platform is distributed into individual clusters at the six different locations within Germany with Kaiserslautern as the main site. The first version of platform was available at March 2009 and first experiments took place at the commencement of April.

The goal if the G-Lab project is that theoretical research and the experimental facility will converge into a Future Internet as depicted in Figure 1. Thus it is important that the experimental facility is flexible enough to adapt to the needs of the experiments and ultimately become a research field itself. With this G-Lab avoids the situation that the platform providers offer their services but nobody is going to use it.

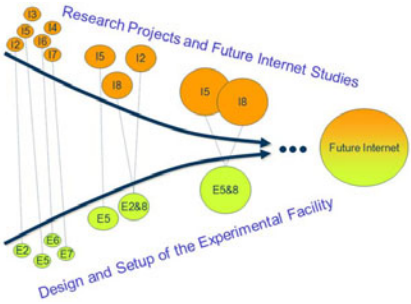


Fig. 1. German-Lab philosophy

2 Hardware Equipment

The hardware equipment consists of three types of nodes and one switch per site. The nodes can be classified in the following categories types:

Normal Node: This is the standard type of node, which can be used to run networking tests and computations.

Network Node: The second node type is designated for special networking tests requiring more network interfaces.

Head Node: The last type is acting as a head node of the local site. It has the task of managing the local site.

After vigorous scrutiny, Sun Microsystems and Cisco have been chosen as hardware provider for the facility. The technical equipment can be seen in table 1.

Table 1. Node hardware

Node Type	Chassis	CPU	RAM	Disk	Network
Head node	Sun Fire X4250	2x Xeon Quad E5450 3.0 GHz	16 GB	16x 146 GB SAS	4x 1 GBit
Network node	Sun Fire X4150	2x Xeon Quad L5420 2.5 GHz	16 GB	4x 146 GB SAS	8x 1 GBit
Normal node	Sun Fire X4150	2x Xeon Quad L5420 2.5 GHz	16 GB	4x 146 GB SAS	4x 1 GBit

Table 2. Node counts

Site	Head Nodes	Network Nodes	Normal Nodes
University of Kaiserslautern	1	2	56
University of Wrzburg	1	2	22
Karlsruhe Institute of Technology	1	2	22
University of Munich	1	2	22
University of Darmstadt	1	2	22
Universitiy of Berlin	1	2	12

All the nodes include a dedicated service processor, i.e. a small computer that allows to control and monitor the hardware remotely with a special management network interface. Each site has one head node, two network nodes and a variable amount of normal nodes as shown in table 2.

The networking equipment consists of a layer-3 switch from Cisco Systems (Catalyst 4500 E Series).

3 Experimental Facility Design

In the design of the experimental facility it has been an important point to use existing solutions, adapt them if needed and integrate them. Thus it was possible to build up a running testbed very quickly. The usage of free and mostly open source software solutions allowed to use the full budget for hardware equipment and also makes it easy to adapt the used software.

3.1 Network Setup

All nodes of a site are located in one network segment interconnected by the switch, which has been split into two virtual switches using VLANs. The public part contains all interfaces of the normal and network nodes and all except one interface of the head node. The private part contains all management interfaces of the service processors and one normal interface of the head node. Both networks are completely separated and only the public network has an uplink to the Internet. With this separation the access to the service processors can be controlled by the head node.

Public IP addresses are needed for all interfaces of each node (except management interface). The addresses are distributed by the head node using DHCP. Global DNS records are managed by the main site (Kaiserslautern), a site-specific zone is delegated to each site to allow decentralized DNS management.

Some sites have policies denying externally controlled nodes with IP addresses in the address range of that site, because some access rules are based on IP ranges. In this situation special firewall rules have been set up that blocks all communication between the nodes and the rest of the site except a few defined proxy hosts.

3.2 Headnode Structure

In the initial design of the experimental facility the head node has an operating system running directly on the hardware, which has early been recognized as being inflexible. Now the head node has been virtualized and separated in a couple of virtual machines. This has some major advantages:

- Different functionality can be separated into separate virtual machines. This even allows for different operating systems (e.g. Fedora Linux and Debian Linux) running on these machines.
- Virtual machines allow easy backups with snapshots of running machines.
- Virtual machines can be cloned and the clone can then be used for development and testing purposes, it can even be sent to other sites.
- The virtualization host provides a remote control (e.g. console login) over virtual machines which is an additional way of access in case the virtual machine is not working properly.

As a virtualization solution VMWare's ESXi 4 is being used but other solutions like Xen and VirtualBox are also being examined. Currently the head node in Kaiserslautern (main site) has virtual machines for monitoring (section 3.5), Planet-Lab Central (section 3.4), a file server, the head node software and various machines for testing purposes.

Headnode Software. The headnode software manages and controls all local nodes at a site. It provides the following services:

- Administration of the local network segment using DHCP
- Provision of boot images for the associated nodes using PXE netboot (see section 3.3).
- Administration of access to the management interfaces of the local nodes via VPN[2].
- Proxy for monitoring that allows the central monitoring server to monitor the management interfaces (see section 3.5).

This system is provided as an ISO image that stores local changes to a disk. So all sites have the same base system with local modifications which allows for easier development.

3.3 Flexible Software Deployment

The headnode software of the local site provides boot images for the nodes via PXE[3] Netboot. Thus any boot image can be booted on any node. In the context of German-Lab we define three categories of boot images:

[2] Virtual Private Network.
[3] Preboot Execution Environment.

1. Planet-Lab boot image (described in section 3.4): This allows a node to boot the Planet-Lab software which is the default. This boot image contains a part that is specific to each node.
2. Virtualization boot image: This kind of boot image provides virtualization with access for all German-Lab users. Thus users can use nodes booted with this image to run custom software images by means of the used virtualization technology. As virtualizers we have developed a boot image using VirtualBox[2] and currently develop a boot image using Xen[3] and KVM[4].
3. Custom boot images: This kind of boot image contains a system designed by a user and only allows access to a limited user group specified by the system itself.

There is a clear trade-off between access for more users and more privileges for users. Planet-Lab provides a very good virtualization when measured in the number of concurrent users that it allows, but it is very limited in the hardware access it provides (e.g. only TCP and UDP sockets, no raw sockets). Custom boot images can provide full hardware access and also allow for kernel modifications but restrict the number of users that can access the node.

The German-Lab experimental facility allows both, access for all users to almost all nodes (Planet-Lab software is the default) and full access to a few nodes if needed. A central management platform for distributing boot images and assigning them to the nodes is being developed.

3.4 Planet-Lab Usage

Planet-Lab[5, 6, 7] is a software, that allows to virtualize nodes using the VServer technology and which provides a central managing and control platform. There is also a testbed called Planet-Lab (for which the software has been designed) with which we do not currently share resources.

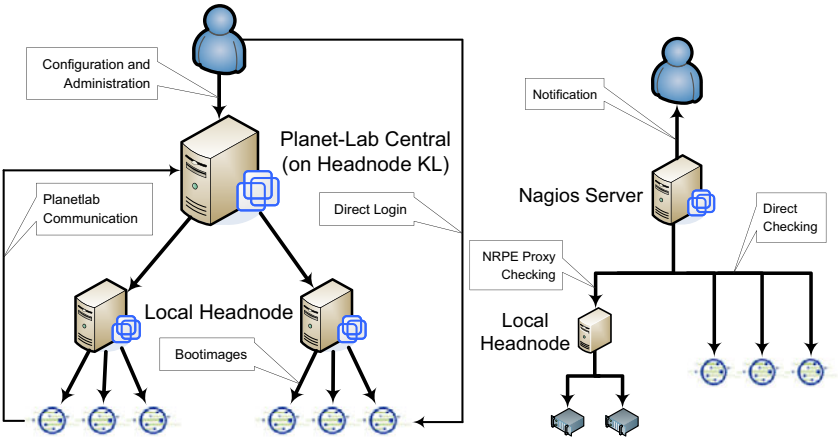
The Planet-Lab software consists of a central server called Planet-Lab Central (PLC) and a boot image for all nodes. On the PLC all sites, users and nodes can be configured and a custom boot image for each node can be generated.

In German-Lab the PLC runs in a virtual machine on the head node in Kaiserslautern. In the Planet-Lab testbed the boot image is booted from a CD or a USB device but in German-Lab that has been modified to be used as a PXE boot image that is provided by the head node software at each site.

Figure 2a shows how the Planet-Lab software is used in German-Lab. The user configures its node on the PLC, which then provides a custom boot image. This boot image is used on the local headnode to boot the node via PXE. Once the node is booted, the node only communicates with the PLC and the user.

3.5 Central Monitoring

The monitoring of the entire infrastructure is also part of the goal. A dedicated virtual server in Kaiserslautern is used for the monitoring infrastructure. The software Nagios[8] is being used to collect monitoring data of individual hosts and services and notify administrators by e-mail when problems occur. Information that is currently monitored is:



(a) Planet-Lab usage (b) Monitoring

Fig. 2. German-Lab Structure

- Resource usage (CPU, memory, disk, etc.) on all virtual machines
- Hardware health of all nodes (using the service processors)
- Availability of all nodes and service processors

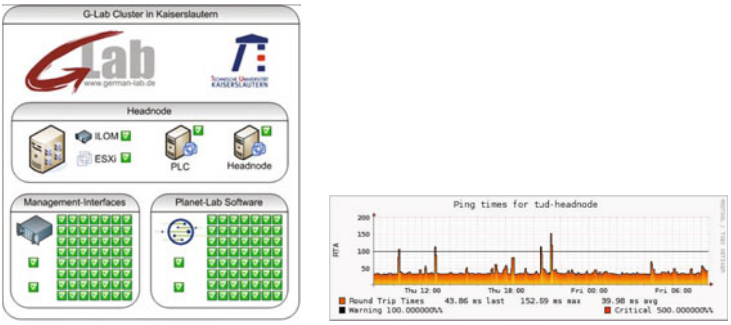
Some of this information is not visible for the monitoring server like resource usage on distant hosts and host and service information of hosts that are not visible from the server like the service processors. To allow the monitoring of these hosts and services the Nagios Remote Plugin Executor (NRPE)[9] software is being used as a proxy. NRPE is a server that allows specified hosts (i.e. the G-Lab monitoring server) to execute preconfigured commands. With this proxy both internal data and hidden hosts can be checked.

To configure the data for the Nagios software (e.g. hosts, services, check commands, users), Nagios Administrator[10] is used. The monitoring information can be visualized in two ways (see Figure 3):

1. A structure diagram gives the current state of each host or host group with green, yellow or red lights. The NagVis[11] software is used for this purpose.
2. Using PNP4Nagios[12] the history of monitored values can be visualized in a timeline graph for each host and each service.

The web-frontends of Nagios, the Nagios Administrator and both visualization tools have been combined in a central website[13]. Of course all monitoring information is also being stored in log files so that future visualization or analysis can work on the history too.

The G-Lab monitoring architecture has been valuable since it was deployed and helps to detect and solve problems quickly. Problems that can be fixed without hardware change have frequently been solved within a few hours.



(a) NagVis (b) PNP4Nagios

Fig. 3. Nagios frontends

3.6 Identity Management

The user management is an important part in an experimental facility supporting project. Especially the organization of the identity of an user and his access rights is a critical issue in public available experimental facility design. In case of the G-Lab project the user management is necessary in two different areas, the infrastructure services, and the testbed platform itself.

The infrastructure services consist of the internal and external project documentation area, mailing lists, help desk, and software management. Whereas the testbed itself can be divided into management and experimenter view. The experimenter requires access to the nodes and testbed resources on several layers. As standard software in G-Lab, the Planet-Lab environment is used, also for the management of access rights. For deploying and operating specialized images a central account management is provided.

The administration of the users and system resources is done by a distributed administration team organized as a sub project of the overall G-Lab project. Each site might have some equipment, but at least users for the facility equipment. The approach distributes the responsibilities for the users assigned to a specific site to a representative of this site. This procedure requires additional role and access rights assignments for an extended group of identities. For example the headnodes, the node management and monitoring, and the private Planet-Lab node administration are typical tasks, which are delegated to site representatives. Also a site representative has to organize the experiments and the resource usage of that site.

Figure 4 shows an architectural overview of the technical structure of the G-Lab identity and role dependency management. In general a central LDAP server stores the users identities in a separate subtree, which is suborganized in subtrees containing the users of a specific site. A basic rule is, that a identity is not associated with any access rights. This is organized in a separate tree, the so called group tree. Each service is represented by a unique group, which grants its members access to this server. A third separate subtree organizes virtual identities on machine level, so that each site has its own system level access user. This enables a fine grained and easy manageable environment on site level, even in case of changes. For services like the private Planet-Lab installation an account synchronization will be realized, so that the central

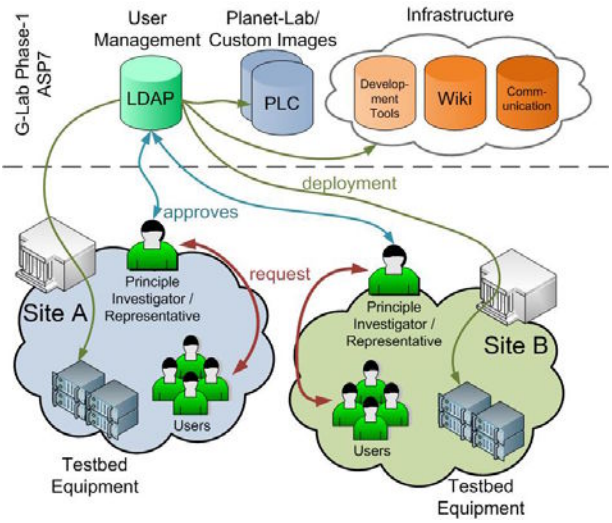


Fig. 4. Identity Management

LDAP database serves as master environment. This can easily be extended to future services, if required. The management of the central database is done by a set of scripts, which respect a set of defined default roles for specific tasks. Also these scripts verify the integrity of the stored user data.

4 Conclusion and Future Work

On a technical level the platform can currently be used to run various software either in the Planet-Lab system, in a virtualized system or in a custom system directly on the hardware. This provides maximal flexibility for experimenters and thus increases the usage of the platform. In the future the platform will be extended by a frontend that allows all users to schedule their experiments and to set up nodes with their boot images. Also the components of the platform will be integrated even more. Monitoring experiments directly will become possible with the CoMon[7] software.

The German-Lab platform has been developed for nearly a year and is running now approximately nine months. At this point an evaluation of some decisions from the very beginning can be done. The first lesson was that virtualization is very important. It provides so many useful features even to a single system that it should be used on servers regardless whether multiple systems are needed or not. The migration from a system that is running on the hardware to a virtualized system can be complicated. An important observation is that monitoring can be very helpful when developing a testbed. So monitoring is not an additional component, it must be integrated into the architecture and should be developed as early as possible. It has also been discovered that VMWare ESXi does not provide all functionality that is needed to develop and run the experimental facility. Especially cloning or snapshotting of running virtual machines or live

migration are not supported by the free version. An extended analysis of existing virtualization solutions can be worth the time because migrations from one virtualizer to another can be complicated.

To ensure the sustainability and continuous development of the platform one G-Lab-Association will be founded. The association will be joined by the partners of the industry and first and second phase of the project though others could also take part in it. The participation in the association was given special attention in the announcement of the second phase. In the past months there were several conversations, especially with industrial partners, in order to clarify whether such a platform could be used under the commercial terms and conditions. It has been experienced that manufacturers are interested and forced by quality control services to test and verify their products in a "real" environment before bringing it into the market. Which gives a developed platform extra importance in commercial market besides many infrastructure providers also shown the interest to test their product in "post-IP" environment.

4.1 Emulations of Network Properties

In the current G-Lab environment the network link characteristics between the clusters are excellent. In the real Internet is this not the case. We can observe different link characteristics, such as packet loss, packet delay and jitter. To provide the experiments of the G-Lab project with realistic environments these network characteristics must be emulated.

As future work we plan to create different emulation scenarios. The first scenario is the simple link emulation between two hosts, the second is the emulation of multi-homing and the third is the emulation of a complex network. We also plan to develop a measurement experiment to measure the three developed scenarios. In the emulation scenarios we want to emulate different datalines like DSL, WLAN or Satellite. The value of the network parameters should be configurable at run time, which might be necessary for some experiments.

Various tools are in use today to design models equivalent or similar to actual network environments. We plan to analyze different tools like the Network Simulator, Traffic Control and Dummynet.

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Design and Evaluation of a Socket Emulator for Publish/Subscribe Networks*

George Xylomenos and Blerim Cici

Mobile Multimedia Laboratory & Department of Informatics
Athens University of Economics and Business, Patision 76,
Athens 104 34, Greece
xgeorge@aueb.gr,blerim153@gmail.com

Abstract. In order for a Future Internet architecture to be globally deployed, it must ensure that existing applications will continue to operate efficiently on top of it. As part of the Publish Subscribe Internet Routing Paradigm (PSIRP) project, we have explored various options for making endpoint centric applications based on the Sockets Application Programming Interface (API) compatible with the information centric PSIRP prototype implementation. We developed an emulator that mediates between the client/server socket calls and the publish/subscribe PSIRP calls, transforming the exchange of packets to distribution of publications. To assess the overhead of our emulator, we measure the execution time of a simple file transfer application in native socket mode, in emulated socket mode and in native publish/subscribe mode.

Keywords: TCP/IP, Sockets, Publish/Subscribe, PSIRP.

1 Introduction

A large fraction of current Internet traffic is due to peer to peer content distribution applications [5], in which participants are solely interested in the exchanged data rather than in the endpoint addresses of their peers. This indicates that the Internet is evolving from a network connecting pairs of end hosts to a substrate for information dissemination. There are many proposals for evolving or redesigning the Internet architecture based on an information centric paradigm, for example, the *Content Centric Networking* (CCN) [3] project and the *Publish/Subscribe Internet Routing Paradigm* (PSIRP) project [2]. The PSIRP project in particular is working on a network architecture based entirely on publish/subscribe principles, and its prototype implementation employs publish/subscribe concepts throughout the protocol stack [4]. In the publish/subscribe model, publishers announce available data, subscribers express their interests, and the network allows them to rendezvous for the exchange of data.

In order to be deployed, an information centric architecture must ensure that it will be possible to efficiently execute existing applications on top of it. While

* Work supported by the ICT PSIRP project under contract ICT-2007-216173.

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 11–19, 2010.
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content distribution applications may be expected to be rewritten so as to operate optimally over an information centric architecture, a vast number of existing endpoint centric applications will have to operate in some kind of compatibility mode. Since most existing Internet applications were written on top of the Sockets *Application Programming Interface* (API) [7], the most direct way to make them compatible with a new architecture is to develop middleware to translate Socket API calls to the information centric calls of the new architecture.

In this paper we describe and evaluate a Socket API emulator for PSIRP, which allows unmodified Internet applications to operate on top of a native publish/subscribe protocol stack. In Section 2 we introduce the basic concepts of the PSIRP architecture and implementation. In Section 3 we discuss the different emulation options available and motivate our selection. In Section 4 we explain how IP addresses and socket calls are translated into PSIRP calls. In Section 5 we evaluate our emulator by comparing the performance of an application in native socket mode, in emulated socket mode and in native publish/subscribe mode. Finally, in Section 6 we summarize our work.

2 PSIRP Implementation Concepts

In the PSIRP prototype implementation, which uses the FreeBSD operating system, publications are handled via a set of calls encapsulated in the `libpsirp` library [4]. To understand how publish/subscribe communication is achieved, in this section we provide an introduction to the `libpsirp` concepts and calls. In the PSIRP architecture, the central entity is a *publication* which is made available by *publishers* to *subscribers*. The network provides mechanisms for publishers and subscribers to rendezvous in order for publications to be transported from the former to the latter. A publication is identified by a *Scope Identifier* (SId) and a *Rendezvous Identifier* (RId) [2]; the SId represents an information collection, while the RId represents an information item within this collection. For example, a user may publish a set of holiday pictures, each identified by an RId, within a scope representing his friends, identified by an SId.

Publications consist of data and metadata; data are mapped to the memory space of the publishers and subscribers. A publisher creates a new publication via `psirp_create()`. This allocates a memory area of a specified size for the publication data, initializes a data structure for the publication and returns a handle to this structure. The publisher can call `psirp_pub_data()` using that handle to get a pointer to the memory block of the publication. When the publication is ready, it is passed to the kernel via `psirp_publish()`, which takes as parameters the desired SId and RId for the publication, as well as a handle to it. The kernel can then decide where to forward the publication to. If a publication with the same SId/RId already exists, the new publication is assumed to be a new version, therefore its version number is increased. A SId or RId in ASCII format is converted to the internal `libpsirp` format by `psirp_atoid()`.

A subscriber calls `psirp_subscribe_sync()` with a specific SId/RId pair to subscribe to a publication. This blocks the subscriber until a matching publication is found or until a timeout expires; in the former case, a handle is

returned to the latest version of the publication. The caller can distinguish new from old versions of a publication by asking for their version numbers via `psirp_pub_version_count()`. To retrieve previous versions, the subscriber must call `psirp_subscribe_versions()`, which returns an array of handles to earlier versions of a specified publication. Finally, `psirp_free()` frees the publication structure and unmaps the memory allocated for the publication.

3 Emulation Options

In the Sockets API, a socket represents a communication endpoint, identified by an IP address and a TCP/UDP port. Communication takes place by having each application attach to a local socket and perform calls on it. The actual communication between sockets is achieved by exploiting the services of the TCP/UDP protocols. As shown in Figure 1.(a), the socket uses either TCP or UDP at the transport layer, the transport layer uses IP at the network layer, and IP uses some lower layer protocol (such as Ethernet) for data transmission [7]. In contrast, in the PSIRP prototype, publish/subscribe applications talk to `libpsirp` which implements its own transport and network layer protocols on top of the lower layers. The goal of the Sockets API emulator is therefore to translate between socket calls and `libpsirp` calls, despite their different approaches.

One emulation approach, shown in Figure 1.(b) is to exploit an existing TCP/UDP/IP implementation to transform the socket calls to IP packets, and then exchange these packets via `libpsirp` calls. The advantage of this approach is that the emulator only has to provide a best effort service, analogous to that offered by IP. For TCP in particular, flow, congestion and error control are

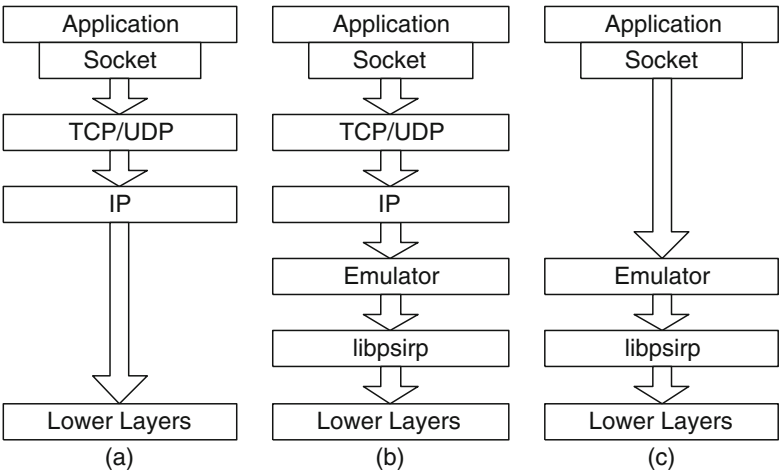

Fig. 1. Socket emulator structure: (a) standard TCP/IP stack, (b) network level emulation, (c) transport level emulation

essentially provided by TCP, and the emulator only sees IP packets. The disadvantage is that by treating PSIRP as a dumb transport, not only do we lose the advantages of its redesigned architecture, we also apply IP specific TCP assumptions to an entirely different architecture. A similar approach has been found to be very detrimental for the performance of TCP applications on top of ATM networks [1]. In addition, going through the TCP/UDP/IP implementation represents a significant communication overhead for the emulator.

The other approach, shown in Figure 1.(c) is to translate each socket call directly to `libpsirp` calls. While this is roughly the same as above for UDP, for TCP it is considerably harder, as the emulator needs to deal with flow, congestion and error control. However, in addition to avoiding TCP/UDP/IP overhead, in this manner the emulator can take full advantage of the facilities provided by `libpsirp`. For example, if the PSIRP prototype provided a reliable transport service for publish/subscribe networks, this transport could be used instead of TCP. Despite the additional complexity, this approach will provide better performance in the long term, therefore we have selected it for the emulator.

4 Emulator Implementation

4.1 Mapping Addresses to Identifiers

Since there is no notion of endpoint identifiers in PSIRP, the Sockets API emulator must translate the TCP/UDP/IP addresses used by sockets to the SId/RId pairs used to exchange publications in PSIRP. The scheme that we implemented is to create an SId for each machine based on its IP address and an RId for each socket in that machine by combining its IP address, its port number and the protocol (TCP or UDP). Therefore publishing to an SId translates to sending data to a machine, while publishing to an RId translates to sending data to a port of that machine.

A client can communicate with a server via the socket emulator based only on the server's endpoint details, exactly as in TCP/UDP/IP. Say that a client

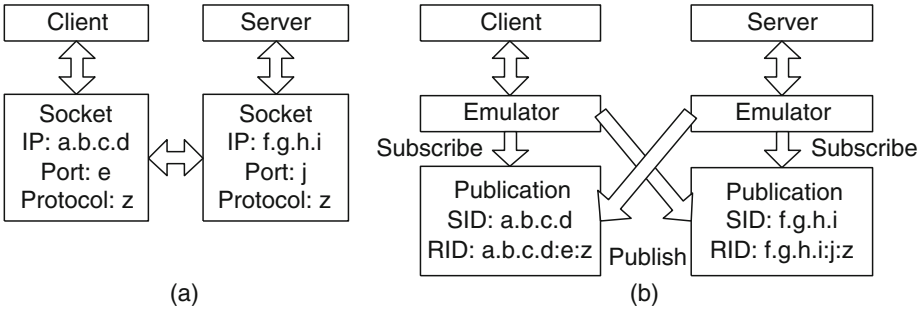

Fig. 2. Address translation: (a) standard TCP/IP socket, (b) emulated socket

with an IP address of `a.b.c.d` uses port `e` of protocol `z` to communicate with a server with an IP address of `f.g.h.i` using port `j` of protocol `z`, as shown in Figure 2.(a). The emulator translates the client to server messages to publications to the SId generated by `f.g.h.i` (the server's IP address) and the RId generated by `f.g.h.i:j:z`. In the server to client direction, messages are translated to publications to the SId generated by `a.b.c.d` (the client's IP address) and the RId generated by `a.b.c.d:e:z`. This arrangement is shown in Figure 2.(b). New messages sent in the same direction are represented by new versions of the same publication, thus allowing sequences of packets to be transmitted.

4.2 Datagram Socket Calls

Sockets come in two varieties: Datagram sockets, implemented on top of UDP, and Stream sockets, implemented on top of TCP. In this subsection we explain how Datagram socket calls are emulated, while the next one deals with Stream socket calls. Figure 3 shows how Datagram calls are emulated; dotted arrows show how Socket calls are mapped to emulator actions, while solid arrows show the publications exchanged between machines. The server first calls `socket()` to create a data structure for its communication endpoint and get a handle to it for later use; this translates to the creation of an equivalent data structure in the emulator. In order for the socket to become accessible to clients, the server calls

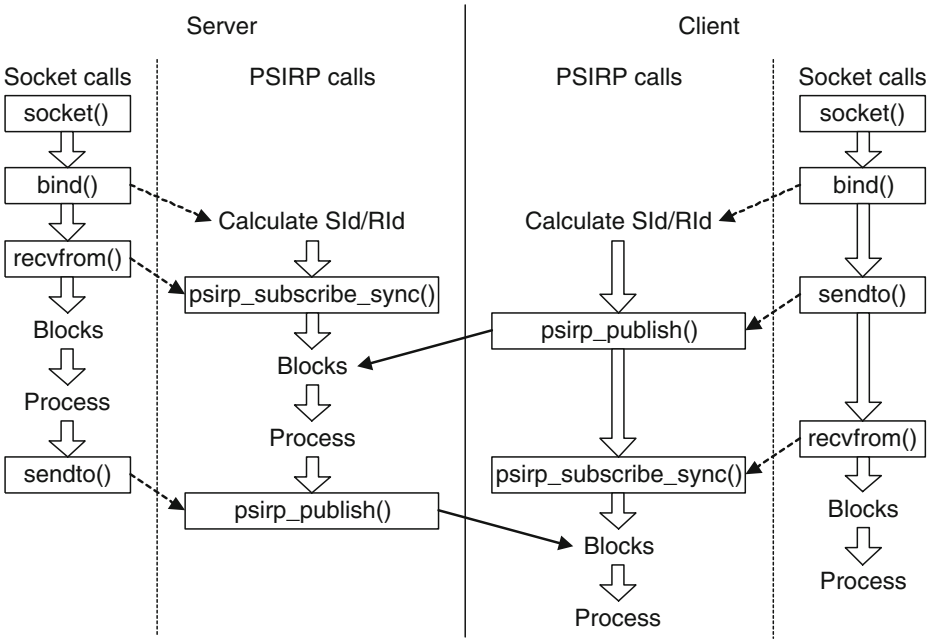

Fig. 3. Datagram socket calls

`bind()` to assign an IP address and a UDP port to the socket; the emulator uses this information to calculate an SId/RId pair for incoming data and stores both the socket address and the PSIRP identifiers in its own structure. The client performs the exact same calls before communication.

In order to receive data, the server issues the `recvfrom()` call on the socket, which is translated by the emulator to a `psirp_subscribe_sync()` call on its incoming SId/RId pair. To distinguish consecutive packets, the emulator ensures that each `recvfrom()` call returns the next version of the same publication; the last version number seen is stored in the socket structure. Each publication contains in its metadata field the IP address and UDP port from which the message was sent. The emulator passes these data to the server via the return parameters of the `recvfrom()` call, so that the server may later use them to send replies. The `sendto()` call is translated by the emulator to a `psirp_publish()` call on the outgoing SId/RId pair generated by the IP address and UDP port provided by the caller in the socket call. In addition, the IP address and UDP port stored in the socket structure of the sender are inserted as metadata in the publication, as explained above. The behavior of the client is symmetric; the only difference is that the client must know in advance the IP address and UDP port of the server to issue the first `sendto()` call.

4.3 Stream Socket Calls

Figure 4 shows how Stream calls are emulated. The `socket()` and `bind()` calls (the latter is optional on the client side) operate exactly as in the Datagram case, leading to the calculation of an SId/RId pair for incoming publications at each endpoint. Only the structure created in the emulator is different: a connected Stream socket must store both local and remote endpoint address and SId/RId pairs, since in Stream sockets data transfer calls do not indicate addresses, unlike in Datagram sockets. The `listen()` call is only used for housekeeping: it creates a list for storing incoming connection requests until the emulator can service them.

The main differentiation from a Datagram Socket however is that in a Stream socket a new socket needs to be created on the server side when a connection is established, leaving the original socket to handle additional connection requests. When `accept()` is called to indicate that the server is ready to receive a new connection request, the emulator calls `psirp_subscribe_sync()` on its incoming SId/RId pair in order to receive the next connection request. On the client side, when `connect()` is called to initiate a connection, the emulator first uses the IP address and TCP port passed to that call, which the client knows in advance, to calculate the SId/RId pair of the server and then calls `psirp_publish()` to send it an empty publication, containing as metadata its own IP address and TCP port. Finally, the client calls `psirp_subscribe_sync()` on its incoming SId/RId pair and waits for a reply from the server.

When the server receives the client's publication, the emulator creates a new socket structure, using the local endpoint address from the existing socket and the remote endpoint address from the publication metadata. The server

calculates the SId and RId for each endpoint as usual, but then it XORs the original local and remote RId and stores the result as its new local RId. As a result, connected sockets are differentiated in the server from unconnected ones as they use both endpoint addresses to calculate the RId for incoming data. Finally, the server calls `psirp_publish()` to send an empty publication to the client's incoming SId/RId pair. When this publication is received by the client, the client's socket structure is also updated by calculating the new incoming SId/RId pair of the server as above and the `connect()` call returns.

At this point connection establishment is complete, and either side can use the `send()` and `recv()` calls to send and receive data, respectively, without

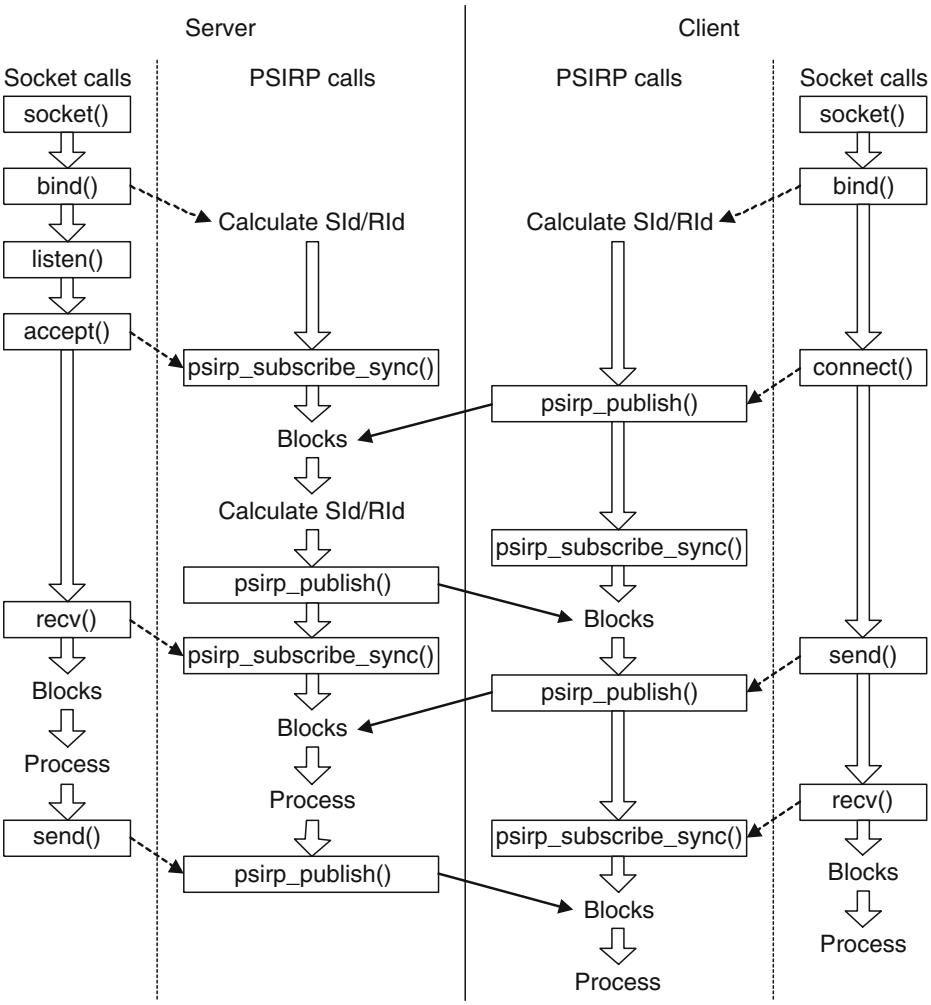

Fig. 4. Stream socket calls

indicating a destination address. Due to the modified server RId used for connected sockets, there is no confusion between publications to connected sockets (data) and unconnected sockets (connection requests).

5 Performance Assessment

In order to assess the overhead of the Socket API emulator, we implemented a *Trivial File Transfer Protocol* (TFTP) [6] application which transfers files using a simple stop and wait protocol over a Datagram socket. This application can execute either over a native UDP/IP implementation or over our emulator. Since the emulator operates over the PSIRP prototype implementation which introduces its own overhead, we also wrote a native `libpsirp` version of the TFTP client and server, by manually replacing socket calls with the corresponding publish and subscribe calls, as explained in the previous section. By executing the same experiments with each TFTP version we can assess the overhead incurred by `libpsirp` when a socket application is ported to it, and the additional overhead incurred by our emulator to run unmodified socket applications.

Due to limitations in the PSIRP prototype, we use the TFTP client and server to transport a small file consisting of 98 data packets with 1 KByte data payloads over an Ethernet. In all versions, a TFTP header is attached to each data packet, and then either UDP/IP headers, PSIRP headers or Socket API emulator metadata and PSIRP headers are added to each packet. We use the `time` command to compute the user space and system space computation time incurred by each version. Due to the coarse (10 ms) granularity of these timers, in each experiment we perform 3 transfers back to back to reduce zero timings;

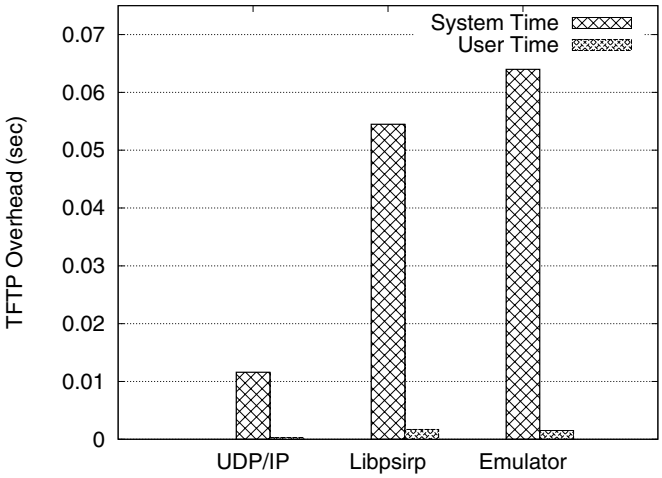

Fig. 5. User and System overhead for each TFTP version

we repeated each experiment 25 times. In Figure 5 we show the average user and system time for the TFTP versions considered: the native socket version, the native `libpsirp` version and the emulated socket version.

Despite the coarse timer granularity, it is clear that the PSIRP prototype is slower than the native UDP/IP stack: even the native `libpsirp` TFTP version is much slower than the UDP/IP version. This is not surprising considering that this is an early prototype, while the UDP/IP stack is being optimized for 20 years. On the other hand, the emulated socket version is only 17% slower than the native `libpsirp` version, indicating that even though the emulator is unaware of the nature of the application, the automated translation between socket and `libpsirp` calls is not very costly. This is very encouraging, as it means that as the PSIRP prototype implementation becomes more optimized, the performance of the emulated socket applications will also improve accordingly.

6 Conclusions

We have presented the design and implementation of a Sockets API emulator for the publish/subscribe oriented prototype implementation of the PSIRP architecture. This emulator translates the socket calls used by existing Internet applications into the calls provided by the `libpsirp` library of the PSIRP prototype implementation. Our preliminary performance evaluation indicates that the performance overhead introduced by the emulator is quite low, on the order of 17%, thus it is a reasonable option for the execution of socket applications over PSIRP.

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Publish/Subscribe on Top of DHT Using RETE Algorithm

Yan Shvartzshnaider[1], Maximilian Ott[2], and David Levy[1]

[1] School of Electrical and Information Engineering
The University of Sydney, Australia
yshv6985@uni.sydney.edu.au,
david.levy@sydney.edu.au
[2] National ICT Australia (NICTA)
max.ott@nicta.com.au

Abstract. This paper discusses the construction of a Global Semantic Graph (GSG) [1] to support future information- and collaboration-centric applications and services. The GSG is a publish/subscribe (pub/sub) based architecture that supports publication of tuples and subscriptions with standing graph queries. We believe that an implementation of an efficient pattern matching algorithm such as Rete [2] on top of a distributed environment might serve as a possible substrate for GSG's pub/sub facility. Rete operates on loosely coupled alpha, beta and join nodes and therefore has been chosen by us for implementation in a distributed setting.

In this paper, we propose a way to perform Rete's pattern matching over a DHT-based Structured P2P network to provide a scalable content-based publish/subscribe service.

Keywords: publish/subscribe system, distributed pattern matching, global semantic graph.

1 Introduction

The distributed publish/subscribe (pub/sub) interaction schema is viewed by many [3, 4, 5] as a more suitable communications paradigm for future Internet architecture as it better reflects the dynamic and asynchronous nature of today's Internet applications and services [6].

A typical pub/sub system offers a loosely coupled, event-based communication schema by offering a spatial, temporal, and synchronisation decoupling between publishers and subscribers, which is claimed to work well in a large-scale distributed environment [6, 5]. There are several variations of pub/sub schemas such as *topic-based*, *content-based* or *typed-based* schemas [6]. In this paper we focus on the implementation of a content-based (or property-based [5]) publish/subscribe schema where the subscription is based on the properties of the published information. To implement such schema on a large and distributed scale there is a need for an efficient and scalable pattern matching system that will support "*event filters*, which are predicates on the content of associated information, and *event patterns*, which are predicates on the relationships among event occurrences" [5].

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 20–29, 2010.
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The Artificial Intelligence (AI) community has widely studied the development of efficient matching algorithms in relation to *production rule systems*. Since their initial introduction in 1943 this work has greatly evolved [7] and is now is widely adopted by various AI systems, such as experts systems. A production rule system, in its simple design, comprises of three main parts: a *set of rules*, a *dataset* and *a rule interpreter* [8]. The rules are evaluated against the dataset by the interpreter and appropriate action is executed. For our purposes, we are primarily interested in the rules evolution part. In particular, our design is based on Rete – an efficient pattern matching algorithm – which is widely employed in production rule- based systems to match data against multiple productions. Rete operates on loosely coupled alpha, beta and join nodes and therefore has been chosen by us for implementation in distributed settings.

The emerging Peer-to-Peer (P2P) networks [9, 10, 11], in particular structured P2P networked overlays have been used as a building block for many global sharing and content distribution systems [12]. In such networks the core characteristics such as efficient routing, key-search, self-organisation, fault tolerance and good load balancing are offered by a Distributed Hash Table (DHT). Thus, in their basic form, they provide a relatively simple "hash-like" interaction interface (e.g., *get(key), put(key, value)*) which is not sufficient for a content-based publish/subscribe system [13] that requires support for more complex and expressive queries. Our system inherits support for such queries from the Rete algorithm.

The rest of the paper is organised as follows. Our motivation for the project is in Section 2. The design overview is covered and discussed in Section 3. In Section 4 we briefly review the Rete algorithm. Section 4.2 provides an overview of related work. Finally, Section 5 summarises our conclusions and gives some notes on future work.

2 Motivation

Our main motivation for this work is the development of a Global Semantic Graph (GSG) [1] to support future information- and collaboration-centric applications and services. The GSG is an Internet-scale tuple store that adopts approaches and methods from the Semantic Web to provide applications and services with the abilty to simply publish their internal state changes via simple tuple insertions, while a subscription is essentially a standing query to a specific pattern that keeps the internal state synchronised with any insertion of new information.

We strongly believe that a global infrastructure like the GSG will provide a convenient, powerful, and sound basis for building novel information-centric applications and services.

3 Design Overview

As mentioned in the introduction, in our approach we port the Rete algorithm onto a DHT-based overlay network to provide a scalable content-based publish/subscribe system.

Our system treats a tuple as a primitive – publications are tuples and subscriptions comprise of rule tuples which are bound by variables into a standing graph query. As a standing query, it remains in the system and continuously monitors the tuple-space to return a set of matched tuples. Each rule tuple is converted to tuple templates by dismissing the binding variables in the rule and replacing them with wildcards – i.e., only rules' constant attributes matter. Hence, for example, the (?a left-to ?b) rule template is represented with a (* left-to *) template. Every tuple template is associated with tuple-storage (also called alpha-memories) in the Rete network. It is worth noting that the *alpha memories* in a Rete network act as a tuple's local cache. Effective caching policies are an open research question in many fields and can significantly improve the performance of the overall system. Although all subscriptions are stored and managed locally by a single Rete network, we achieve scalability by distributing tuple templates among the DHT nodes. The classic DHT architecture is extended to support a separate storage for published tuples and tuple templates. This includes: mapping of every tuple and tuple template to unique DHT keys, similar to RDFPeers [14] and introduction of a new insert_condition(key, tuple pattern, subscribing node ID) method. Thus, as depicted by Figure 1, during publication, all tuples will be routed to a rendezvous node and matched against the tuple templates stored there. In case of a match, the tuple is forwarded to the subscribing node to complete the matching process.

3.1 Implementation

Subscription. As mentioned above, each subscription is comprised of rule-tuples that are converted to tuple templates and distributed over the DHT. In more

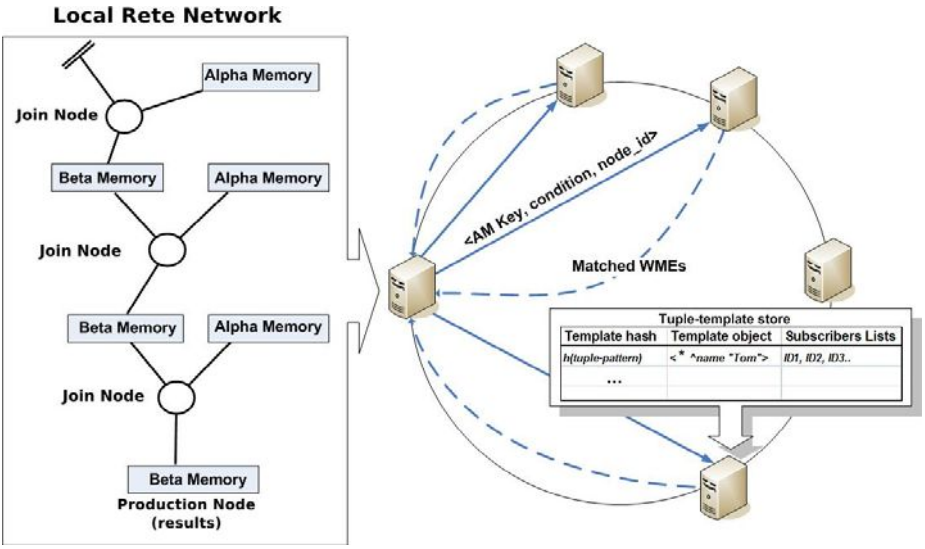

Fig. 1. An extremely simplified illustration of our implementation approach

practical terms, we pick any constant from a tuple template associated with the rule and hash it to create <key,tuple template>. The tuple template is then routed to a designated rendezvous node where it is stored in a "Tuple template Store". See Figure 1 for a simplified illustration of this process.

Publication. During the publication of a tuple, we hash each of the fields separately (e.g., *h(subject), h(predicate), h(object)*) to create three unique DHT keys for the same tuple, each key is used to route a copy of the tuple to a different rendezvous node. By using the same hash function for both subscription and publication, we guarantee that tuple and relevant tuple templates will rendezvous. The tuples matched at the *rendezvous node* are then delivered to relevant subscribers nodes.

```
SUBSCRIBE {
       (?x, ^on, ?y),
       (?y, ^left-of, ?z),
   NOT {
           (?z, ^color, red),
           (?z, ^color, green),
           (?z, ^color, blue)
       }
      }
```

(a)

```
INSERT  {
     (B1,^on,B2),
     (B3,^color,green),
     (B1,^on,B3),
     (B1,^color,red),
     (B7,^color,red),
     (B2,^on,table),
     (B2,^left-of,B3)
    }
```

(b)

Fig. 2. Script Example (a) create a subscription, (b) inserts (publishes) collection of tuples

For our implementation of the Rete algorithm we have used the newly emerging Scala[1] language. Scala is a general purpose, object oriented and functional hybrid programming language. The Scala compiler produces Java binary code, and hence can be seamlessly integrated with existing Java solutions. This allows us to use our implementation with openChord[2], a Java implementation of the Chord [9] algorithm. Our implementation also includes a small Backus–Naur Form (BNF) grammar parser for a scripting language, similar to SPARQLs graph pattern query[3] syntax, to allow creation of more complex subscriptions and publication commands. See Figure 2 for a sample syntax of the script.

[1] http://www.scala-lang.org/
[2] http://open-chord.sourceforge.net/
[3] http://www.w3.org/TR/rdf-sparql-query/

3.2 Discussion

We have implemented our approach successfully and we are able to facilitate a distributed content-based publish/subscribe with support for expressive and complex querying over a structured overlay network. We recognise, however, that there is a place for future improvement.

One problem is *load-balancing*. We generate multiple key pairs for the same tuple, hence tuples with identical fields would be always routed to the same node and can cause significant overload when a particular tuple's attribute is shared by many (e.g., predicate: *suchAs* or object: *class* in RDFS scheme).

Our initial ideas is to tackle the *load-balancing* problem by introducing built-in time stamps with every tuple to indicate the time and date range of the publication, for example, *<10/3/2010-23/3/2010,subject, predicate, object>*. So, when hashing the tuple's fields we are insuring a different hash every time. Consequently, when creating a subscription, we need to indicate what is the tuple's valid query range – the range limit must be set in advance to insure proper matching. If a user specifies a range bigger than the limit it would be split into several ranges and subsequently create a separate subscription query per range.

The other issue that is open for debate is *push* versus *pull*-based approach. The current implementation is push-based system, that is, the matched tuples are pushed to the subscribing node. The benefit of this approach is that there is no the need to constantly query for updates, however the subscribing node can find itself overloaded with updates in a case of multiple and/or complex subscriptions. An alternative pull-based system avoids the overloading problem by giving control over the flow of updates back to the subscriber. That is, a subscriber node must periodically query the data store for updates. For our future work, we will explore the possibility of an hybrid querying model: by default the updates will be "pushed" to subscribing nodes, however, if a node becomes overloaded, some update streams will be blocked and queried later.

Finally, we are extending the current implementation and slightly changing the behaviour of a classical DHT-based overlay network. Our ultimate goal is work towards a "lightweight approach" [15] that is, to implement a pub/sub service without changing the DHT-generic algorithm. One option that we are considering is replacing the current hash function that is used to generated DHT keys with an alternative mapping function that better supports range and multi-attribute queries.

In Figure 3 above, subscription 1 returns matches of *all the Movies produced by Steven Spielberg* and Subscription 2 *matches all the Movies produced by Steven Spielberg with Tom Hanks as an actor*. The results of Subscriptions 1 are a subset of the total results returned by Subscription 2, therefore some of the *alpha memories* can be re-used and shared.

Ultimately, we envision the Rete network at the subscribing node utilising the DHT nodes as *alpha memories*, so that similar subscriptions from different subscribers will be able to reuse already available *alpha-memories* and also include other subscriptions as part of their Rete network. For example, as shown in Figure 3, the Rete network for Subscription 2 can comprise of new conditions in

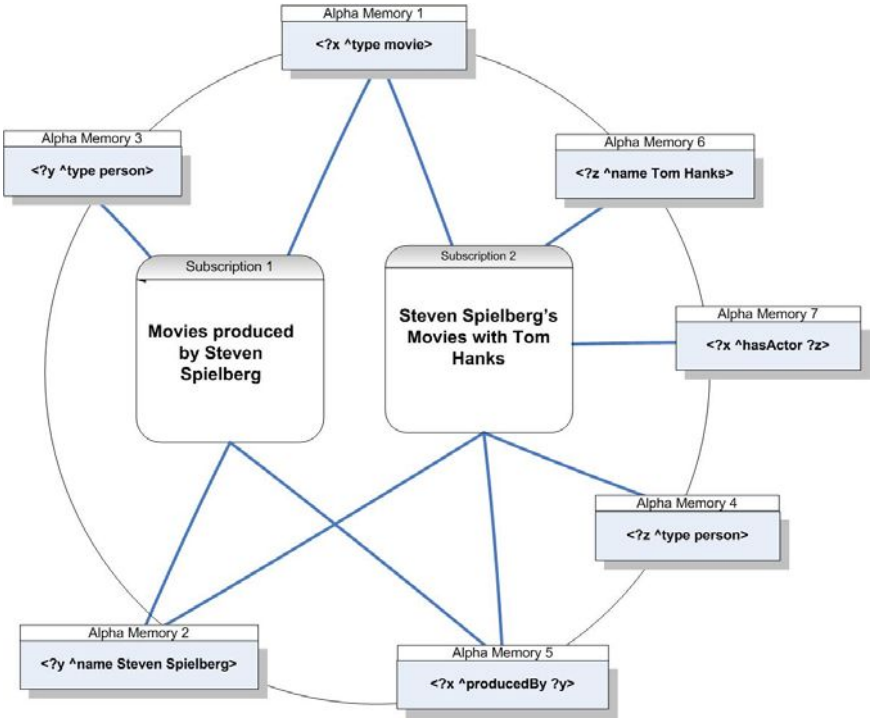

Fig. 3. An intuitive example showing the matching of subscription

conjunction with those previously defined Subscription 1. Such a configuration would allow more efficient utilisation of the Rete algorithm and provide a better platform for our future work [1].

Evaluation. As discussed in Section 3.2, the RETE-based pub/sub service allows the reuse and sharing of the results from past subscriptions. The system will perform better with time as more subscriptions are created. The GSG can take full advantage of this as it will deal with large number of simultaneous users, each with many complex and expressive subscriptions. To our knowledge, benchmarks available to date do not take into account these operational conditions. Hence, there is a need for a new benchmark to properly evaluate these types of systems. A proper evolution is part of our future work and is out of scope of this paper.

4 Rete Algorithm

This section provides a brief summary of the Rete algorithm from Chapter 2 of Doorenbos' [16] thesis and the motivation for using it in our implementation.

Rete [2] is an efficient and well-adapted pattern matching algorithm that is widely employed in production-based expert systems. It reduces the matching process time, by exploiting the "temporal redundancy" and "structural similarity" of the data; each state of a matching cycle is saved separately in the loosely coupled Rete dataflow network and can be reused in future pattern matches [17].

4.1 Rete Primitives

Rete operates on *productions* and *working memory elements* (WME). A *production* is defined by set of *conditions* (also called rules) that are evaluated against the WME dataset and *actions* which are executed when these conditions have been met.

4.2 Rete Dataflow Network

As depicted in Figure 4, the Rete-dataflow network comprises of *alpha memory*, *beta memory* and *join* nodes. The *alpha memory* (AM) node is part "alpha network" that acts as a predicate on the WMEs. For example, the AM node in Figure 4, that is defined by Condition 1 (<?x> ***on*** <?y>) contains WMEs that match the corresponding template (i.e., WME with attribute **on**). The AM treats *?x, ?y* as wildcards.

The *beta memory* (BM) nodes contain the result of *join* nodes. The input to "join" is the previous *beta node* and the corresponding *alpha* node. In other words, the *join* node filters through all the WME joined on variable values to the related *beta* nodes in a form of a join tuple; this process is called *left activation*.

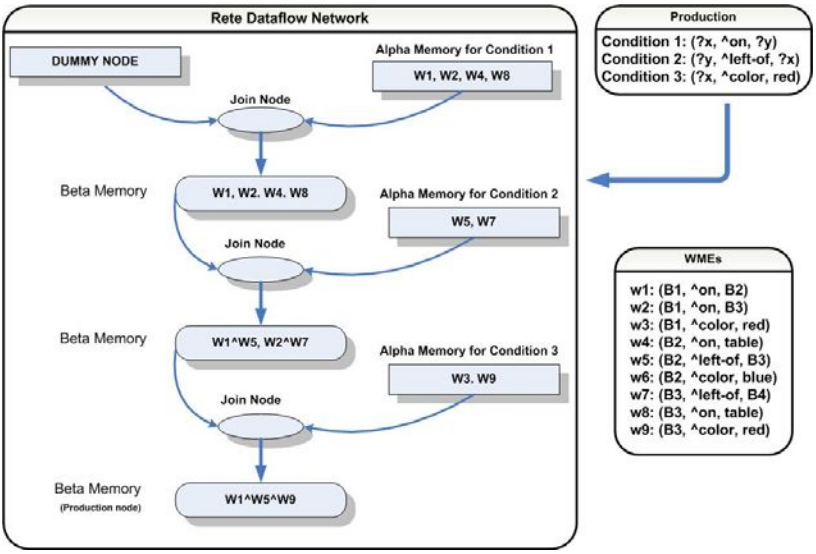

Fig. 4. Rete dataflow network acts a filter, inspired by [16]

On other hand, *right activation* is triggered by feeding a new WME into an appropriate AM node.

The matching process is a chain reaction of either *right* or *left* activations by the end of which all the matched WMEs are found in the *production node* which are essentially any last BM node in the chain flow.

5 Related Work

Several research efforts that have looked into implementing a content-based publish/subscribe communication schema over a DHT overlay.

In [18] the authors propose a rendezvous-based approach to publish/subscribe over the DHT overlay. The *rendezvous nodes* that are responsible for matching events, start the notification process. The system comprises of a three-layer design: *application*, *CB-pub/sub* and *overlay network*. The *CB-pub/sub* layer utilises the underlying overlay network, to provide generic applications that run on it with publish (pub), subscribe (sub) and notification (notify) functionalities. The system performs attribute-split, key space-split and selective-attribute mappings of subscriptions and published events onto the key space. The underlying overlay network is extended with an implementation of an additional one-to-many send primitive (*m_cast*) to optimise the overall system performance.

A distributed query evaluation is offered in [19]. The underlying DHT overlay is extended with *multiSend()*, *index_query()* methods and a new "extended matching" message format to support *Single Query Chain (SQC)* and *Multiple Query Chain (MQC)* algorithms. In SQC and MQC for each conjunctive multi-predicate query a single query and multiple query chains are created, respectively, in order to distribute the processing load between the nodes. Each node matches the received triples against its local queries and sends intermediate results to the next node in the query processing node chain. The final notification to the subscriber node is generated by the last node in the chain. The MQC and SQC, however, significantly increase the overall network traffic which in a large-scal environment will undermine the performance of the system.

More recently, in CAPS [15, 20] a content-based publish/subscribe architecture is proposed for event dissemination infrastructure on top of DHT. The system also employs a *rendezvous* model to facilitate meetings between events and subscriptions. The selection of the *rendezvous nodes* is governed by the internal DHT routing protocol and hence, the chosen node would be known by other nodes in the network. CAPS does not extend the underlying generic DHT distributing algorithm and this makes CAPS adaptable to other DHT-based overlays.

The above approaches put their focus on scalability and efficient retrieval capabilities of a distributed content-based pub/sub system. Much less attention is given to the ultimate expressiveness of the queries. Our approach addresses this problem, which we feel is equally important to the effective operation of the applications that will use our infrastructure.

6 Conclusion and Future Work

We have described our implementation of a content-based publish/subscribe system based on the Rete algorithm. Our design touches on new and less explored fields of combining AI and Peer-to-Peer related research approaches. We use Rete, an efficient matching algorithm that is widely used in AI's production rule based expert systems. The constructed Rete network exploits the "temporal redundancy" of the previously matched information to provide an efficient matching of data (working elements) to predefined conditions. In our case, a Rete network is generated to manage subscriptions. The network keeps matches for specific subscription's condition and shares them with other subscriptions containing the same condition. The efficiency of the Rete algorithm comes from "matching only the changed data elements against the rules rather than repeatedly matching the rules against all the data " [21]. In addition, the conditions' "structural similarity" (i.e., the n-tuples) makes the Rete approach even more effective because many of the condition tuples share same attributes.

The main motivation behind our work is the Global Semantic Graph project [1], which is an Internet-scale persistent publish/subscribe system. The applications and services "publish" their internal state changes via simple *n-tuple* insertion while a "subscription" is essentially a standing query that keeps the internal state synchronised with any new information provided by others. In our future work we plan to examine approaches mentioned in Section 3.2 to improve the the overall system and utilise it as substrate for GSG's pub/sub facility.

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Experimental Testing in the Future Internet PERIMETER Project

Eileen Dillon, Gemma Power, and Frances Cleary Grant

Telecommunications Software and Systems Group, Waterford Institute of Technology, Cork Road, Ireland
{edillon,gpower,fcleary}@tssg.org

Abstract. The Future Internet faces many challenges for new Internet architectures, protocols and services requiring early experimentation and testing in large-scale environments. Furthermore, the complexity of the software and technologies running on the Future Internet will require more innovative and different approaches to the testing process. This paper examines these requirements and challenges and demonstrates how they are dealt with in the Telecommunications Software and Systems Group using a case study of the Future Internet project; PERIMETER. Details of the testing and experimental methodologies and the role of testbed activities involved in this Quality of Experience network mobility project to develop and adapt to the needs of the Future Internet are provided. The paper concludes with observations of the further developments and innovations required to achieve an end-to-end solution for experimental testing in PERIMETER, and more generally for other Future Internet projects.

Keywords: Future Internet, Testbeds, Experimental Testing and Methodologies, PERIMETER.

1 Introduction

The current Internet is fundamental for the functioning of our modern economy and society, in domains as diverse as education, health, defence, commerce, travel and entertainment. It is expected that by 2011 about 3 billion hosts will be connected to the Internet, which is a 500% increase compared to 2008 [1]. It is also estimated that the annual global Internet traffic will reach a half zettabyte by 2012, which is 250,000 times more than in 2003 [1].

The Internet was not designed for its current level of usage and it is apparent that the existing Internet architecture will need to be adapted to deal with the increasing demands made upon it. The Future Internet will have to evolve to meet numerous challenges such as scalability, mobility, manageability, trust and openness whilst still assuring that the proposed solutions are governable, sustainable and usable.

Significant amounts of research and resources have already been allocated to the Future Internet research area. In Europe alone, more than 90 Information and Communication Technology (ICT) projects, with an overall budget of more than 400

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 30–39, 2010.
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million Euros, are addressing several challenging dimensions of the Future Internet under the Seventh Framework Programme (FP7) [2]. The Future Internet Assembly (FIA) [3] gathers the major European industrial stakeholders and researchers in an effort to foster a strong European footprint in the Future Internet. The Future Internet Research and Experimentation (FIRE) Initiative [4] endeavors to address the emerging expectations of the Future Internet by providing a research environment for investigating and experimentally validating highly innovative and evolutionary ideas. The currently emerging Public Private Partnership (PPP) European Future Internet Initiative (EFII) aims to complement existing initiatives to provide a more holistic approach to addressing the challenges of the Future Internet in Europe [5]. All of these initiatives are working hard at defining the aforementioned challenges that lie ahead for the Future Internet and for implementing solutions that meet these challenges.

The Telecommunications Software & Systems Group (TSSG) research centre in Waterford Institute of Technology (WIT) Ireland is an active participant of the aforementioned European initiatives. The TSSG push the boundaries of the state of art in Future Internet research through its past and present participation in many FP7 Future Internet projects (including 4WARD, AutoI, Efipsans, Panlab(PII), PERIMETER, Persist, ThinkTrust, Vital++) [2] in areas such as security, network management, mobility and Quality of Experience (QoE) and the verification and validation (V&V) of these Future Internet thematic areas through their dedicated experimental facilities group and federated testbed activities.

This paper examines the requirements and challenges of the Future Internet and demonstrates how they are dealt with in the TSSG using a case study of the Future Internet project; PERIMETER [6]. Section 2 provides an overview of this FP7 project before providing an analysis of the testing and experimental processes that are applied. The specific role that the testbed activities have in meeting some of the challenges of the Future Internet is provided. The section concludes with an analysis of the effectiveness of methodologies employed. The observations of further developments and innovations that are required to achieve an end-to-end solution for experimental testing in PERIMETER, and more generally for other Future Internet projects, are provided in Section 3, before the paper concludes with some summarising remarks.

2 PERIMETER: A Case Study

This section provides the details of a case study of the Future Internet related FP7 PERIMETER project conducted by the TSSG. The PERIMETER project is deemed a suitable project for this case study as the project incorporates many of the challenges of the Future Internet, including QoE, security, end user impact and the establishment of a new paradigm for seamless mobility. The TSSG are one of ten European partners involved in this project and have official roles as software developers, official testbed providers and validation engineers in the project. TSSG's previous experience in V&V, experimental facilities and federated testbed activities has provided a solid basis toward selecting and applying suitable methodologies to apply to the PERIMETER project.

The case study begins by giving an overview and some details of the PERIMETER project. Next, the testing and experimental methodologies that are applied to the

PERIMETER project and the role that testbeds have are detailed and examined to illustrate how some of the challenges of the Future Internet are met by the approach employed. The case study concludes with an analysis of the effectiveness of methodologies employed.

2.1 PERIMETER Overview

The PERIMETER project [6] entitled "User-Centric paradigm for Seamless Mobility in Future Internet" is a three-year project funded by the FP7 programme. PERIMETER is listed as one of eight 'Experimentally driven advanced research" projects in the first wave of the FIRE Initiative [4].

PERIMETER's main objective is to establish a new paradigm of user centricity for advanced networking. Putting the user in the centre rather than the operator enables the user to control their identity and preferences (e.g. cost, security, network selection). This enables the mobile user to be "Always Best Connected" (ABC) in multiple-access multiple-operator networks of the Future Internet.

To achieve this, PERIMETER develops and implements protocols and a middleware to address requirements for privacy, security, resilience and transparency. The network selection is done based upon a parameter called Quality of Experience (QoE) [7]. For the Future Internet to succeed and to gain wide acceptance of innovative applications and service, QoE objectives have to be met. Analysis of QoE parameters allows the evaluation of network connections by user perceivable indexes and parameters (i.e. cost, energy efficiency, security, privacy and connection performance). Network selection can be done manually by the user or can be handled automatically by PERIMETER's QoE management system for generic QoE definition, QoE signaling, and QoE based content adaptation [8] thus creating a single-sign-in like experience. Fig. 1 provides an illustrative diagram of the user centric approach employed by PERIMETER.

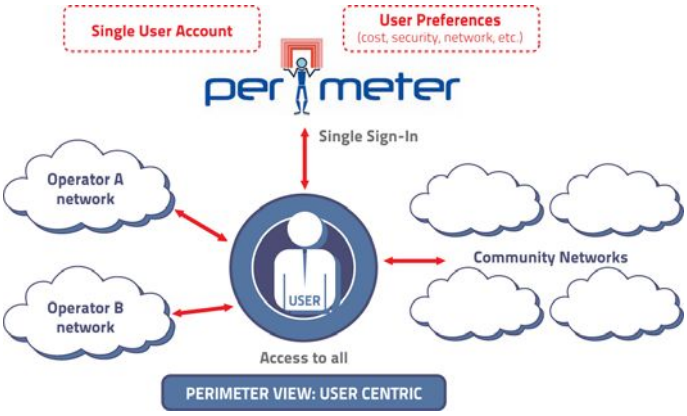

Fig. 1. PERIMETER's User Centric View in the Future Internet

The architecture of PERIMETER consists of Terminal Nodes and Support Nodes. The PERIMETER Terminal Nodes are installed on devices which are operated by the user and which have some storage and computational restrictions. Currently mobile phones and netbooks based upon Google's Android platform [9] are supported. The Support Nodes are physical devices that can act as data sources and gateways and are distributedly connected using a peer-to-peer approach both to other PERIMETER Support Nodes and to other PERIMETER Terminal Nodes. The Support Nodes do not have any storage or computational restrictions and are therefore used as the main storage for the system and for expensive computational activities. The Support Nodes reside in PERIMETER's state of the art federated QoE testbed between Waterford Institute of Technology (WIT), Ireland and Technical University of Berlin (TUB), Germany while the Terminal Nodes are installed on mobile devices which run on the federated testbed infrastructure. The PERIMETER system is supported by a number of identified applications which also run across the federated testbed and support the verification of the user centric scenario based approach adopted in the PERIMETER project.

2.2 Testing and Experimental Methodologies

The PERIMETER project develops and implements protocols and a middleware for a new user-centric paradigm for Future Internet mobility. This experimentally driven research presents a new set of challenges to traditional and existing processes and to testing in large-scale environments across a number of testbeds. For PERIMETER, two key issues must be considered. Firstly, the real world and digital world must be interconnected. Thus, the services must be aware of the users' environment and preferences. Secondly, the user should be ABC where "Best" is defined by the user. As the two issues imply rather vague than more formally definable quality indicators; testing and experimentation, with user involvement, is necessary for the project in addition to standard testing mechanisms. Therefore, for PERIMETER, it is possible to draw a distinction between testing and experimentation. Testing is a continuous process needed in software development. Experimentation, on the other hand, is needed to guarantee the success of the research, innovation and usability aspects of the project. The implementation of each of these will be elaborated upon in turn.

2.2.1 Testing Methodologies

A number of existing development and testing methodologies (for example: Waterfall, Spiral, DSDM, SCRUM, XP and Agile) were evaluated in order to find one that could meet the specific challenges of determining the functional correctness of the final PERIMETER system. The chosen methodology needed to be adaptable, fluid and allow for all the different components and technologies of the project and for its phased iterative delivery.

The Agile software development methodology [10] was chosen as the most suitable existing methodology for use in the project. This methodology was deemed suitable as Agile software projects are characterised by iterative and incremental development, accommodation of changes and active customer participation [11]. Adopters of Agile report that this development process leads to higher software quality and customer satisfaction ratings when compared to using traditional methods, with more

productive and motivated developers [11]. Whilst smaller developer teams have cited higher success rates than larger teams [12][13], agile principles can and have been applied successfully to large scale projects and distributed teams (for example [14 and 15]). In PERIMETER, the use of Agile is combined with a Test Driven Development (TDD) [16] approach to provide the adaptable, fluid and iterative development and testing cycle that is required [17].

The need for processes, supporting tools and automation is necessary for the success of most software projects, but is essential for the success of large scale experimental ventures. PERIMETER employs a number of processes and tools [18] to aid the processes of team collaboration and communication, testing, continuous integration (Hudson) and structured software development.

In this stage of development and testing, i.e. in a continuous build environment, a number of testing processes including unit (white-box), functional (black-box) and integration testing, are conducted to ensure that the developed software is brought to a level where it achieves its functional objectives.

2.2.2 Experimental Testing

To effectively validate and demonstrate the results of the PERIMETER project, the innovative aspects of the project are verified in a suitable large scale, and state of the art, testbed demonstration facility. This end validation goal is implemented by the interconnection of two state of the art testbeds, one in WIT and one in TUB over Layer 3. This interconnection will be matured to a Layer 2 federation in the next phase of the project. PERIMETER has also applied for, and has been successful in procuring a FEDERICA slice (5 virtual nodes) [19] for use by the project. The federated testbed embraces the principles of openness, governance, excellence and management in order to make the testing and experimental infrastructure as efficient as possible, in keeping with experimental infrastructures 'best principles' [20].

Conformance tests are performed on the PERIMETER system installed and running on the federated testbed to ensure that key components of the system are functioning as expected. These tests are complemented with interoperability tests to ensure the end-to-end functionality of the system is as expected. Both sets of tests are tested against the scenario under analysis, in the scenario-driven approach used in this project in order to demonstrate that the testing process is robust and to ensure the verifiability and reliability of the results. It is essential that both testing processes are repeatable and reproducible in order to achieve this. Application testing for the specific applications running on the federated testbeds is also performed.

This process is further complimented by the employment of a user-driven approach to the requirements specification and the determination of features and subsequent testing phases with the use of Living Labs [21] and dedicated usability sessions. Performance and scalability issues are addressed with the introduction of emulation and simulation of network conditions.

Finally, the entire experimental process (and the testing process cf. Section 2.2.1) is realised by means of several iterative cycles, where refinements are made, further functionalities are implemented and where testing and experiments become more stringent as the project progresses and naturally matures.

2.3 Role of the Federated Testbed

The federated testbed used in PERIMETER plays a vital role in the experimental testing phase of the project. The federated testbed is used to conduct a number of testing processes including the scenario conformance tests, interoperability tests, application tests, Living Labs tests, performance and scalability tests. These testing processes could not be performed in as realistic an environment, achieving such a level of realism, without the use of the federated testbed. Further, without the federated testbed, the cost of achieving this level of realism would be greatly increased. Fig. 2 [22] provides an illustration of the cost versus realism model, where cost is determined as a function of complexity, dimension (scale) and environmental conditions and the level of experimental realism, being an inverse function of the abstraction level. [22] suggest that the federation of testbeds (whether heterogeneous or homogeneous federation) could be used as a mechanism to control this cost.

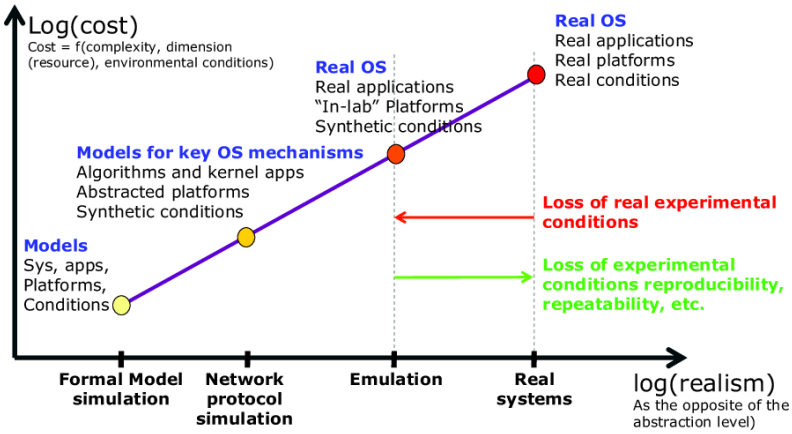

Fig. 2. Cost Versus Realism Model [22]

The findings from this case study go a step further than this. The PERIMETER experience indicates that not only does the use of the federated testbed control the cost; it actually does this without actually increasing cost but whilst increasing the level of realism achieved. This finding, and the role of the federated testbed in the Cost Vs Realism model for PERIMETER, is illustrated in Fig. 3. Furthermore, the use of the federated testbed allows the experimental testing to proceed using real platforms, real operating systems and real applications in a realistic environment, which is closer to the actual target system. Another advantage is that the tests conducted can be repeated and reproduced as often as is required without affecting many aspects of the cost, as the tests are not conducted on an actual system, but rather in the testbed environment. In addition, issues and experiments which depend upon external environmental factors, such as those that involve end users, can be conducted in this environment before the system is released to the actual target system. This is an important advantage in ensuring the final system's usability and acceptability in the real world.

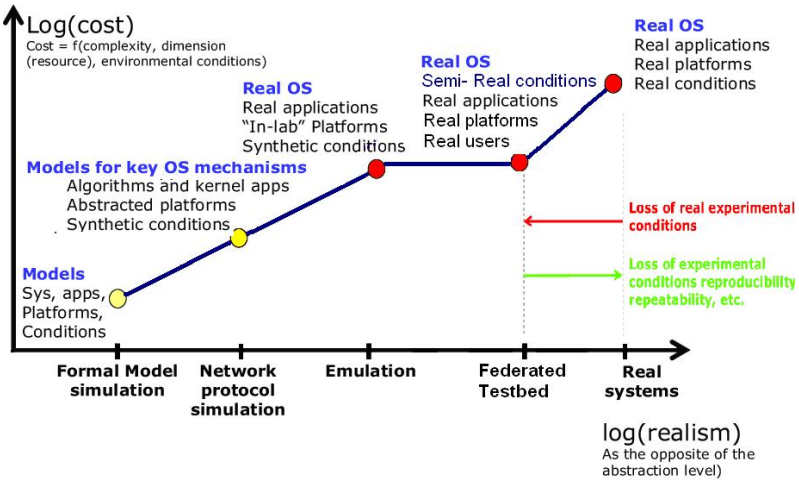

Fig. 3. Role of the federated testbed in the Cost Vs Realism Model for PERIMETER

2.4 PERIMETER Case Study - Methodologies Analysis

This case study has described the testing and experimental methodologies employed, and the role of the testbed activities, for one Future Internet project; PERIMETER.

Whilst the described processes and methodologies suit this particular project, they do not follow, from start to finish, a stringently defined, end-to-end solution which could be used for any experimentally driven project. Because of this, a number of different practices and processes are used to achieve the required results. These were carefully chosen for the project to ensure that they were as compatible as possible. However, in some instances some of the processes were not completely automated and required manual intervention. Furthermore, it did require additional effort to ensure the completeness of the testing and experimental processes and to determine how stringently these processes were employed.

On the other hand, the advantages gained by the use of testbeds within the project cannot be overstated. The federated testbed has allowed the PERIMETER system to be tested within a very realistic environment while still keeping cost at a manageable level.

3 Experimental Testing Requirements for PERIMETER and the Future Internet

As the Future Internet evolves it faces many new challenges for Internet architectures, protocols and services. The software and technologies that run on top of the Future Internet will be more complex and innovative than ever. To meet these challenges new approaches to the testing process, to experimentation and to the use and reuse of testbeds will have to be addressed, particularly in large scale environments.

The complexity of the software and technologies of the Future Internet will require innovative and different approaches to the existing testing processes and tools

available. In PERIMETER, the lack of full automation of the testing process did hamper these processes. We deduce that the use of automated tools will be essential in order to conduct robust and repeatable testing and experimental analysis on even larger scale projects. An end-to-end robust, adaptive and scalable testing and experimental process could be of benefit to all Future Internet projects. However, we found that no such complete end-to-end solution appears to exist, and therefore it is difficult to determine the completeness and stringency of the methodologies employed. The Future Internet community should give some consideration to the fundamental requirement for an adaptive and fluid model, which can evolve and meet the specific challenges that the innovation of the Future Internet brings. The standardisation and the certification of such a model would also be an important factor in ensuring that a complete and stringent practice is adhered to. This adoption of standards and use of a certified process could be applied as a *'trust factor'* to the delivered service or application, thus providing an element of confidence in the adherence to the recommended methodology.

The role that testbeds and the federation of these testbeds can have in the Future Internet is extremely important. The previous section demonstrated the role they play in just one Future Internet project; PERIMETER. Here, it has been demonstrated that the federated testbed used allowed a level of realism to be achieved that would otherwise not have been possible except for testing in the actual target environment. However, testbed environments are expensive to create and maintain and the level of expertise that is required to use and run these is substantial. The work of initiatives, such as FIRE [4], in promoting the use/re-use of existing testbeds and the federation of these testbeds in order to provide a richer environment for experimentation and testing should be encouraged. Further endeavours to reuse and to federate existing facilities (for example, using FEDERICA slices [19]), to open existing facilities for use (for example, using the Panlab Teagle tool [23]) and to facilitate experimentally driven research (for example, the initiatives of the second wave of FIRE FP7 projects [4]) must be endorsed in the Future Internet.

Testbeds also have a considerable function to play with regard to the employment of user-driven approaches in the Future Internet. The Future Internet will be an Internet for the people where user needs, user involvement and user acceptability will be of utmost importance. The employment of testbeds in the experimentally driven domain that is the Future Internet will provide a realistic, but safe, testing environment in which to test how these factors are met, when used in conjunction with a Living Labs co-creation process.

4 Summary

The challenges which lie ahead in the area of experimental testing in the Future Internet are many. This paper has examined these and the current and upcoming initiatives that exist across Europe in dealing with this complex thematic area. The paper examines PERIMETER as a case study of how testing and experimental processes, and the use of testbeds, are applied to one Future Internet project. The observations gleaned from this project are used in an analysis of what is required for a robust adaptive and scalable testing and experimental process in the Future Internet. Here it is

observed that no complete end-to-end solution exists that can handle the experimental nature of Future Internet projects and a suggestion is made that this is an area that should be considered by the Future Internet community as a whole. The many advantages of using testbeds in experimental testing processes and experimentally driven research are emphasised and the work in relation to testbed use, reuse and federation is greatly encouraged. Finally, the role that users themselves play in the requirements and features of the services and applications and in the testing processes themselves should not be overlooked.

This paper concludes by remarking that though many challenges do lie ahead in the Future Internet, none are insurmountable. Much worthy and innovative work is already underway or just begun (for example, the second wave of FIRE FP7 projects [4]) to meet these challenges. Measures should be in place to ensure that the current work leads to tangible and usable outputs that can then be built upon in future works. This will require much collaboration by all parties in the Future Internet community as it strives to meet all the challenges that arise going forward.

Acknowledgments. This research activity is funded under the EU ICT FP7 project, PERIMETER (Project No.: 224024).

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An Economic Case for End System Multicast

Morteza Analoui and Mohammad Hossein Rezvani

School of Computer Engineering, Iran University of Science and Technology (IUST)
16846-13114, Hengam Street, Resalat Square, Narmak, Tehran, Iran
{analoui,rezvani}@iust.ac.ir

Abstract. This paper presents a non-strategic model for the end-system multicast networks based on the concept of replica exchange economy. We believe that microeconomics is a good candidate to investigate the problem of selfishness of the end-users (peers) in order to maximize the aggregate throughput. In this solution concept, the decisions that a peer might make, does not affect the actions of the other peers at all. The proposed mechanism tunes the price of the service in such a way that general equilibrium holds.

Keywords: End-system Multicast, Non-strategic behavior Modeling, Walrasian Equilibrium, Replica Exchange Economy.

1 Introduction

End-system multicasting is a solution concept in order to disseminate digital content (such as videos and teleconferences) in large scale decentralized peer-to-peer (P2P) networks. The end-user nodes, which are referred to as peers, receive the content by using IP unicasting and then forward it to the other peers of the multicast group.

Since the peers are selfish users who typically belong to various administrative domains, they do not tend to cooperate with each other. Thus, it is more likely that a peer downloads the content from its upstream peer rather than uploading to the downstream peers. This behavior can potentially degrade the overall throughput of the end-system multicasting. In this context, the most important question is the following; "how should we exploit the inherent selfishness of the peers, so that the aggregate outcome of the activity of individual peers behaving toward their own self-interests still leads to the network's overall utility maximization?" We believe that this problem could be investigated by microeconomics. In microeconomics, the goal is to maximize the aggregated utility of the consumers in conjunction with the maximizing the profit of the firms as the selfish agents of the system. In our mapping, the bandwidth of the service provided by the network can be viewed as good and the behavior of the overlay peers can be mapped to that of the consumers in the economy. Also, the origin server and the peers who forward the content to their downstream peers can be thought as the firms of the economy.

To address the selfish behavior of the overlay peers, some works have used game theoretic approaches [1, 2, 3, 4, 5, 6, 7] which fall into the category of *"strategic behavior modeling"*. Here, the players seek to maximize their utility regard to the decisions that the

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 40–48, 2010.
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other players might make. Also, another category of proposals exist in which the selfishness of the peers is controlled by means of distributed pricing [8, 9].

A major contribution of this work is that unlike the game theoretic approaches, here the decision that a given peer might make, does not depend on the decisions taken by the other peers. Roughly speaking, our work falls into the category of *"non-strategic behavior modeling"* for which we have adopted the model of replica exchange economy developed in microeconomics. In this solution concept, the peers are classified into some distinct *"types."* The criterion used for classifying the peers may be either the service level agreements, or the amount of available bandwidth of the peers. This can make sense in real P2P networks such as the end-system multicasting, for example, if the peers are classified into types such as Gold, Silver, Premium and, etc. Also, the classification of the peers can be done upon receiving the request in admission control unit based on the amount of downloading/uploading capacities. Clearly, the peer who has the higher downloading/uploading capacities is more likely to have better conditions in order to forward the service to the other peers. We shall not discuss the problem of user classification here. Instead, we will assume a given set of user types and will try to model the multicast network by using the concept of the replica exchange economy in such a way that the allocations are Pareto optimal (PO). Also, by using the concept of *"Walrasian general equilibrium"*, we have developed a mechanism to find the equilibrium price in such a way that the total demanded bandwidth of the system equates its total supplied bandwidth. To the best of our knowledge, there has been no investigation on modeling the end-system multicast networks based on the concept of replica exchange economy.

The remainder of this paper is organized as follows: we discuss the related works in Section 2. Section 3 presents the formulation and basic definitions concerned to the replica exchange economy without production. Section 4 presents the replica exchange economy with production. Finally, we remark the conclusions in Section 5.

2 Related Work

There already exists a significant body of research work toward self-organization in the end-system multicast networks. Some of the projects which have been carried out in the area of the end-system multicast include ZIGZAG [10], Narada [11], and NICE [12].

Finding the optimal bandwidth-efficient allocation in the end-system multicast can be understood as a utility-based resource allocation problem. Here, each user is associated with a utility, which is defined as an increasing, concave, and twice continuously differentiable function of the user's receiving rate. Cui et al. [13] have shown that finding among all possible trees the one with the optimal *"maximum utility"* rate allocation is a NP-hard problem. Also, Zhu et al. [14, 15] have shown that finding *"maximum-bandwidth"* multicast tree in the end-system multicast network with linear capacity constraints is a NP-complete problem.

To address the selfishness of nodes, some works have applied game theory. In game theoretic methods, the overlay nodes are treated as game players with conflicting interests regarding the allocated resources [1, 2, 3, 4, 5, 6, 7]. The cheating about node throughput, and link cost have been investigated in [1] and [2] respectively. The selfishness of the overlay nodes in tree construction process has

been investigated in [3]. The relation between the users' selfish behavior and the network routing has been investigated in [4]. In Wu *et al.* [5], an auction-based game theoretic approach is proposed in which the downstream peers submit their bids for the bandwidth at the upstream peers. They show that the outcome of this game is an optimal topology for the end-system multicast network and it also minimizes the streaming costs. Also, the authors of [6] have proposed an auction-based model to improve the performance of BitTorrent. They have investigated the use of a proportional share mechanism as a replacement to BitTorrent's current mechanism and have shown that it achieves improvement in terms of fairness and robustness without any significant modification to BitTorrent protocol.

A few other works have also proposed to adjust the selfish behavior of the peers within distributed pricing. Cui *et al.* [8] are the first to apply the priced-based resource allocation method in the end-system multicast. The significant advantage of their work over previous works is that it purely relies on the coordination of the peers. This property makes the protocol deployable to the existing network infrastructures and accords fully with the initial design objectives of the end-system multicast network, which is to avoid any change to the existing infrastructure by leaving the required functionalities to the peers. Another priced-based work in the context of the end-system multicast networks is [9]. Wang et al. in [9] propose an intelligent model based on optimal control theory. They, also propose mathematical models that emulate the selfishness of the overlay peers, captured by some forms of utility function. They show that, even when selfish peers all seek their maximal utilities, the resultant network still achieves near-optimal performance.

3 Replica Multicast Exchange Economy

The distributed nature of the non-strategic behavior modeling approach lends itself to design price-based protocols in such a way that may require less information in comparison with the strategic modeling approaches [16, 17]. In this section, we will present an analytical model for the end-system multicast network by leveraging the rich theory of replica exchange economy as a non-strategic solution concept. For the sake of simplicity, we first consider a basic exchange economy without the complications of production. We call the proposed economic system *"Replica Multicast Exchange Economy"* or *RMEE*. We will define the problems of the consumer and the firm in detailed and also will state the constraints that arise due to the dual role of the peers in the multicast networks. Interested readers can find the theorems and their proofs in chapter 5 of [16].

The end-system multicast network consists of an origin server and V peers denoted as $\mathcal{V} = \{1, 2, ..., V\}$. We start with the idea of a basic exchange economy consisting of a finite but arbitrary large number V of consumers, each with his or her own preferences and initial endowments. Now think of each consumer's preferences or endowments as making that consumer a different *type* of consumers from all the rest. Thus, we now can think of there being an arbitrary finite number of different *types* of consumers, and the basic exchange economy consists of one consumer of each type.

Now imagine that each consumer is marked with an exogenous or endogenous label indicating his or her *type*. The new economy is clearly different from the basic exchange economy. We call this economy the *"replica exchange economy."* Roughly speaking, a replica exchange economy is a special case of the basic exchange economy which has a finite number of *types* of consumers, and all individuals of the same type are identical in that they have *identical* preferences and identical endowments. Let there are $\mathcal{I}$ types of consumers in the basic exchange economy and index these types by set $\mathcal{I} = \{1, ..., I\}$. For any type $i \in \mathcal{I}$, all consumers of that type share the common preferences $\overset{i}{\geq}$ on $\Re_+$ and have identical endowments. Let e_v^i denotes the initial endowment of consumer $v \in \mathcal{V}$ who has type $i \in \mathcal{I}$. We define the initial endowment of RMEE as a vector $\mathbf{e} \equiv (e^1, ..., e^{\mathcal{I}})$, where $e^i = \sum_{v \in \mathcal{V}} e_v^i$ denotes the aggregate amount of initial endowment for consumers of type i.

We now proceed to define the consumers in RMEE. The consumers of RMEE are the peers either who relay the service to their downstream peers or who are the leaves in the tree. We further assume for $i \in \mathcal{I}$ and $v \in \mathcal{V}$ that preferences $\overset{i}{\geq}$ can be represented by a utility function u_v^i consisting of an economic part and an empirical part. The former component can be simply expressed as a function of allocated bandwidth of the service, whereas the latter accounts for the node's empirical utility for receiving the service. This part of the utility may be characterized by various characteristics arising from the underlying physical network routings such as loss rate, delay, and etc.

Now, we proceed to state the optimization problem of consumer v in RMEE. Each consumer v upon joining to RMEE is endowed with an income e_v^i which actually is the amount of credit, enabling it to enter in RMEE and take share in the future transactions. In other words, one can imagine that consumer v is provided with the income in terms of the amount of the service. Also, note that in competitive markets, every consumer is price-taker, in the sense that he or she takes the prices as given, whether acting as a buyer or a seller.

Let p denotes the price of each unit of bandwidth in RMEE. Thus, the quantity $p.e_v^i$ can be thought as the *"initial monetary income"* of consumer v. Each consumer in RMEE must solve the following optimization problem:

$$\max_{x^i_{Par(v),v}(t,.)} u_v^i(t,.) \tag{1}$$

$$\textbf{s.t.} \quad p(t).x^i_{Par(v),v}(t,.) + \sum_{l \in \mathcal{L}(f_{Par(v),v})} q_l(t-1) \leq p(t).e_v^i \tag{2}$$

$$x^{i}_{Par(v),v}(t,.) \leq CD_v \tag{3}$$

$$b \leq x^{i}_{Par(v),v}(t,.) \leq B \tag{4}$$

Where, $f_{Par(v),v}$ denotes the flow that is originated from parent of v and is terminated in node v. Similarly, $x_{Par(v),v}$ denotes the rate of flow $f_{Par(v),v}$. Also, $\mathcal{L}(f_{Par(v),v})$ denotes the underlying physical links from which the flow passes. Condition (2) states the *"budget constraint"* of peer v who is under investigation as a consumer. The sum on the left-hand side of Eq. (2) is just the expenditure of consumer v in which the first term represents the price of the demanded service and the other term (the sigma) is the sum of the prices of the all physical links that the flow $f_{Par(v),v}$ has passed through in the previous time slot. Note that at the time that slot t starts, the quantity of the new allocations has not been determined yet. That is why we have used the prices of the links in round $t-1$ for the optimization of the consumer in round t. The right-hand side of Eq. (2) simply is the initial monetary income of peer v. Constraint (3) implies that the rate of the service that is demanded by consumer v should not be greater than his or her downlink capacity, namely CD_v. Constraint (4) is an application-specific constraint and states that the rate at which the service is received should be in the interval between its minimum and its maximum allowed bandwidths.

We denote the solution to type-i consumer v's problem in RMEE model in time slot t as follows

$$x^{i}_{Par(v),v}(t, p(t), p.e^{i}_{v}, \sum_{l \in \mathcal{L}(f_{Par(v),v})} q_l(t-1), CD_v, b, B) \tag{5}$$

The bundle $x^{i}_{Par(v),v}(t,.)$, known as *"Walrasian Demand Function"* or *WDF*, is the consumer's demanded bundle which depends on market price and the consumer's endowment income. Note that, all of the parameters of WDF in Eq. (5), apart from price $p(t)$, are a-priori known.

We can interpret the consumer's endowment e^{i}_{v} as giving the quantity of the good that he or she supplies in the market. In RMEE model, the total demand for the bandwidth (the good) will simply be the sum of every individual consumer's demand for it. Similarly, market supply will be the sum of every consumer's supply. It is common in microeconomics literature to describe the equilibrium state of the economy by a function named *"aggregate excess demand function"* or *AEDF*.

Definition 1 (AEDF). *The AEDF of RMEE model is a real-valued function as following:*

$$z(p) \equiv \sum_{i \in \mathcal{I}} \sum_{v \in \mathcal{V}} x^i_{Par(v),v}(p,.) - \sum_{i \in \mathcal{I}} \sum_{v \in \mathcal{V}} e^i_v \tag{6}$$

When $z(p) > 0$, the aggregate demand for the service exceeds its aggregate endowment; resulting to an excess demand for it. On the other hand, when $z(p) < 0$, there is excess supply for the service. If at some prices p, we had $z(p) = 0$, i.e., the demand becomes equal to the supply, then we would say that the entire RMEE is in general equilibrium. The price that equates the demand and the supply is called *"Walrasian equilibrium price"* or *WEP*.

Definition 2 (WEP). *A price* $p^* \in \Re_{++}$ *is called a WEP if* $z(p^*) = 0$.

Definition 3 (Walrasian Equilibrium Allocations (WEA)). *Let* p^* *be a Walrasian equilibrium for RMEE with initial endowment vector* $\mathbf{e} \equiv (e^1, ..., e^{\mathcal{I}})$, *and let*

$$\mathbf{X}(p^*) \equiv (x^1(p^*, p^*.e^1), ..., x^{\mathcal{I}}(p^*, p^*.e^{\mathcal{I}})) \tag{7}$$

Where $x^i = \sum_{v \in \mathcal{V}} x^i_{Par(v),v}(p^*, p^*.e^i_v)$ *and* $e^i = \sum_{v \in \mathcal{V}} e^i_v$. *Then* $\mathbf{X}(p^*)$ *is called a WEA.*

Let us now introduce a significant metric toward welfare measurement of economic systems, namely *"Pareto optimality"* or *PO*. An allocation is PO if it is not possible to make someone strictly better off without making someone else strictly worse off.

4 Replica Multicast Exchange Economy with Production

In this section, we will expand RMEE model to include production as well. We call the system *"replica multicast exchange economy with production"* or *RMEEP*. Most of the important properties of RMEE model continue to hold. However, one of the main differences between RMEEP and RMEE is that here the profits earned by each firm must be distributed back to the consumer who owns it. The consumers of RMEEP seek to maximize their utility subject to their constraints whereas the firms seek to maximize their profits. Also, note that both the agents are price-takers in the sense that introduced in Section 3.

The RMEEP includes two sources of firms. The firm can be either the origin server which originally offers the service to the network or can be a peer who relays the service to the corresponding downstream peers. We now let $y^i_v \in \Re$ be a *"production plan"* for a type-i firm v. If, for example, firm v uploads (sells) 7_{bps}

of the service to its downstream peers, then it turns out that $y_v^i = 7$. Clearly, since the leaf nodes do not relay the service to the other nodes, their production plan will be zero. To summarize the technological possibilities in production, we suppose each firm v possesses a *"production possibility set"* $\mathcal{Y}_v^i$, $i \in \mathcal{I}$, $v \in \mathcal{V}$.

Each intermediate peer (actually the firm) seeks a production plan to maximize its profit. The optimization problem of a type-i firm v in time slot t can be formalized as follows

$$\max_{y_v^i(t,.)} \; p(t).y_v^i(t,.) \tag{8}$$

$$\textbf{s.t.} \qquad y_v^i(t,.) \le CU_v \tag{9}$$

$$y_v^i(t,.) \le K_v \,.\, x_{Par(v),v}^i(t,.) \tag{10}$$

$$y_v^i(t,.) = \sum_{k \in Chd(v)} x_{v,k}^i(t,.) \tag{11}$$

Where K_v denotes the number of children of type-i node v and CU_v denotes its uploading capacity. Constraint (9) states that the amount of the bandwidth that is sold by peer v should not exceed its uploading capacity. Constraint (10) implies that the amount of the bandwidth that is sold by peer v can not exceed the rate at which it has been bought by peer v. This constraint, referred to as *"data constraint",* is unique in the overlay multicast networks, mainly due to the dual role of the non-leaf peers who are both the receivers and the senders. In other words, the intermediate peer v currently uploads to its downstream peers the service which it has downloaded from its upstream peer. Since peer v sells the service to its downstream peers, one can imagine that the production plan of firm v equates the sum of those rates. Constraint (11) just states this fact.

The solution to Eq. (8) is in fact the production plan of type-i firm v. If this production plan is multiplied by the equilibrium price, will result in the profit of the firm which is denoted by $\Pi_v^i(p)$.

In RMEEP model, the optimization problem of the consumers is the same as that of RMEE model, except for the *"budget constraint"*. In RMEEP model, the constraint (2) must be replaced with the following constraint:

$$\textbf{s.t.} \qquad p(t).x_{Par(v),v}^i(t,.) + \sum_{l \in \mathcal{L}(f_{Par(v),v})} q_l(t-1) \le p(t).e_v^i + \sum_{\tau=0}^{t-1} \Pi_v^i(p(\tau)) \tag{12}$$

We explained the components of the *"budget constraint"* of a type-i consumer v in Section 3. The right-hand side of Eq. (12) simply is the initial monetary income of the type-i consumer v plus its share in the profit earned by selling the service to its

downstream nodes during the previous time slots, denoted as $\sum_{\tau=0}^{t-1} \Pi_v^i(p(\tau))$. Note that due to the dual role of the intermediate peers, one can imagine that they are acting as both consumers and firms in RMEEP. Hence, the budget of each consumer can arise from two sources- from the monetary endowment of the service already owned, and from the profit shares that it earns by selling the service to its children.

Let $m_v^i(p)$ denotes the right-hand side of (12). Similar to Section 3, we can denote WDF of the type-i consumer v's problem in RMEEP model in time slot t as follows

$$x_{Par(v),v}^i(t, p(t), m_v^i(p), \sum_{l \in \mathcal{L}(f_{Par(v),v})} q_l(t-1), CD_v, b, B) \tag{13}$$

Altogether, we can represent RMEEP model as the collection $(u_v^i, e^i, \mathcal{Y}_v^i)_{i \in \mathcal{I},\ v \in \mathcal{V}}$. Similar to the case with RMEE, we can again define AEDF of the system for RMEEP model as follows

$$z(p) \equiv \sum_{i \in \mathcal{I}} \sum_{v \in \mathcal{V}} x_{Par(v),v}^i(p, m_v^i(p),.) - \sum_{i \in \mathcal{I}} \sum_{v \in \mathcal{V}} y_v^i(p,.) - \sum_{i \in \mathcal{I}} \sum_{v \in \mathcal{V}} e_v^i \tag{14}$$

5 Conclusions

In this paper we have presented an analytical model for the multi-rate end-system multicast networks based on rich theory of replica exchange economy of microeconomics. In this model, the content provided by the origin server can be viewed as good and the origin server and the peers who multicast the content to their downside are considered as the firms. On the other hand, due to the dual role of the peers in this network, they can be considered as the consumers as well.

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Towards a Secure Rendezvous Network for Future Publish/Subscribe Architectures

Nikos Fotiou, Giannis F. Marias, and George C. Polyzos

Athens University of Economics and Business, Athens, Greece
{fotiou,marias,polyzos}@aueb.gr
http://mm.aueb.gr

Abstract. Publish/Subscribe is often regarded as a promising paradigm for Future Internet architectures. Its information oriented nature and its particular security features have stimulated current research efforts which aim at applying publish/subscribe principles to a clean-slate Internet architecture. One of the core components of publish/subscribe architectures is the rendezvous network. Any security failure that a rendezvous network may face will probably jeopardize the operation of the whole (inter-)network. In this paper we highlight security requirements and potential security issues of rendezvous networks and we present security solutions that can be applied in order to shield them.

Keywords: Future Internet, Publish/Subscribe, Security.

1 Introduction

Publish/subscribe is regarded as a promising paradigm for the development of future Internet applications and even as a candidate for a clean slate future Internet architecture. Publish/subscribe is currently under investigation in a variety of research efforts-such as the PSIRP [1] and CCNx [2] projects. Its information centrism manifests a significant shift from the traditional endpoint-centric Internet paradigm. In addition, the interest-based decision on accepting data or actively looking for information at the subscriber (recipient) side, shifts the power towards the receiver and thus restoring some balance, compared to the commonly used send-receive paradigm that empowers the sender.

Publish/subscribe architectures are built with three basic components: publishers, subscribers, and an event notification infrastructure also known as the *rendezvous network* [3]. Publishers are information providers who advertise their information by virtue of publications. Subscribers on the other hand are information consumers, who express their interest on specific pieces of information by issuing subscriptions. The rendezvous network is responsible for matching subscriptions with publications and for the initiation of the data transfer from the information publisher to the information subscriber(s) when both are present and ready. Publication and subscription operations are decoupled in time and space, and the fact that the publication-subscription matching takes place at an independent point in the network allows for efficient deployment of multicast, e.g. [4], mobility, as described in [5], as well as multihoming and indirection, e.g. [6] [7].

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 49–56, 2010.
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The rendezvous network is probably the most critical part of a publish/subscribe architecture. Apart from matching subscriptions with publications, the rendezvous network is the place where access control, policy implementation, and publication scoping can take place [8]. The rendezvous network may be composed of separate, dedicated nodes, such as in [1], or it may be part of the core network, as it happens in [2]. In the former case the rendezvous nodes are usually organized in one-dimensional DHTs–such as Chord [9] and Pastry [10]–when the subscription-publication matching takes place using keys, or in multi-dimensional DHTs–such as CAN [11]–when it comes to semantic-based subscription-publication matching.

Being the core component of publish/subscribe architectures, the rendezvous network is expected to be the target of various security attacks. The purpose of this paper is to describe possible threats against the rendezvous infrastructure by taking into consideration previous experience with DNS and DHTs, to identify the role of the rendezvous network in the already defined security weaknesses of publish/subscribe architectures, as well as to present a set of security solutions that can be applied in securing the rendezvous network. The remainder of this paper is organized as follows. Section 2 presents the rendezvous network's security requirements, while Section 3 outlines some potential security threats. Section 4 overviews security solutions that can be used in order to secure a rendezvous network and finally Section 5 presents our conclusions with discussion and ideas for future work.

2 Security Requirements

General security requirements for publish/subscribe architectures have been defined by Wang et al. [12] as well as by Lagutin et al. [13]. The role of the rendezvous network in order to achieve these requirements is of critical importance.

Publish/subscribe architectures are required to provide publication confidentiality, i.e., publications should not be revealed to unauthorized subscribers, as well as subscription privacy, i.e., users' subscriptions should be kept secret. Rendezvous networks have to provide *access control* mechanisms in order to achieve publication confidentiality, as well as encryption and trust mechanisms in order to provide subscription privacy.

Integrity is also a security requirement for publish/subscribe architectures. Integrity can be classified into three categories: Information integrity, which concerns the exchanged messages, subscription integrity, which concerns users' subscriptions, and finally service integrity, which concerns the protection of the service from malicious faults, such as from malicious nodes inserting bogus messages. The rendezvous network should provide supporting mechanisms that will enable integrity in publish/subscribe systems. Such mechanisms include encryption, digital signatures, as well as access control.

Publish/Subscribe architectures are also required to have high availability. Being the most critical component of a publish/subscribe architecture, the rendezvous network should be fault tolerant and it should also support fair

distribution of the load to all rendezvous nodes. Moreover the rendezvous network should be robust and particularly resistant to Denial of Service attacks.

Apart from these traditional security requirements, publish/subscribe architectures should provide authentication, anonymity, accountability, as well as scoping. The rendezvous network may–or may not–participate in the authentication mechanism. Nevertheless it should be able to properly authenticate publishers, subscribers, as well as rendezvous nodes. Moreover, rendezvous nodes should be able to identify themselves to publishers as well as to subscribers. On the other hand, the rendezvous network should support anonymity, accountability, and scoping, as it is responsible for managing user subscriptions, which should be kept secret, and for matching subscriptions with publications, therefore it is the proper place for applying accounting mechanisms and for limiting publication dissemination.

3 Security Threats

Most of rendezvous network functionality resembles to DNS, as it resolves subscription requests to publications and publishers. Nevertheless rendezvous networks are mainly implemented using DHTs–or DHT-like architectures–since DHTs have various good properties, including good load distribution. Security lessons learned from both systems should be considered.

From its inception DNS suffered from some severe problems. Poor implementation, the absence of a trust management architecture and the lack of security mechanisms led to DNS spoofing and DNS cache poisoning attacks [14]. Attackers were able to alter DNS messages or even hijack DNS sessions by taking advantage of the lack of cryptography as well as of the small set and non-randomized transaction identities. Even in 2005, after many years of DNS usage, it was found that reliance on poor transitive trust relations can lead, in many cases, to failures–as they occur in different administrative domains than the attacked DNS server [15]. Rendezvous networks should be designed with trust in mind, moreover as we learned from the DNS paradigm, no matter how well-designed a protocol is, its implementation may suffer from security vulnerabilities, therefore a reliable rendezvous network should be fault tolerant. Finally, the distributed nature of rendezvous networks and the fact that a subscription message may traverse various rendezvous nodes before it reaches its final destination, requires particular care towards resilience to (partial, intermediate) failures.

DHTs allow for efficient key lookup and load balancing among peers. Nevertheless unreliable nodes may lead to a series of problems. Sit et al. [16] classify DHT security attacks into three broad categories:

- Routing attacks: these are attacks in which a malicious node tampers with routing, e.g., it forwards lookups to an incorrect node, it issues wrong routing updates, or misleads other nodes to join a fake network, leading to network partition.
- Storage and retrieval attacks, in which a malicious node denies the existence of a piece of information, or it refuses to serve particular requests.

– Miscellaneous attacks, such as byzantine nodes, nodes that overload the rest of the network with bogus traffic leading to denial of service attacks as well as unstable nodes.

Moreover DHTs have been found to be susceptible to *Sybil* attacks, i.e., attacks in which a malicious node may spoof multiple identities [17], as well as to Eclipse attacks, i.e., attacks in which multiple malicious nodes cooperate and hide a set of a DHT network from a legitimate node [18]. The security threats that DHTs pose make the role of trust in a rendezvous system even more dominant. Trust mechanisms that promote the cooperation between reliable nodes while isolating the unreliable ones should be used. Moreover the attacks on DHTs show us–among other things–the importance of identity management; in a reliable rendezvous network efficient identification and authentication mechanisms should exist. These mechanisms will guarantee the authenticity of rendezvous nodes (and their functionality).

The current Internet is continuously threatened by Denial of Service (DoS) attacks. Wun et al. were the first that identified and classified this kind of attacks in content based publish/subscribe networks [19]. Interestingly enough, all kinds of attacks identified are based on the rendezvous network; the attackers may flood the rendezvous network with publications, causing the creation of an intolerable number of notifications towards subscribers, or if the rendezvous network allows parameterized subscriptions–e.g., all publications that concern Future Internet published between 2007 and 2009–attackers may use complex expressions and consume the resources of the rendezvous nodes. Finally, attackers may issue a large number of subscriptions, saturating rendezvous nodes memory, since state has to be kept for each subscription. It is observed that DoS attacks against rendezvous networks originate from malicious publishers and subscribers, therefore rendezvous networks should be supported by mechanisms that are able to detect, react, and mitigate them. Nevertheless, if we observe the current Internet it can be seen that users can easily hide their identity, or even have multiple identities. As a result a single mechanism that will simply identify and isolate misbehaving users will probably be ineffective. It will be necessary to identify messages, i.e., publications and subscriptions, that are suspect to lead to a DoS situation and manage them accordingly.

Spam is another security threat that is expected to target publish/subscribe architectures. Although publish/subscribe architectures are considered to be less vulnerable to this kind of attack, as subscribers have to explicitly express their interest on a specific piece of publication before they receive it, Tarkoma has shown that spam is feasible in publish/subscribe architectures [20]. Spam in these architectures can be achieved by learning user preferences. A malicious publisher that is able to predict subscription messages can easily craft publications that will match these subscriptions. The role of the rendezvous network in learning user habits is very important. Therefore, a reliable rendezvous network should guarantee user anonymity and privacy. Moreover a rendezvous network should have mechanisms that will allow the isolation of malicious publications.

4 Security Solutions

Various security solutions have been proposed for securing publish/subscribe architectures, each one satisfying some of the desired rendezvous network security requirements and offering some protection against the described security threats.

EventGuard [22], is a mechanism that aims at providing security for content-based publish/subscribe systems. Its goal is to provide authentication for publications, confidentiality, privacy, and integrity for publications and subscriptions, as well as to assure availability while keeping in mind performance, scalability, and ease of use. It uses six "guards," that secure six critical publish/subscribe operations (subscribe, advertize, publish, unsubscribe, unadvertize, and routing) as well as a meta-service that generates tokens and keys. All operations involve communication with the meta-service before sending any message. The approach that EventGuard uses for achieving subscription privacy is not so effective, as two–encrypted–subscriptions to the same publication will be the same, and by taking into consideration the number of publications that may exist in a specific rendezvous node, brute forcing subscription messages is a realistic threat. Moreover EventGuard involves many message exchanges with the security meta-service and its re-keying functionality is not well described.

Pallickara et al. [23] achieve message confidentiality and integrity by using *key management centers* (KMC). KMCs are rendezvous nodes enhanced with cryptographic mechanisms. Each of KMC can be responsible for multiple topics. However, each topic can be handled by a single KMC. Publication and subscription operations involve message exchanges with the KMCs, which lead to the creation of a symmetric key that is used for encrypting messages. Although this approach appears to be scalable, the fact that each topic is handled by a single KMC makes the system less tolerant to faults.

The aforementioned security solutions, apart from confidentiality and integrity services, provide the means for identifying nodes in a publish/subscribe architecture. Node identification can be the basis of an anti-spam mechanism–as described in [20]– in which malicious users are blacklisted. User blacklisting may also be a first step towards a DoS resistant rendezvous network.

Miklos was one of the first that discussed access control in publish/subscribe architectures [24]. His approach is based on assigning positive access rights according to a policy list that exists in every rendezvous node. This policy list is used in order to examine whether a user has the credentials to perform an operation. A more sophisticated solution was presented by Belokosztolszki et al. [25] that is based on the OASIS role-based access control system [26]. Attribute based encryption (ABE) [21] is another powerful solution that can be used for access control in a rendezvous network. Badet et al. use ABE to create a social network in which users can safely publish personal data in their profile [27], which in terms of publish/subscribe is the rendezvous network.

Subscription privacy is–to our knowledge–still an open issue in publish/subscribe architectures. This happens because from the one hand subscriptions should be kept secret in order to assure subscriber privacy, but on the other hand rendezvous nodes should be able to handle the encrypted subscription

messages in order to compare them with publications. Moreover due to the, not so big, number of publications that may exist in a rendezvous node, it is probably computationally feasible to decrypt the encrypted subscription messages by brute force. New trends in security such as homomorphic encryption [28] or even schemes that are used in distributed databases for private information retrieval, e.g., the scheme described by Ambainis [29], may be the building blocks for a subscription privacy mechanism.

Denial of Service attacks are, probably, the most difficult to be tackled, as their causes, effects, or even the layer of execution, may vary. Although it seems impossible to create a DoS attack-free rendezvous network, authentication mechanisms, challenge-based and CPU-intensive operations, as well as micro-payments, may possibly offer a level of protection for the rendezvous network.

5 Conclusions

The rendezvous network is a critical part of a publish/subscribe architecture as it is responsible for its core functions. As a result, the rendezvous network has an important role regarding the satisfaction of publish/subscribe security requirements. Therefore, at the outset, the rendezvous network seems to be the Achilles heel of a publish/subscribe architecture, since a successful security attack targeting it may jeopardize the whole operation of the system.

In this paper we highlighted the role of the rendezvous network in fulfilling the security requirements of publish/subscribe architectures and we overviewed some security mechanisms that can enable the rendezvous network in achieving this target. Future work in this area includes the incorporation of a selection from all these mechanisms into a single scheme, which will lead to the creation of a secure and robust rendezvous network.

Since most of the proposed solutions are seen basically in the context of the application layer, serious investigations into their feasibility, effectiveness, and efficiency for the core of a clean-slate architecture for a general purpose future internet are required. Such investigations are in pur future research plans in the context of the PSIRP [1] and PURSUIT (EU FP7 funded) projects.

Acknowledgment

The work reported in this paper was supported by the ICT PSIRP project under contract ICT-2007-216173.

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Mobile Botnet Detection Using Network Forensics

Ickin Vural and Hein Venter

Department of Computer Science
University of Pretoria, Pretoria 0002
Tel.: +27 12 420 3654; Fax: +27 362 5188
Ickin@tuks.co.za, hventer@cs.up.ac.za

Abstract. Malicious software (malware) infects large numbers of computers around the world. This malware can be used to promote unwanted products, disseminate offensive content, or provide unauthorized access to personal and financial information. Until recently mobile networks have been relatively isolated from the Internet, so there has been little need to protect them against Botnets. Mobile networks are now well integrated with the internet, so threats on the internet such as Botnets have started to migrate onto mobile networks. Botnets on mobile devices will probably appear very soon, there are already signs that this is happening. This paper studies the potential threat of Botnets based on mobile networks, and proposes the use of computational intelligence techniques to detect Botnets. We then simulate anomaly detection followed by an interpretation of the simulated values.

Keywords: Botnet, mobile, malware, computational intelligence, network forensics.

1 Introduction

Unsolicited bulk sms, otherwise known as spam, is an unsolicited sms sent to a large number of sms addresses, where the owners of those addresses have not asked for or consented to receive the sms [1]. Spam is used to advertise a service or a product, much like receiving junk mail through the post.

Spim is the version of spam that targets instant messaging (IM) services [2]. It has increased in prevalence along with the use of mobile phones and other hand held devices to convey short messages rapidly.

Within a few years conveniences that people access today on a desktop will be available on mobile devices. Greater adaptation of these devices will encourage more users to access personal and financial data on their phones. This means that the current threats to the internet such as spyware, Phishing and spam will move to mobile devices.

Desktop operating systems are getting harder to exploit [3], but devices such as cell phones have code bases that are largely unexplored, and updates to their software occur less frequently. The increasing processing power of mobile phones and growing features and apps included make mobile phones an ideal candidate for exploitation by malware.

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 57–67, 2010.
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Malware writers are already attacking commercial programs for mobile devices. SymbOS/Mobispy was the first mobile spyware [3]; it could remotely activate infected phones and turn them into eavesdropping devices, sending secret copies of text messages to the author.

Another issue is that mobile phone users have to pay the costs so financial crime has become a new driver for malicious software.

Malicious programs such as J2ME/RedBrowser [4] which is a Trojan horse program that pretends to access WAP web pages via SMS messages. In reality instead of retrieving WAP pages, it sends SMS messages to Premium Rate numbers thus costing the user more than intended.

The following section describes how spammers make use of Bot-nets to send spam (thus, not compromising their own identities).

2 Botnets

A Bot-network (Bot-net) consists of a set of machines that have been taken over by a spammer using special software sent over the Internet, typically in the form of a Trojan horse. This Bot software hides itself on its host machine and periodically checks for instructions from its human Bot-net administrator. Bot-nets, or Bot-nets for short, are often controlled today using Internet Relay Chat (IRC) [5]. The owner of the computer usually has no idea that his machine has been compromised until its Internet connection is shut down by an ISP. As most ISPs block bulk sms if they suspect it is spam, the spammers who control these Bot-nets typically send low volumes of sms from numerous infected computers. It is done in this fashion so as not to arouse suspicions of bulk sms originating from a single computer. Thus, the spam sms can be traced to an innocent individual's network address and not the spammer's network address.

While the number of Bot-nets appears to be increasing, the number of bots in each Bot-net is actually dropping. A Bot is defined as an infected machine in a Bot-net. In the past, Bot-nets with over 80 000 bots were common [5]. Currently, Bot-nets with a few hundred to a few thousand of bots are common. One reason for this is that smaller Bot-nets are more difficult to detect. When Procter & Gamble ran a security check of its 80,000 PCs, it found 3,000 were infected with Bot-net software [6]. The following section describes Botnets that have spread to mobile devices.

3 Botnets on Mobile Networks

Mobile devices are capable of accessing the internet through technologies such as HSDPA, EVDO, UMTS, EDGE and GPRS [7].

The connection between the internet and mobile devices acts as a gateway for malware to move from the internet to mobile networks.

Mobile devices do not use IP addresses when sending sms or mms. Every device on the mobile network has a International Mobile Subscriber Identity (IMSI) and MSISDN [8]. IMSI can be used to identify, authenticate and register the device on the mobile network. MSISDN is the number that is used to call another mobile device or

send a message [9]. Proxy servers must contain MSISDN numbers together with IP addresses if the Bot wanted to connect to the Internet.

Domain names do not exist on mobile networks; this makes it impossible to use techniques such as fast flux and multihoming on devices in the mobile network. The consequence is that mobile devices cannot act as proxy servers as they can be easily detected [10]. The following section describes Botnet characteristics.

4 Mobile Botnet Characteristics

This section describes Botnet characteristics that could be used by a Botnet evaluation engine to detect Botnets.

The characteristics identified are the following:

- Number of recipients per sms: For this, a number between 0 and 1 is assigned to each sms. A value of one indicates that the sms is sent to one recipient. A value close to zero means the sms was sent to many recipients.
- smses per second: The frequency of outgoing smses.
- A high ratio of out going to incoming smses is a characteristic of a spamming Bot-net [11]

We believe that the identification of these characteristics would make Botnet identification possible. The authors of this paper propose to implement a network forensics system to identify Botnets.

The following section introduces the concept of network forensics which is the basis of this paper.

5 Network Forensics

Network forensics is the capture, recording, and analysis of network events in order to discover the source of security attacks or other problem incidents [12]. Capturing network activity for forensic analysis is simple in theory, but relatively trivial in practice. Not all the information captured or recorded will be useful for analysis. Identifying key features that reveal information deemed worthy for further intelligent analysis is a problem of great interest to the researchers in the field [13].

- Network forensics systems can be one of two kinds: “Catch-it-as-you-can” systems, in which all packets passing through a certain traffic point are captured and written to storage with analysis being done subsequently in batch mode. This approach requires large amounts of storage [14]
- "Stop, look and listen" systems, in which each packet is analyzed in a rudimentary way in memory and only certain information saved for future analysis. This approach requires less storage but may require a faster processor to keep up with incoming traffic [14].

One concern with the “catch-it-as-you-can” approach is one of privacy since all packet information (including user data) is captured. Internet service providers (ISPs) are

expressly forbidden by the Electronic Communications Privacy Act (ECPA) from eavesdropping or disclosing intercepted contents except with user permission, for limited operations monitoring, or under a court order [12].

The following section discusses the use of advanced Botnet detection techniques.

6 Mobile Botnet Identification

This section discusses anomaly detection and how anomaly detection techniques can be applied to sms traffic to detect spamming Botnets. The purpose is to identify mobile devices that may have been infected with Botnet software. The purpose is not to identify the identity of the Botnet controllers, otherwise known as spammers. Using a combination of statistical and social network analysis techniques it is possible to build behaviour models of individual mobile phone users and groups of mobile device users. These behaviour models are used to establish the "normal" expected behaviour of the mobile users and compared with current or recent usage data to detect "abnormal" changes in communication behaviour.

The shortcoming of using an abnormal behaviour model is deciding what constitutes abnormal behaviour. One way of detecting abnormal behaviour is by using mathematical models. The problem with using mathematical techniques is that they impose strict boundaries around the profiles of what is considered ‘normal’ and ‘abnormal’ behaviour [15]. The slightest deviation from the ‘normal’ profile, while the actual network activity is still considered to be normal, will cause the anomaly detector to trigger a false alarm. Because of the uncertainty in defining the boundaries between what is ‘normal’ and ‘abnormal’ user behaviour when analyzing command sequences or mobile traffic behaviour of users, this is an ideal area for fuzzy systems to be applied [16].

Another option is the use of artificial neural networks [17]. The advantage of using neural networks in anomaly detection is that features of ‘normal’ and ‘abnormal’ behaviour can easily be learned by the neural network, as opposed to applying mathematics to describe the features of the data to the anomaly detector.

The following section describes how anomaly detection could be applied to detecting Bot-nets on a network by analysing mobile traffic behaviour.

7 Mobile Botnet Identification with Network Forensics

The authors of this paper believe that an anomaly detection system, that monitors mobile accounts to determine if they have been compromised by Botnet software, could be useful for mobile communications network provider. The mobile networks server could monitor the mobile accounts to learn “normal” behaviour and detect any significant deviations from the baseline. This is similar to how a bank would monitor ATM usage to detect fraud.

The initial step would be to build a profile of each mobile user on the test network’s behaviour, or so called “normal” behaviour. The behaviour model is represented by building a profile of the mobile user’s sms-sending behaviour.

A list of the metrics chosen for building such a behaviour profile for each mobile user is as follows:

- Volume of weekly out going sms traffic
- Consistency of weekly out going sms volume
- sending delays of sms
- Median of sending delays of sms
- Volume of weekly incoming sms sms
- Consistency of weekly incoming sms volume

This information would be gathered by a network forensics tool for further analysis. The volume of weekly out going sms traffic was chosen as a metric as it is easy to measure, because it is based on a count of the number of sms generated by an individual in a given time period. Therefore, it provides information on traffic volume generated by a particular mobile user. Consistency of weekly sms traffic volume is computed to determine how "consistent" or "reliable" a mobile user (individual) is with the weekly volume of sms traffic sent over a time period. The aim is to determine the consistency of the measurements that was collected of the previous metric (volume of sms traffic). The autocorrelation, which is a mathematical tool used to find repeating patterns that have been buried under "noise" [19], is used to determine the consistency of weekly sms traffic volume. The autocorrelation r in equation 1 below is computed using statistical methods derived from [20] and [21]. This autocorrelation produces a number between -1.0 and 1.0 to indicate the relationship between each time series point in the weekly sms volume data. This is used to describe the evolution of the data captured over time so as to determine trends in the analysis of the sms traffic volume. This is useful in determining randomness as well as to attempt to predict future sms traffic volumes. This technique basically attempts to reduce the "noise" in the data collected by filtering out "random" data. Given N observations of the volume of weekly sms traffic $x_1 \ldots x_N$, we can form N-1 pairs of observations (x_1, x_2) (x_2, x_3)… (x_{N-1}, x_N) where each observation is separated by one time interval [20] as follows:

$$r = \sum_{t=1}^{N-1} (x_t - x_1)(x_{(t+1)} - x_2) \div \sqrt{\left[\sum_{t=1}^{N-} (x_t - x_1)^2 \sum_{t=1}^{N-1} (x_t - x_2)^2\right]} \tag{1}$$

Where $x_1 \ldots x_N$, are a set of N observations.

$$x_1 = \sum_{t=1}^{N-1} x_t/(N-1) \tag{2}$$

is the mean of the first N-1 observations and

$$x_2 = \sum_{t=2}^{N-1} x_t/(N-1) \tag{3}$$

is the mean of the last N-1 observations.

Delays between smses sent is a measure of the time delays between each sms message sent by an individual. This provides information on expected delays between each message sent by a mobile user (individual).

The median of sending delays is used to compute the commonly occurring time delays between smses sent by using the statistical median. The purpose, in this case, is also being able to remove the "noise" associated with the sending delay data collected.

The volume of weekly incoming sms traffic has been chosen as a metric because spamming Botnets send many smses but receives no smss because it does not have a domain which can be used for receiving smses and is only designed to send smses. Thus a high ratio of out going to incoming smses is a characteristic of a spamming Botnet [22]

The next step is to apply the use of fuzzy techniques to sms behaviour analysis. Fuzzy techniques have already been applied to various anomaly detection problems, for example in intrusion detection [23], [24] and diagnosis of traffic anomalies in network traffic management [25]. The use of fuzzy techniques for sms traffic anomaly detection, offers a number of advantages. The first advantage is that fuzzy techniques have been shown to be useful for overcoming the rigid boundary or "sharp boundary problem" found in anomaly detection [26].

Fuzzy techniques help to provide a softer distinction between "normal" and "abnormal" behaviour and can be used to rate changes in behaviour by degrees of abnormality. The second advantage to using fuzzy techniques with sms behaviour analysis is that fuzzy techniques have been proven to be useful for dealing with uncertain or imprecise information [24]. This is ideal when working with data extracted from observations of human behaviour. The third advantage of using fuzzy techniques is that they have been shown to be useful for fusing together information from various inputs and applying heuristics to determine the overall status of the inputs [23]. The anomaly detection process follows a set of steps which are outlined as follows. First we calculate the "normal" or baseline behaviour for the selected mobile users in a network. This is done by selecting the mobile users that will be monitored, and then decide on a period of time that will sufficiently be used as the profiling period, i.e. to "learn" a "normal" behaviour profile.

Using the following formula, the baseline behaviour is computed.

$$x_{ij} = f_j(\mu\ i), \text{ for } i=1\ldots, k;\ j=1\ldots6 \tag{4}$$

Where i is the iteration number of each mobile user, k is the total number of mobile users monitored, μ i is the set of observations of interaction with mobile user i during the time period, and *f*j is the function used to compute metric j during the time period. We then derive xij, which is the normal "behaviour" measurement for the interaction with mobile user i, using metric j.

Then we select another time period to examine the mobile users on the selected networks to be used later to detect abnormalities from the "normal". This current behaviour is computed as follows.

$$y_{ij} = f_j(\ i), \text{ for } i=1\ldots,k;\ j=1\ldots6 \tag{5}$$

Where i is the iteration number of each mobile user, k is the total number of mobile users monitored, i is the set of observations of interaction with mobile user i during the time period and f_j is the function used to compute metric j during the time period. We then derive y_{ij} which is the current "behaviour measurement" for the interaction with mobile user i using metric j. The difference between the two measurements (equations 4 and 5) is computed as follows in order to detect if there was an abnormality.

$$d_{ij}=y_{ij} - x_{ij}, \text{ for } i=1 \ldots k; j=1\ldots6 \tag{6}$$

The difference for each mobile user (d_{i1}, d_{i2}, d_{i3}, d_{i4}, d_{i5}, d_{i6}) is used as input for the computational intelligence agent and an abnormality rating is output as follows.

$$z_i = (d_{i1}, d_{i2}, d_{i3}, d_{i4}, d_{i5}, d_{i6}) \text{ for } z_i \text{ in } [0, 1] \tag{7}$$

Where z_i is the overall change in sms traffic behaviour for the mobile users on the selected network and where e_i 0.0 indicates little change in sms behaviour and e_i 1.0 indicates a large change in sms behaviour.

The above concept can be better explained by the use of an example. Suppose we had our computational intelligence agent monitor the behaviour of two sms accounts from a mobile user. If, as in this example, the behaviour of the user's sending patterns were learnt (after accounting for "noise" and randomness as explained earlier) and we were able to build up a sending profile of these sms accounts, then the next step would be to monitor their current behaviour as part of a network forensics information capture and plot the change in sending behaviour.

The following section uses simulated examples to explain how the metrics listed above are used to detect abnormalities is a network.

8 Simulations

To simulate abnormality in a mobile network we first measure the values of the metrics listed in the previous section, namely, volume of outgoing sms traffic (metric 1), consistency of outgoing sms traffic volume (metric 2), sending delays for smses sent (metric 3), median of sending delays for smses sent (metric 4), volume of incoming sms traffic(metric 5) and consistency of the outgoing sms traffic (metric 6) for the monitored mobile accounts bob and alice during the observation period. Using simulated values for the metrics listed above, we measure the values for x_1 (which is bob) and x_2 (which is alice) for each metric j= 1….6 Metric j=2 is computed by applying equation 1 to the values measured in metric j=1 over the observation period. The measurements for each mobile account monitored are tabulated as in equation 4.

Metric 1 = the volume of weekly smses sent by the mobile user monitored on a scale between 0 and 1 where a value close to 0 indicates a large amount of smses sent; a value close to 1 indicates that few smses are sent.

Metric 2 = autocorrelation of weekly smses sent by the mobile user monitored on a scale between -1 and 1 where the closer the value is to 0, indicates the greater the consistency of the values measured of metric 1.

Metric 3 = sending delays of smses sent by the mobile account monitored on a scale between 0 and 1 where a value closer to 0 indicates a small delay and a value of 1 indicates a large delay.

Metric 4 = median of sending delays of smses sent by the mobile account monitored on a scale between 0 and 1 where a value closer to 0 indicates a small delay and a value of 1 indicates a large delay.

Metric 5 = the volume of weekly smses received by the mobile user monitored on a scale between 0 and 1 where a value close to 0 indicates a large amount of smses received; a value close to 1 indicates that few smses are received.

Metric 6 = autocorrelation of weekly smses received by the mobile user monitored on a scale between -1 and 1 where the closer the value is to 0, indicates the greater the consistency of the values measured of metric 1.

Table 1. Metrics measured during observation period

Metric	bob	alice
Metric 1	0.3	0.2
Metric 2	0.2	0.1
Metric 3	0.4	0.2
Metric 4	0.4	0.2
Metric 5	0.3	0.2
Metric 6	0.2	0.1

The above measurements help us to train our computational intelligence agent over a period of time so that the computational intelligence agent can lean the mobile account's behaviour. This learnt behaviour forms a baseline against which all future observations are measured. The next step in detecting abnormality in a network is to measure the current behaviour of the sms accounts and compare this against the baseline values captured above. Using simulated values for the metrics listed above, we measure the values for y_1 (which is bob) and y_2 (which is alice) for each metric j= 1…6. The metrics are tabulated as in equation 5 and represent the current state of the sms accounts monitored sending behaviour.

Table 2. Metrics measured during current period

Metric	bob	alice
Metric 1	0.7	0.3
Metric 2	0.8	0.3
Metric 3	0.7	0.3
Metric 4	0.7	0.3
Metric 5	0.3	0.2
Metric 6	0.2	0.1

The above measurements are used to compute equation 6.

First we compute the absolute value of the difference between values in y and values in x, so the values would be calculated as follows:

$d_{11} = y_{11}-x_{11} = 0.7 – 0.3 = 0.4$ (difference in volume of weekly smses sent by bob on a scale between 0 and 1). In the same way the values for $d_{11}…d_{24}$ are calculated, assume the values were calculated as follows: $d_{12} = 0.6$, $d_{13} = 0.3$, $d_{14} = 0.3$, $d_{15} = 0.0$, $d_{16} = 0.0$ $d_{21} = 0.05$, $d_{22} = 0.2$, $d_{23} = 0.1$, $d_{24} = 0.1$, $d_{25} = 0.0$ and $d_{26} = 0.0$

From these above computed values of d, e_1 and e_2 are computed from equation 7 as follows:

z_1= (0.4, 0.6, 0.3, 0.3, 0.0, 0.0) = 0.4 for $z_1$in [0, 1]
z_2= (0.1, 0.2, 0.1, 0.1, 0.0, 0.0) = 0.125 for $z_2$in [0, 1]

The following table (table 4.3) shows the two example sms accounts and the overall change in sms sending behaviour for each one as computed in equation 7.

Table 3. Change in behaviour for monitored sms addresses

Sms Address	change in sms traffic behaviour (z)
bob	0.4
alice	0.125

The results tabulated above show the change in sms traffic sending behaviour. The following section interprets the tabulated values.

9 Interpretation of Results

In this section we will attempt to interpret the results of our simulation in the previous section. The values tabulated in table 4.3 indicate that bob has experienced a large change in sms sending behaviour as indicated by z 1.0 and alice has experienced little change in sending behaviour as evidenced by z 0.0.

Thus, by monitoring sms sending behaviour we hope to detect abnormalities in sending behaviour over time that could possibly indicate a Botnet infected mobile device. This can then be used by a network forensics tool to identify Botnets on a mobile network.

The following diagram, figure 1, shows a network forensics system with its inputs. Each input is a metric chosen to build a behaviour profile of a mobile users account. These metrics are computed and filtered for randomness and “noise” and then fed into the fuzzy system as inputs. The fuzzy system then computes the overall change in sms traffic behaviour and produces the abnormality output z as in (7) for each sms account monitored.

Where z_i 0.0 indicates little change in sms behaviour and z_i 1.0 indicates a large change in sms sending behaviour for each sms account i. A high ratio of outgoing (metric 1) to in coming (metric 5) sms traffic may also indicate a spamming Botnet.

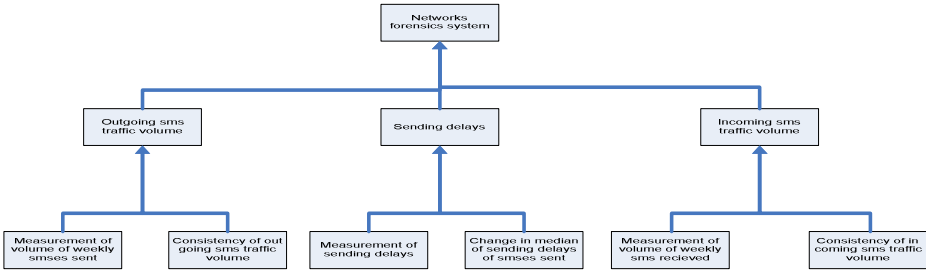

Fig. 1. Network forensics system with its inputs

10 Conclusion

While mobile Botnets are not yet a great threat, the authors of this paper believe they will soon become threats to the mobile networks and the mobile operators. The implementations proposed will it is hoped enable a mobile operator to detect Botnets on a network. This would obviously require some scaling. Thus the authors are looking at graph analysis techniques to detect abnormality on a larger scale. When applied to a live system this would also enable us to better identify Botnet characteristics and thus build better Botnet identifiers.

Acknowledgments

This work is based on research supported by the National Research Foundation of South Africa (NRF) as part of a SA/Germany Research cooperation programme. Any opinion, findings and conclusions or recommendations expressed in this material are those of the author(s) and therefore the NRF does not accept any liability in regard thereto.

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Towards Linked Open Services and Processes

Reto Krummenacher[1], Barry Norton[2], and Adrian Marte[1]

[1] Semantic Technology Institute, University of Innsbruck, Austria
{reto.krummenacher,adrian.marte}@sti2.at
[2] Karlsruhe Institute Technology, Germany
barry.norton@kit.edu

Abstract. The combination of semantic technology and Web services in form of 'Semantic Web Services' has until now been oriented towards extension of the WS-* stack with ontology-based descriptions. The same time, there is a strong movement away from this stack — for which the 'Web' part is little more than branding — towards RESTful services. The Linked Open Data initiative is a keen adopter of this approach and exposes many datasets via SPARQL endpoints and RESTful services. Our developing approach of 'Linked Open Services', whose current state is described in this paper, accommodates such Linked Data endpoints and general RESTful services alongside WS-* stack-based services with descriptions based on RDF and SPARQL. This capitalises on the Linked Data Cloud and makes service description and comprehension more easy and direct to the growing Linked Data community. Along the way, we show how the existing link between service messaging and the semantic viewpoint, commonly called 'lifting and lowering', is usually unduly restricted to ontology-based classification and misses how the effect of a service contributes to the knowledge of its consumer. Our SPARQL-based approach helps also in the composition of services as knowledge-centric processes, and encourages the development and exposure of services that communicate RDF.

1 Introduction

In the existing literature the combination of semantic technologies and Web services to form 'Semantic Web Services' has been oriented towards extending the WS-* stack (based on the XML-based languages WSDL and SOAP, and built on to form UDDI and BPEL) with ontology-based descriptions [3,4,15]. Their claimed benefits are centred on an increased degree of automation in high-level tasks such as the discovery, composition and mediation of services that are a priori already explicit aims of the Web service approach. At the same time, however, there is a strong movement away from this underlying stack – for which the 'Web' naming is acknowledged as little more than branding – towards a resourceful ('RESTful' [6]) model of services [12]. The REST model directly adopts the Web resource model of HTTP and eschews the WS-* stack languages as cumbersome and redundant. Among the prominent adopters of this approach is the Linked Open Data (LOD) community who seeks to increase the amount of Semantic Web-style content, in the form of RDF, online.[1] As well as dedicated RESTful services, Linked Data exposes many datasets via passive SPARQL endpoints.

[1] http://www.linkeddata.org

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 68–77, 2010.
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Our approach of 'Linked Open Services' (LOS) combines LOD endpoints and RESTful services in general, with services based on the WS-* stack, and makes service descriptions based primarily on RDF and SPARQL.[2] These descriptions show how each service is a consumer and producer of RDF data, and thereby are Linked Data citizens. We thus leverage widely-adopted Linked Data vocabularies and the wealth of data currently exposed. We believe this is much closer to the original vision of agents on the Semantic Web, without the unproductive bias of enterprise dreams of big SOA. Indeed only by such an accessible integration of service functionalities can the Web of Data realise its promises. This requires more than purely data-oriented mash-ups and must involve heavy on-the-fly computation and the possibility to trigger real-world side effects in back-end systems.

To this end, we reuse existing work on linking service messaging with semantic representations, commonly referred to as 'lifting and lowering' [7]. We explicitly address, however, their usual undue orientation towards simple ontology-based classification, which misses the opportunity to encode how a service contributes to the knowledge of its consumers.

In this paper we propose further a SPARQL-driven approach to service *composition* that puts RDF at the fore and employs SPARQL also in the definition of processes that combine LOS. For LOD developers this brings first the advantage of familiarity, since SPARQL is the de facto language for interaction with Linked Data. Secondly, whereas current Linked Data endpoints are, at this point, mostly conduits for static, precomputed data, our approach brings side-effecting reactivity to Linked Data compositions (or 'mash-ups'), without requiring either the heavyweight XML-based WS-* stack or the complicated and, despite RIF standardisation, generally poorly accepted rule languages employed by traditional Semantic Web Services.

The paper reviews the state-of-the-art in describing and composing Semantic Web Services in Section 2, and compares this with LOS composition in Section 3. After the presentation of the basic concepts and technicalities of our approach, we depict in Section 4 a first implementation and discuss some insights gained from initial experimentation with WS-* stack-based and RESTful services, as well as Linked Open Data. Conclusions are drawn from this work in Section 5.

2 Semantic Web Services and Their Composition

Traditional Semantic Web Services aim to apply ontology- and rule-based semantics to enhance the Web service description stack to increase automation in their consumption and composition. OWL-S[3] is an ontology for the description of Web services that has been primarily applied to SOAP services with WSDL descriptions. The OWL-S Service Profile increases the amount of static information available about a Web service from that associated with WSDL and UDDI in the form of OWL-DL descriptions. In particular the taxonomic classification of services that UDDI adds to WSDL is captured via so-called 'service categories'.

[2] `http://www.w3.org/TR/rdf-sparql-query/`
[3] `http://www.w3.org/Submission/OWL-S/`

Input and output messages of service operations — more precisely, in combination with OWL-S' Service Grounding, and the WSDL 1.1 bindings, the 'parts' of messages — are mapped onto OWL concepts. This is a direct map that can be supplied with a transformation between the physical representation, for instance in XML, and the semantic representation in OWL. The concepts attached to parameters express what is communicated by the messages *in themselves*, not what implicit knowledge can be inferred from executing the service, in particular in relationship to the knowledge represented in the input. The OWL-S Process Model allows a composite Web service behaviour to be derived from 'atomic processes', which are primarily SOAP operations. This process model defines control-flow in a strictly block-oriented fashion, but dataflow is specified in a graph-oriented manner, i.e., by connecting parameters of atomic and contained composite processes. This assumes that outputs, once transformed to semantic representations, directly fulfil the required inputs. In order to express conditions for the decision points in processes, the OWL languages are insufficient and SWRL is recommended, though this non-standard language has limited tool support and adoption potential from the communities composing services.[4]

The Web Services Modeling Ontology (WSMO, [13]) is a similar, but broader scoped, extension of Web services with ontology- and rule-based modelling and reasoning. 'Grounding' within service models similarly allows a correspondence between physical messages, primarily the WSDL description of SOAP messages, and ontology concepts. 'Lifting' and 'lowering' are the respective terms for the resulting inter-transformations. Rather than mixing an ontology language and a separate rule language, as in OWL-S, the WSMO meta-model is implicitly included in a family of ontology languages, called WSML for Web Service Modeling Language [2], which includes the ability to define rules. For composition, an ontologized form of abstract state machines [1] has been proposed for the definition of orchestrations [5]. These build on the rule of the WSML syntax to form implicitly stateful behaviours. The control flow, in terms of propagation of control between states, is thus implicit in rule-oriented form.

The W3C submissions of OWL-S and WSMO led not to standardisation of either, but rather to the definition, and subsequent recommendation, of SAWSDL, 'Semantic Annotations for WSDL and XML Schema'.[5] SAWSDL does not define any specific ontological service model at all, but allows 'model references' from parts of the WSDL service schema into arbitrary ontological models. Since, however, model references are shown to explicitly apply to WSDL messages, and 'lifting and lowering schema' references are also provided, it is clear that SAWSDL preserves the simplistic view that the concepts targeted are mere representations of the messages and are not extended, as in our approach, to the implicit knowledge associated with to service execution. One response to the publication of SAWSDL has been to view it as an extra (dual) link to the existing models. Another, that of WSMO-Lite [16], has been to produce a reduced ontological model in direct correspondence with WSDL. Since the 'procedure-oriented service model'[6] (formerly 'minimal service model') ontology associated with WSMO-Lite is close to our requirements, we choose to reuse this for RPC-oriented LOS.

[4] `http://www.w3.org/Submission/SWRL/`
[5] `http://www.w3.org/TR/sawsdl/`
[6] `http://www.wsmo.org/ns/posm`

3 Linked Open Services and Their Composition

So far we have claimed limitations of existing SWS approaches with respect to the loss of implicit knowledge associated with service invocation, and the use of non-standard and unfamiliar languages for description. The above described approaches also fall down in the developing 'Web 3.0' context with respect to two increasingly common types of service. Firstly, a significant rejection of the WS-* stack in 'Web 2.0' and derivative extensions, is due to the RESTful services movement. In REST arbitrary operations are not overloaded (usually over HTTP POST) on single endpoints in SOAP RPC style, rather the HTTP operations (GET, PUT, POST and DELETE) are applied to resources addressed via individual URLs. Secondly, many Linked Data providers simply provide SPARQL endpoints, and therefore one standard RESTful interface. Furthermore there are an increasing number of RESTful services that already directly produce and consume RDF.[7]

For all of these reasons we propose a model for Linked Open Services in which services are viewed primarily as RDF 'prosumers'. LOS may wrap SOAP, non-RDF communicating RESTful services, or SPARQL endpoints. They can be understood, however, and combined using SPARQL, and their composition is open to include further Linked Data sources to extend a process's knowledge base. The core concept of our approach is thus to have Linked Open Services take RDF graphs as an input and produce RDF as output.[8] This idea, related to the service descriptions proposed in [14], and its novel extensions for composition are depicted in Figure 1.

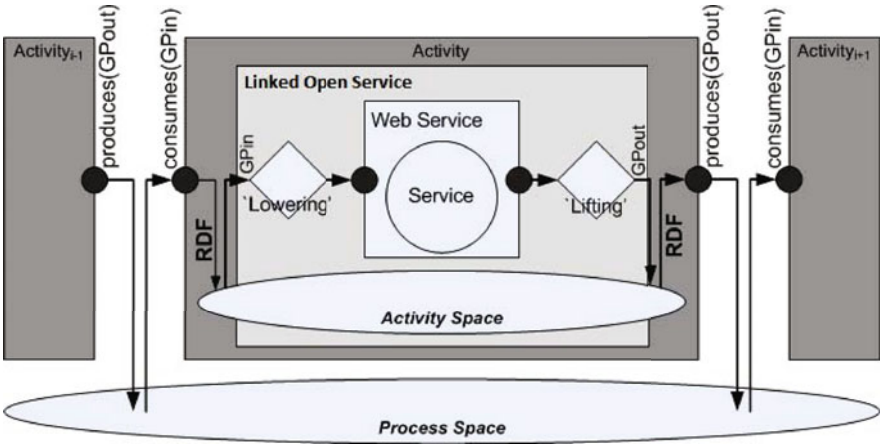

Fig. 1. Linked Open Services composition

[7] Witness the recent acceptance as an Apache Incubator project of Clerezza: `http://incubator.apache.org/clerezza/`

[8] The input is consumed and the output added to the process's knowledge base, which in our implementation is hosted in the Process Space.

No matter what type of underlying service is used, at the level of LOS, all communication is conducted at the semantic level by means of RDF. The description of the services' input and output are no longer given by linking some value to some concept in the ontology, but rather by a (SPARQL) graph pattern which precisely describes the expected input graph, respectively the guaranteed knowledge content of the output statements. For example, a graph pattern that describes the input to the GeoNames 'search' RESTful service[9] is shown in Table 1, as part of the semantic service description of the example process in Section 4.

Table 1. Lightweight service description for GeoNames' search service

```
@prefix sparql: <http://www.w3.org/TR/rdf-sparql-query/#> .
@prefix sawsdl: <http://www.w3.org/ns/sawsdl#> .
@prefix wsl: <http://www.wsmo.org/wsmo-lite#> .
@prefex posm: <http://www.wsmo.org/ns/posm/0.1#> .
@prefix geo: <http://ws.geonames.org/> .
geo:search rdf:type posm:Service;
 posm:hasInputMessage [sawsdl:modelReference _:input];
 posm:hasOutputMessage [sawsdl:modelReference _:output];
 sawsdl:loweringSchemaMapping
  <http://www.LinkedOpenServices.com/ex/geo/search.sparql> .
_:input rdf:value "?x geo:name ?name ."^^sparql:GraphPattern .
_:output rdf:value "?x wgs84:lat ?lat; wgs84:long ?lng ."^^...
```

Having the corresponding RDF data available, Web dereferencable or in a process-specific semantic space (termed *Process Space* on Figure 1), allows for the creation of SPARQL CONSTRUCT queries that fetch and prepare the required RDF statements. A semantic space is a semantic-minded evolution of tuplespaces into a blackboard-style communication and coordination platform for the realization of collaborative problem-solving activities [9]. The fundamental principle on which computation over semantic spaces is based is that participating agents or services do not communicate with each other directly, but instead collaboration is guided by the coordinated access to and the (concurrent) modification of a shared set of RDF statements. In the simplest case, the one depicted in Figure 1, all the required and produced RDF data is shared in a process-owned space against which the SPARQL queries are executed. Generalizing this setting, without altering any of the core concepts, the input data could also be derived by more complex CONSTRUCT queries over public data sources, or by means of Semantic Web Pipes [8] that aggregate data from various sources on the Web or the LOD cloud. As a direct consequence of this implicit invocation of services within a process, there is no need to ship data around between Linked Open Services. In fact, any service selects precisely the data from the space that is required, and as such, the input to one service is a priori independent of the output of the predecessors. In other words, the execution of a process does not really require the specification of any data flow.

[9] `http://sws.geonames.org/search`

The last missing piece to Linked Open Service composition relates to the fact that most Web services still do not accept or produce RDF data.[10] Although more and more services rely on RDF directly, as well as on REST principles in invocation, XML, as well as simplifications in JSON and YAML[11], is still the predominant format for data transfer over the Internet. This is oddly even the case for some services that are associated with RDF data, such as for example the GeoNames services that are assumed part of the Linked Open Data cloud. In order to invoke such a Web service, it is still necessary to transform from RDF to the expected data format of the service implementation, and to map the output of the service back into RDF. Consequently a LOS implementation may, in general, specify not only what graph patterns it consumes and produces, but also how the input RDF is 'lowered' to the expected data format of the service, respectively how the output of the service is 'lifted' back to RDF. On the other hand, if the service directly consumes RDF, the input is read directly from the activity space in which an activity's knowledge (triples) is maintained during its lifetime. In all other cases the lowering box prepares the appropriate call. This can be done by help of a simple SELECT query that binds values to the query variables in a URL of a simple HTTP GET, or by more heavyweight XSLT or XSPARQL[12] transforms for XML-based services; e.g., services expecting SOAP messages.

To this end, activities, which are stages of a process, consume RDF too, and through the execution of a service contribute to the overall process knowledge maintained in the space. Hence, before carrying out the lowering, to derive the input message for the invocation, the declared 'consume' CONSTRUCT is carried out over the process space, and any required external Linked Data sources, populating the activity's space. After the invocation the returned message, lifted if necessary, is added first to the local activity space, the 'produce' CONSTRUCT is then carried out over the activity's space in order to derive triples that are injected into the process space. In this way, implicit knowledge about the links between the input and output, as well as provenance and context related to the authority (i.e., the service) can be included in the activity's contribution to the process' knowledge base.

4 Evaluation

In order to evaluate our conceptual approach, we have carried out a proof-of-concept implementation. To illustrate the implementation we designed a hypothetical scenario that relies on the composition of three services: a real existing RESTful service that produces RDF data, a real one that returns an XML message and a locally implemented service that relies on SOAP messages; we chose not to use an existing service in this latter case only to avoid real-life side effects. A formal definition of our evaluation process[13] is given in Table 2.

[10] Though it should be noted that the content negotiation aspect of REST allows this to be seamlessly added and, in general, we support this.

[11] See `http://www.json.org/` and `http://www.yaml.de/en/`

[12] `http://www.w3.org/Submission/2009/01/`

[13] According to the simple process schema published at `http://www.linkedopenservices.org/ns/process/simple/0.1#`

Table 2. Process formalization

```
@prefix act: <http://www.[los].org/ns/activity/1.0#> .
@prefix proc: <http://www.[los].org/ns/process/simple/0.1#> .
@prefix geo: <http://ws.geonames.org/> .
@prefix sparql: <http://www.w3.org/TR/rdf-sparql-query/#> .
_:p1 rdf:type proc:Sequence;
   proc:first _:a1; proc:next _:x;
   proc:consumes "CONSTRUCT ..."^^sparql:construct .
_:a1 act:performs geo:search;
   act:consumes
     "PREFIX geo: <http://www.geonames.org/ontology#>
      CONSTRUCT {?x geo:name ?o1 .}
      WHERE {?x geo:name ?o1 .}"^^sparql:construct;
   act:performs
     "PREFIX wgs84: <http://www.w3.org/2003/01/geo/wgs84_pos#>
      CONSTRUCT {?x wgs84:lat ?lat; wgs84:long ?lng .}
      WHERE  {?x wgs84:lat ?lat;
                 wgs84:long ?lng .}"^^sparql:construct .
_:x proc:first _:a2; proc:next _:ch .
_:a2 act:performs geo:findNearByWeatherXML;
   act:consumes "CONSTRUCT ..."^^sparql:construct;
   act:performs "CONSTRUCT ..."^^sparql:construct .
_:ch proc:hasBranch _:cbr, _:br .
_:cbr proc:hasCondition "ASK ..."^^sparql:ask;
   proc:choosesFor _:p2;
   proc:asserts "CONSTRUCT ..."^^sparql:construct .
_:br proc:choosesFor _:p2;
   proc:asserts "CONSTRUCT ..."^^sparql:construct .
_:p2 proc:hasFirst _:a3 .
_:a3 act:performs <http://localhost:8080/axis/YP.jws>;
   act:consumes "CONSTRUCT ..."^^sparql:construct;
   act:performs "CONSTRUCT ..."^^sparql:construct .
```

The process composes three activities that represent each of the three services, and represents control flow between activities. Note, however, that the LOS approach is agnostic to the choice between block- and graph-oriented control flow; it is also suitable for data-driven approaches, wherein control may implicitly be defined according to some global scheme.[14] For the purposes of illustration, we present a simple block-oriented representation of processes with explicit control flow. One reason to do this is to demonstrate the use of SPARQL ASK queries to define conditions that affect the process control. We thereby show that our scheme is more expressive than existing data-driven 'mash-up' approaches to the composition of services as RDF prosumers, but still avoiding non-standard rule languages in favour of a standard whose use is natural and familiar to the community to whom we intend to appeal.

[14] Such data- and control-driven schemes are surveyed in [10].

The first service is from GeoNames. It takes the name of some geographic feature, and a return type as query variables, e.g.: `http://ws.geonames.org/search?type=rdf&name=Innsbruck`. When the client (HTTP) accepts RDF, the service will return RDF/XML, rather than some proprietary XML document, which can be written directly back into the activity space without the need for lifting. A semantic description of this service has already been shown in Table 1. There are references given to the condition on the input of the service, the effect of the output, and a link to a lowering schema mapping; there is not need for a lifting schema, as the service natively returns RDF. In our SPARQL-minded approach, the condition and effect of a service are given by graph patterns, as they are defined by the SPARQL specification. In fact, the graph patterns that are part of the service description relate directly to the consumes and produces part, respectively, of the activity description – compare, for example, the value of the input in Table 1 with the consumes statement of the first activity in Table 2.

The lowering for the 'search' service is very simple in the given case. All that is needed is the name of the location to *search* which becomes part of the service URL as query string. Hence, the sole variable binding of the SPARQL SELECT query "`SELECT ?name WHERE {?x geo:name ?name .}`" becomes the ?name variable of the service URL. The SELECT query is executed against the activity space which is populated by the aforementioned CONSTRUCT query.

The second service is GeoName's 'findNearByWeatherXML' service.[15] It takes latitude and longitude coordinates as input and returns the observations of the nearest weather station as an XML document, including temperature, humidity, clouds, atmospheric pressure, and wind direction and speed amongst other things. The service description is similar to the one presented in Table 1, and we omit it here.

The third service that is invoked depends on the weather information returned by GeoName's weather service. For this reason, the process contains a 'choice' construct over the knowledge in the process space that was asserted by the preceding activities. Choices and loops are both based on branches; a choice is expected to have exactly one non-conditional (default) branch and any number of conditional ones, based on SPARQL ASK queries, whereas a loop is expected to have exactly one such conditional branch only. Each branch is allowed to assert further knowledge, via a CONSTRUCT query, into the process space. The choice exemplified in Table 3 is realized by means of a temperature threshold.[16]

Table 3. The 'choice' construct ASK query

```
PREFIX wth: <http://www.example.org/weather#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

ASK {?x wth:temperature ?t .
FILTER (?t < "10"^^xsd:integer) .}
```

[15] `http://ws.geonames.org/findNearByWeatherXML`

[16] The predicate 'temperature' matches the predicate of the lifted output of the weather service.

Consequently, either the SOAP-based yellow pages service is invoked to query the phone number of a local restaurant, or under better weather conditions the contacts of some tennis club in the chosen city. As both branches of the choice trigger the same activity that wraps the service `http://localhost:8080/axis/YP.jws`, the difference in execution is given by the asserted triples of the branches. As stated above, the conditioned branch causes the request for a restaurant's contact information, and hence the type of business to look for that has to be communicated to the yellow pages service is "Restaurant", otherwise "Tennis".

The triples which are asserted by the choice construct match the condition graph pattern of the yellow pages service: "`{?x yp:loc ?l; yp:type ?t.}`". The choice ensures that the input data which is required by the last activity of our composition is available in the process space, and can be moved into the activity space. From there the activity's consumes CONSTRUCT is invoked and the semantic data is lowered to the required SOAP message, which is sent to the yellow pages service. The response XML is analogously lifted back to RDF and written back to the process space as final output of the process. In our example a graph is constructed that describes a restaurant or tennis club entry with a name, phone number and address.

5 Conclusion

Recapulating the paper, we have presented the notion of Linked Open Services, which leverage established semantic technologies of the greatest familiarity to the Linked Data community, such as RDF(S) or SPARQL. As well as creating useful descriptions, these same technologies are used to create processes, and to determine their control- and data-flow. This is complementary to the combination of Linked Data and SWS presented in [11] but goes beyond mere identification of descriptions as LOD into executable processes that interact with the Linked Open Data cloud. In particular, we intend to establish a composition approach that is more intuitive and appealing to the Linked Data community. Indeed, instead of logic-heavy descriptions, via the use of complicated and non-standard rule languages, as proposed by various SWS frameworks, we describe services as prosumers of RDF data by graph patterns that are directly applied for the construction of the input and output (RDF) messages of the involved services. In that way, services emerge from Linked Data and, after execution, explicitly feed RDF data back to the Semantic Web.

Although we have not yet been able to evaluate our approach at scale, to ensure that Linked Open Service composition can be efficiently realised, we are very confident that it matches up to the needs of more complex services and processes too. Experience with state-of-the-art semantic repositories has shown that they are performing enough to realise semantic spaces for compositions that rely on several millions of triples. Future work includes the realization of more complex evaluation scenarios, and the inauguration and evaluation of the approach in the context of knowledge-intensive processes on top of large knowledge bases of private linked data and (public) Linked Open Data.

Acknowledgement. The work is supported by the EU FP7 projects (IP) SOA4All and (e-Infrastructures) SEALS. We thank our colleagues from these projects for their valuable discussions and related work, in particular Jacek Kopecky and Carlos Pedrinaci.

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Energy Consumption Information Services for Smart Home Inhabitants

Michael Schwanzer[1] and Anna Fensel[2,3]

[1] University of Applied Sciences Technikum Wien, Austria
michael.schwanzer@gmail.com
[2] FTW Telecommunications Research Center Vienna, Austria
[3] Semantic Technology Institute (STI) Innsbruck, Austria
fensel@ftw.at

Abstract. We investigate services giving users an adequate insight on his or her energy consumption habits in order to optimize it in the long run. The explored energy awareness services are addressed to inhabitants of smart homes, equipped with smart meters, advanced communication facilities, sensors and actuators. To analyze the potential of such services, a game at a social network Facebook has been designed and implemented, and the information about players' responses and interactions within the game environment has been collected and analyzed. The players have had their virtual home energy usage visualized in different ways, and had to optimize the energy consumption basing on their own perceptions of the consumption information. Evaluations reveal, in particular, that users are specifically responsive to information shown as a real-time graph and as costs in Euro, and are able to produce and share with each other policies for managing their smart home environments.

Keywords: Energy Awareness, Smart Home, Social Networks, Energy Efficiency, Visualization, User Interfaces.

1 Introduction

With the growing concerns of climate change, society, in particular, the European Union, shows increasing interest in energy efficient living that would lower carbon dioxide emission [10]. Many energy-related projects have been dealing with lower layer infrastructures, however very few addressed creation of information services delivering valuable energy consumption data to the service providers and end-consumers.

Now there is a large number of Internet of Things enablers for smart metering, home automation (sensors and actuators installed in buildings) and communication opportunities allowing creation and affordable deployment of energy efficiency services addressing smart home inhabitants. Automation control systems and energy optimization services enable the users to co-control their environment (switching on/off of devices, configuring actuators, etc.) according to their policies and together with the energy supplier, for achievement of a higher level

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 78–87, 2010.
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of energy optimization. *Smart Meters* are devices to read the energy consumption data more frequently (typically, every 15 minutes) and automatically, being internationally on the rise, and the expectation is that they will be installed ubiquitously in the EU within a number of years [3,12,13]. Approaches on how this information can be used to help the consumers actually save energy, have been investigated [1,4,2,6] and show that giving the users a feedback about energy consumption definitely has a positive impact. Also until Smart Meters are widely installed, various end-user devices are already on the market to give the consumer insight on consumption patterns [5,8,7]. Due to the novelty of such technologies, there is still insufficient knowledge about how the consumption data needs to be visualized to motivate the user to be most energy efficient.

Furthermore, the new generation of the energy efficiency user interfaces can benefit from modern content-rich communication technologies, such as social networking platforms where the consumers can interact with each other while using the interfaces. While people never really had a chance to interact with each other about their energy consumption, now Social Networks and the Web give this opportunity, and their potential benefits need to be investigated.

The paper is structured as follows. Section 2 defines the objective of the work and the solution approach. Section 3 provides the technical implementation details. The user study and experimental settings are discussed in Section 4. The results are presented in Section 5. Section 6 summarizes and concludes the paper.

2 Objective and Approach

The overall objective of this research is to facilitate home owners and building managers in saving energy within their environments and in optimizing their energy costs, while actively controlling and maintaining their preferred quality of living.

Therefore, to approach the energy efficiency goal, we leverage on a technical solution that integrates smart metering and building automation in order to offer energy-optimization capability for both the energy consumer and provider. This solution enables both parties to profit from the deregulated energy market, by leveraging information about the energy usage, about the user needs, and about the potentials for optimization through a smart environment control. For the energy end-consumers we assume a knowledge acquisition solution that supports creation and maintenance of policies describing preferences in energy use (e.g., green energy), as well as the rules for controlling the devices on the basis of the real-life sensor data from actuation of appliances.

Energy information services automatically channel relevant energy information (e.g., pricing, real time consumption) over the Internet into the user policy framework for employment in the reasoning mechanisms. Our goal in this research is to understand and analyze the effect of different energy consumption and home automation visualizations and rich communications on the energy-effectiveness of smart home inhabitants.

To achieve the realization of the optimal energy information services, a prototype of an energy awareness service was set up within the Social Network

Table 1. Selection of possibilities to visualize consumption data

Data	Entity	Visual	Explanation
Trees	Number	Number	Number of trees needed to absorb carbon dioxide (CO2) emitted by producing consumed energy
Energy	Watts	Graph	Realtime energy consumption graph in Watts
CO2	Number	Number	Amount of carbon dioxide (CO2) emitted by producing consumed energy
Energy	Watts	Graph	Calculated Watt usage per hour as a bar graph
Cost	Euro	Number	Calculated cost of energy used in Euro
Energy	Watts	Graph	Consumption graph in comparison with other users
Energy	Watts	Number	Watt usage as a number

Facebook[1] to visualize consumption data for participants while their interactions with the service where monitored by *Google Analytics*[2].

To promote the prototype to the users of Facebook the name *GreenHome-Challenge*[3] was invented. Players have had their own virtual flat that they are in charge of, and the goal of the game has been to keep the energy consumption of this virtual place as low as possible. Therefore players can either apply *SmartRules* [9] (Energy Saving Approaches) or turn off appliances that other players turned on, as soon as noticed. A visualization on the start-page of the game shows the user the current energy consumption in his or hers own virtual flat. Within the prototype, every user got randomly assigned one of the shown visualization approaches (Table 1) while their interaction with the interface was monitored within the span of one week. Furthermore, a follow-up survey with 115 participants was conducted.

3 Implementation

The experiment was implemented as a public Facebook application GreenHome-Challenge where the user had to add the application to his or her profile before starting the game. By adding the application a record for the user was generated automatically, including a random number between 1 and 7 to decide which visualization will be presented to the user within the game. Furthermore, this number decided which SmartRules the user found pre-defined within the virtual flat. The energy consumption graph from the game is shown in Figure 1, and a list with the most current activities happening in the smart home has been presented to the user: in this example, someone is charging a laptop.

All predefined SmartRules were based on a publication by Greenpeace[4]. All participants had the opportunity to create their own SmartRules by posting them as a natural language comment within the application. The direction 'to

[1] http://www.facebook.com - Facebook is a registered trademark of Facebook Inc.
[2] http://www.google.com/analytics - Google Analytics is a registered trademark of Google Inc.
[3] http://apps.facebook.com/greenhomechallenge/
[4] http://marktcheck.greenpeace.at/1537.html

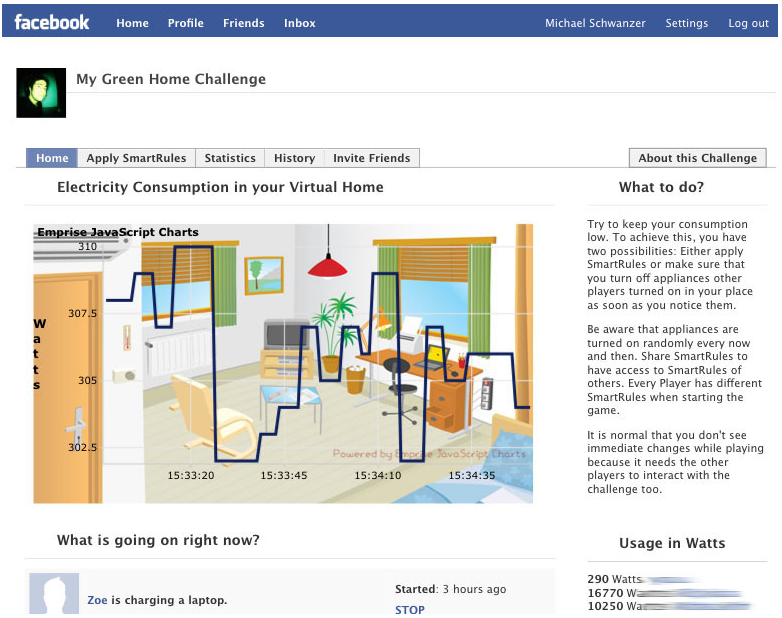

Fig. 1. Prototype of an energy awareness information service within *Facebook*, showing energy consumption visualized as a realtime graph

turn off the oven automatically if it is working for more than 2 hours without a human intervention' or 'turn off all the lights in the house between midnight and 7am' are examples of such user-created rules. Within 24 hours those rules got manually translated to a machine-readable format for the virtual flats, so the participants could add the existing SmartRules to their profiles and see the outcome. Every sub-page of the application included a Google Analytics Snippet to monitor usage of the GreenHomeChallenge.

To keep participants active within their virtual flats, a script started by a cronjob (scheduled start time on the server) turned on appliances in the name of other players (i.e. friends visiting your virtual home). This was necessary to keep the game steadily updating itself and more interesting to the users.

To harvest the real-time consumption data to be shown within the mentioned virtual flat, a market-ready device giving a direct feedback on a display (CurrentCost[5]) was set up and connected to computer. Through an inbuilt interface, data collected by the device could be preprocessed by the computer and uploaded to an online database. High density of data-reading (every 3 seconds) was necessary to provide flexibility for the visualization approaches tested within the virtual environment. These data were used as a basis for the consumption rates in all virtual flats but were preprocessed with appliances that were turned on or off by the users in their virtual home before it got visualized.

[5] http://www.currentcost.com

4 Evaluation Set Up

After the GreenHomeChallenge game was added to the Facebook Application Directory, it could be potentially reached by any of its 350 million users. It was promoted on the SESAME project weblog[6] as well as on Facebook itself (mainly as postings to friends of friends).

Furthermore, interactions with the prototype where monitored automatically with Google Analytics. As the call to take part was public on the Internet, there was no access control for any particular demographic group and furthermore, this information was not recorded on purpose to keep the participants' privacy intact.

In order to evaluate the approach, to complement the Facebook game as an experiment, a 12 questions web-survey user study was conducted comparing the effectiveness of different consumption visualization approaches and additional aspects of energy efficiency.

The survey questionnaire was sent to all participants of the experiment as well as an open call via email to random other people known by the authors. Beside that, the survey was promoted on Facebook by sharing the link to the application and furthermore the call to take part was replicated on the fast growing social network Twitter (which is a free social messaging tool that lets people stay connected through brief text message updates up to 140 characters in length).

5 Results

Here we present the finding received from observing the Facebook game interactions and the user survey.

5.1 Facebook Game Interaction

Overall, 41 participants added the GreenHomeChallenge to their Facebook profile. The participants interaction within the virtual flat, trying to keep energy consumption low, were monitored and analyzed depending on the visualization method the user was presented to. Altogether 41 users played the GreenHomeChallenge on Facebook and caused more than 900 unique visits monitored by Google Analytics.

The interaction within the virtual flats (applying SmartRules, turning off appliances) to keep energy consumption low where monitored and analyzed depending on the visualization method the user was assigned to. Figure 2 thus shows that the approach to show consumption cost in Euro caused the highest rate of interaction (twice as much as any other approach) within the prototype, followed by the realtime graph showing immediate changes (20% more than others that caused almost the same interactions).

[6] http://sesame.ftw.at/2009/12/30/sesames-greenhomechallenge-play-it-now-on-facebook/

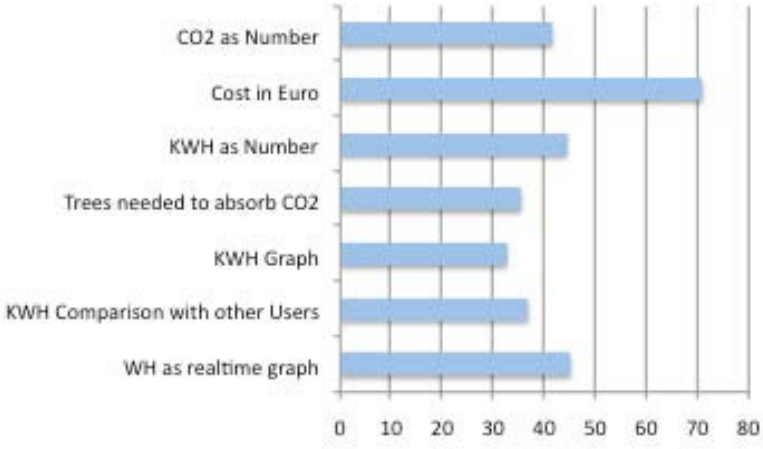

Fig. 2. Average user interactions depending visualization

5.2 Survey Results

The survey results have produced additional data, complementing and at times confirming the data obtained via the Facebook game. The survey itself was completed by 105 participants fully and 15 participants partially, including 24 people that identified themselves as participants of the GreenHomeChallenge experiment as well.

Regarding the user awareness about energy efficiency, only 39 of the 115 participants know what a Smart Meter is, but 97 are ready to save energy by changing habits. Results also show that only 19% are informed online by their

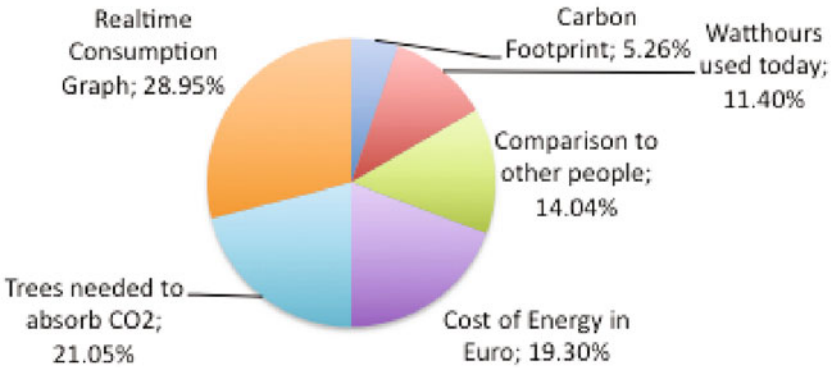

Fig. 3. Diagram showing which visualization approaches users think convey them the most.

energy provider about their consumption, while 33% inform themselves more often than once a year about their energy consumption.

The survey with 115 participants gave insights about the effectiveness of the alternative energy consumtion visualisations: 28.95% of respondents see their consumption the clearest and most meaningfully shown as a realtime graph, followed by 21.05% seeing their motivation most conveyed by a visualization of trees needed to absorb CO2 produced by the consumed energy. Further results are shown in Figure 3.

Asking participants for their motivation for saving energy, Figure 4 shows clearly that cost savings are the most important criterion while just a few are trying to save energy because of reputation, for example. For different visualization methods, 28.95% of the users see their consumption the clearest and most meaningfully shown as a realtime graph, followed by 21.05% seeing their motivation most conveyed by a visualization of trees needed to absorb CO2 produced by the consumed energy. Only 5.26% answered that carbon footprint would motivate them.

Giving the same answer options (regarding visualization methods) as when asking which methods express the participants' consumption most, they answered very different to the question which method shows the biggest energy waste. Here the visualization of carbon footprint shows for 23.76% the biggest waste of energy, while only 14.85% think that the cost of energy shows the biggest waste.

Being explained what Smart Meters are and how they can be combined with home automation systems, between 105 and 107 participants answered the questions about SmartRules and gave very positive answers about using them in the future if their home automation system would support them (see Figure 5). Out of the 24 participants that identified themselves as players of GreenHome-Challenge, 52.38% think they will become more energy aware because of smart visualization and information systems like GreenHomeChallenge showed them. 42.86% answered neutral and only 4.76% disagreed.

ITEM	1	2	3	4	5	Total Stars	Average Rating
Cost Savings	1.8% 2	7.0% 8	7.9% 9	21.9% 25	61.4% 70	114	4.3
Reduction of CO2 emission	9.7% 11	8.0% 9	19.5% 22	25.7% 29	37.2% 42	113	3.7
Social Responsibility	13.8% 16	14.7% 17	26.7% 31	19.0% 22	25.9% 30	116	3.3
Reputation	44.0% 48	18.3% 20	19.3% 21	10.1% 11	8.3% 9	109	2.2
	77	**54**	**83**	**87**	**151**	**452**	**3.4**

Fig. 4. Users motivation

Item	Yes sure	Maybe	No	Total
Use pre-installed Smart Rules to save energy?	67.3% 72	30.8% 33	1.9% 2	107
Download new Smart Rules (free)?	73.6% 78	22.6% 24	3.8% 4	106
Buy Smart Rules that would save you money?	16.8% 18	53.3% 57	29.9% 32	107
Share Smart Rules with Friends?	62.6% 67	34.6% 37	2.8% 3	107
Share Smart Rules with the Community?	52.3% 56	41.1% 44	6.5% 7	107
Adjust existing Smart Rules?	50.5% 54	46.7% 50	2.8% 3	107
Develop your own Smart Rules?	37.1% 39	48.6% 51	14.3% 15	105
Average %	51.5%	39.7%	8.8%	746.0

Fig. 5. SmartRules Usage

According to the participants answers, more than 70% want realtime or at least timely consumption data as necessary for the perfect energy awareness information system. With over 60%, extrapolation of possible cost savings is a mandatory feature for the participants. More than 50% see the perfect energy awareness system online through a website while only 15% want that information offline through a monthly bill.

6 Conclusions and Future Work

To investigate how different visualization approaches impact the end-user, an energy awareness information service was designed and a prototype implemented within the social network Facebook. Real-time consumption data have been collected through a market ready device set up in a flat and streamed to the Internet. Users of the prototype were shown randomly assigned visualizations and had the opportunity to turn on and off virtual appliances in a virtual place and define policies to influence the consumption visualization. Interactions were monitored and showed that the approach to show consumption cost in Euro caused the highest rate of interaction (twice as much as any other approach) within the prototype, followed by the realtime graph showing immediate changes (20% more than others, that caused almost the same interactions).

The contributions of this work comprise advancement in understanding the users' motivations for being more energy-efficient. A prototype for the users of

Facebook, named GreenHomeChallenge, has been developed and employed for evaluation, where players have their own virtual flats that they are in charge of, with the goal of keeping the energy consumption of this virtual place as efficient as possible. A visualization on the start-page of the game shows the user the current consumption in his or her own virtual flat. The participants interaction within the virtual flat, trying to keep energy consumption low, were monitored and analyzed depending on the visualization method the user was assigned to. Also the players showed themselves capable to define and apply SmartRules (Energy Saving Approaches) as well as managing the smart home by turning off appliances that other players turned on, as soon as noticed. The results from the conducted survey complement and support the findings derived from monitoring of the game, and confirm the consumers' interest in the topic and the approach.

In real life, the described services would work on an ontology-based models describing the environments and situations that can be detected [11]. These models are being complemented with the service access points that encapsulate different interactions: data harvesting or control - between the users and the system. The preferences of the users in respect to the system behavior are modeled with a set of rules. The application of information services eventually merging both physical and virtual environments contribute to the pan-European vision of the Internet of Things.

For the future work, with the rise of smart meters usage and therefore accessibility of energy consumption data as well, energy awareness information systems have the opportunity to become handy enablers for consumers to save energy by changing their consumption habits. Bringing these data into social networks, people can team up and help each other detect energy waste and develop methods to save energy as a community. It would be necessary to investigate how users will react on social encouragement or 'shame' as soon as their data in the GreenHomeChallenge are real and not processed for a virtual place. Naturally, privacy would then also become an issue to be addressed.

Acknowledgments. This work is partially funded by FFG (Austrian Research Promotion Agency) in the framework of SESAME project (SEmantic SmArt Metering: Enablers for Energy Efficiency, http://sesame.ftw.at). The Telecommunications Research Center Vienna (FTW) is supported by the Austrian government and the City of Vienna within the competence center program COMET.

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Knowledge Management in Sensor Enabled Online Services

Dominick Smyth*, Paolo Cappellari**, and Mark Roantree

Interoperable Systems Group, Dublin City University, Dublin 9, Ireland
{dsmyth,pcappellari,mark}@computing.dcu.ie

Abstract. The Future Internet, has as its vision, the development of improved features and usability for services, applications and content. In many cases, services can be provided automatically through the use of monitors or sensors. This means web generated sensor data becoming available not only to the companies that own the sensors but also to the domain users who generate the data and to information and knowledge workers who harvest the output. The goal is improving the service through better usage of the information provided by the service. Applications and services vary from climate, traffic, health and sports event monitoring. In this paper, we present the WSW system that harvests web sensor data to provide additional and, in some cases, more accurate information using an analysis of both live and warehoused information.

Keywords: Data warehouse, data mining, enrichment, sensor networks.

1 Introduction

The use of sensors to monitor subjects and instruments generates a continuous flow of information which, in turn, provides information services that are current and accurate. Recently the city of Dublin, like many other European cities, deployed a bike sharing scheme for the rental (and return) of bike from stations throughout the city centre. Stations are equipped with sensors that monitor bike availability and publish such data to the DublinBikes website (www.dublinbikes.ie). Consumers can connect to the website (either through PC or mobile application) to find station locations, bikes and return space availability and available payment options. This data is also utilised by the provider to identify where bikes can be removed from for maintenance. This, however, is the extent of what can be accomplished using the live data provided by sensors.

Motivation. There are many requirements where it is necessary to access historical data or look for trends and patterns over time. Automating these needs can be made possible by sensor networks but websites providing this sensed data are focused on current information needs only. Currently there is no system in place that utilises or manages historical data. In this paper we present the Web Sensor Warehouse (WSW) system for harvesting sensed data from the web, transforming this data and providing an interface to express data mining queries, as well as continuing to support existing

* Funded by Enterprise Ireland Grant No. IR/2009/0001.
** Funded by Enterprise Ireland Grant No. CFTD/08/231.

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 88–96, 2010.
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"live" queries. This will make customers better informed as they plan their pick up and drop off schedule, while providers can plan for expansion based on usage trends over periods of time.

We consider the following queries, which can not be solved using existing sensed data alone:

1. If there are no bikes at 'Golden Lane', how long until there will be some?
2. Will there be space at 'Mountjoy Square' in 15 minutes?
3. If there are no bikes at 'Smithfield', are there any bikes nearby?
4. At what time of day can the provider take bikes for maintenance?
5. When a station was full, were neighbouring stations also full?
6. Which stations are reporting erroneous information?

Contribution. We provide a framework for harvesting web sensor data to facilitate enhanced decision making. The system is fully automated, from online data harvesting, through transformation and on to presentation for mining operations using standard query languages, XPath and XQuery. Details of the system's performance is provided through a series of experiments. We wil also show that our level of detail and performance is not offered by any other, similar, system that we have encountered.

Paper Structure. The paper is structured as follows: in §2, we provide a discussion of similar research in this area; in §3, a description of the Web Sensor Warehouse architecture is provided; in §4, we discuss our methods for transforming the sensed data so as to prepare it for data mining queries; in §5, we provided details of the levels of information that can now be extracted and how the system performs; and finally in §6, we provide some conclusions.

2 Related Research

Extensive work has been conducted in the field of information extraction from the web. Examples include *Stalker* [9] and *TextRunner* [2]. A comparative survey of a number of methods was conducted by Chang et al. [3]. These works focus on wrapper induction and generation for extracting information from semi-structured data, but they do not address enrichment, persistent storage or data mining queries as we do in this work.

While Wang et al. [17], in their discussion of data transformation in the context of RFID data streams, focus on semantic enrichment of data by exploiting contextual information and complex ECA rules, we are interested in storing historic data while harvesting from web sources. Works such as [11] and [6], in considering the transformation of XML data in the interest of interoperability, propose generic constructs and operations to transform XML data, yet they leave data enrichment and mining capabilities out of the picture.

A previous paper at this symposium proposed a system of *Stream Reasoning* [5] for real time reasoning that combines static knowledge with continuous data streams. We expand on these principals by deducing our knowledge from the corpus of archived data and not simply from observed occurrences.

Balazinska et al. [1], provide extensive information on storing web sensor data, previous trend analysis and emerging trend recognition. Additionally, Etzioni et al. [7]

provide a system for monetary savings based on mining a warehouse of web sensor data. This method combines time-series analysis [4] and optimal-stopping [13] with rule learning techniques to generate its results.

A substantial amount of research has also been conducted into mining for *association rules* in XML documents ([10], [16] and [15]), while Nayak et al. [14] surveyed general data mining techniques that can be applied to the structure or content of XML documents. All these approaches focus on discovering relations between data in XML format but they do not deal with the task of collecting, enriching and structuring the data as we do in this proposal.

3 Web Sensor Warehouse System and Components

In this section, we describe our Web Sensor Warehouse (WSW) architecture that incorporates the physical world of bicycle hardware and station sensors; the harvesting, enrichment and transformation processors, and the query and mining interface. This architecture is illustrated in Fig. 1.

Dublin Bikes, a self-service bike rental system, was introduced to the city centre of Dublin, Ireland, in September 2009. The service comprises 450 custom bikes and 40 stations, positioned in close proximity around the city.

Each station consists of a number of sensor equipped parking stands (that report empty or occupied) to store bikes, and the bicycles themselves. For each station, the sensor information is made available through dublinbikes.ie, which specifically provides: station ID, number of bikes available, number of free bike stands, total number of bike stands, number of tickets (this data is stored but not used in our work as its purpose is not yet known). Since no timestamp information is available we provide our own when harvesting the above data.

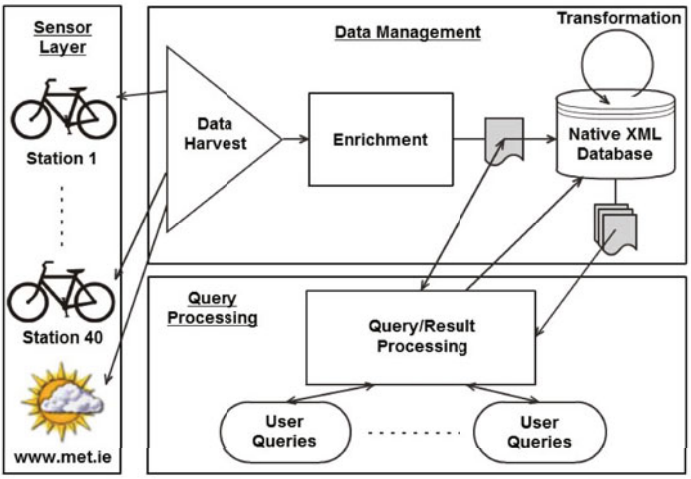

Fig. 1. Web Sensor Warehouse Architecture

Data Management Component. The Data Management component collects and organises data. It is composed of three processors: Data Harvest, Enrichment and Transformation.

The Harvest processor, a website wrapper, retrieves data from the DublinBikes website every 60 seconds. With input of a list of station URLs, output is provided as a single XML sensor document that combines the results from each station. The full harvest process generally takes less than one minute with failed retrievals recorded as an error. Since bikes can not be hired between 12:30am and 5am, this process is run from 5am until 1am the following day. Harvests outside this time would be redundant.

The Enrichment processor provides context for current station sensor data. Time of retrieval, based on GMT synchronised time in processor, along with the time taken to retrieve data from each station is added to the harvested data. In addition, the most recent weather conditions, as reported by met.ie, are integrated into the sensor output file. This enrichment process runs in series with the harvesting phase, hence at same rate and intervals throughout the day.

The schema defined with our database groups files into collections based on changes to the date, ie. "Dublin" → Year → Month → Day. Within each 'Day' collection we store our enriched documents. The XML schema at this point can be seen in Fig.2 - Original Schema.

The Transformation processor is composed of two parts. The first transformation reorganises data from the minute-by-minute view output of the collection to a station-by-station view; the second transformation provides a summation of all station data on a minute-by-minute basis. These transformations facilitate the query and mining processes as will be detailed in §4. The transformation process take place once a day after the last data collection (1:00am) in order to eliminate latency within the system.

Query Processing Component. The Query Processor, thanks to the enrichment and transformation processors, is able to express even the most complex of our queries through XQuery expressions alone. This exposure allows queries to be modified and customised without any alteration of the system implementation. Queries are issued on the most recent sensed data and on the transformed documents within the warehouse. Warehoused information allows us to identify previous occurrences and trends that match the status of the recent sensed data. Through this combination, we provide results to the user and provider, both on what is currently being experienced and what we project will be experienced.

4 Transforming the Sensor Output

Every 24 hours, the WSW system harvests 1,200 sensor output documents, each representing the enriched data from all stations at the given time stamp. For live queries, this information is sufficient to determine bike or free slot availability. Our data mining queries, however, can not readily be executed using standard XML query languages on the data in this format. To prevent the addition of a computationally intense programming layer, we used a number of algorithms to preprocess the data prior to executing our queries.

Transform 1 - Station Information. Considering each station in order, and each harvest chronologically, the XQuery code listed in Algorithm 1 produced 40 *station information* documents. Each document contains a minute-by-minute 'status' of one station for an entire day. This status is limited to one of 5 options:

- Full - All bike stands occupied by a bike.
- Empty - All bike stands empty.
- Error - An error occurred during data collection from the web site.
- Bad Sum - Sum of free spaces and available bikes did not equal total stands.
- Normal - Data collected and processed as expected (default value).

The average harvest time for the given station over the course of the day was also recorded. The station information documents generated by this transformation are used to answer queries 1, 5 and 6 and we will expand on how their use facilitates our evaluation of events within time windows in §5.

Transform 2 - System Information. Again using all 1,200 sensor output documents for the 24-hour period but operating independently of transform 1, we now generate an output document with the total number of bikes available, total stands available and a count of stations recording an error across the entire system on a minute-by-minute basis. This *system information* document is used to answer query 4 and will also be used in further analysis of overall usage or errors.

Transformation Costs. These transformations require roughly 30 minutes to execute. Since our transformations take place at 1am, after harvesting has terminated, we have not investigated the optimisation techniques [8, 12] that could improve on this time frame. As before, the XQuery statements used to generate the transformed documents can be edited without any changes to the current WSW system.

Algorithm 1. Station Transform

```
for i in 1 to 40
    for $s in /station/station[id=i]
        let $start := concat( '<result><hour>', $s/../time/hour,
            '</hour><minute>', $s/../time/minute, '</minute>' )
        let $result := if ($s[number(./error)=1])
            then '<status>error</status>'
            else if ($s[number(./bikes) + number(./spaces)
                    ne number(./total)])
                then '<status>bad sum</status>'
                else if ($s[number(./spaces) eq 0])
                    then '<status>full</status>'
                    else if ($s[number(./bikes) eq 0])
                        then '<status>empty</status>'
                        else '<status>normal</status>'
    order by $s/time/hour, $s/time/minute
    return concat( $start, $result, '</result>' )
```

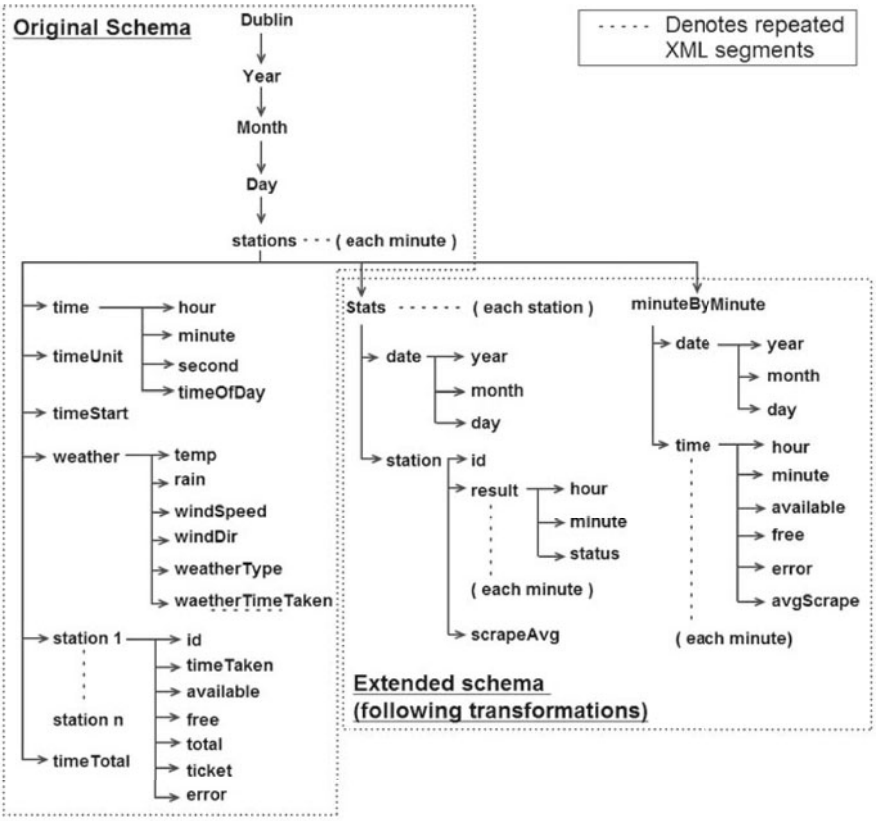

Fig. 2. XML Schema

In terms of storage space, each sensor output document contains 12kB of data which equates to total of 14400kB per day. Each station information document is 144kB, thus a total of 5760kB per day, and the single system information document is 256 kB. Thus, these transformed documents added 42% to the daily memory requirements of the system. The full storage schema following these transformations is shown in Fig. 2.

5 Experiments

Table 1 shows the queries that form our experiemental work, along with their configuration parameters. Queries were expressed in XQuery, used simple statistical methods to calculate results and utilised a combination of stored sensor documents, transformed documents and the most recent data from dublinbikes.ie.

Our experimental machine was a Pentium 4 PC with a 3.20 GHz processor and 1 GB of memory. It ran Ubuntu with a Linux kernel of 2.6.31-19-generic. For storage purposes, we used eXist version 1.4.0.

Table 1. Query parameters and execution times

Query	Parameters	Execution Time (ms)
1 There are no bikes at 'Golden Lane', when will there be some?	Time frame = 14 days	12263
2 Will there be space at 'Mountjoy Square' in 15 minutes?	Time frame = 14 days	1021
3 There are no bikes at 'Smithfield', are there any bikes nearby?		323
4 At what time of day can we take bikes for maintenance?	Time frame = 14 days Time window = 1 hour	8284
5 When a station was full, were neighbouring stations also full?	Time frame = 14 days	48332
6 Which stations are generating erroneous data, either repeatedly or continuously?	Time frame = 14 days Percentage required for 'Repeated Error' = 10 Continuous $\geq$ 15 mins	Repeated = 51619 Continuous = 520619

Query Processing Methodology. Apart from target user, queries may also be categorised by complexity. Queries 2, 3 and 4 are simplistic requiring only basic calculations for results. Queries 1, 5 and 6, however, are more complex requiring a number of stages of calculation.

In these queries, a time window forms an integral part of our evaluation.

- Query 1 - Window of how long the station was empty on previous days.
- Query 5 - Windows when any given station was full.
- Query 6 - Windows when a station continuously reported erroneous data.

Using the original sensor files to calculate these time windows would have required a computationally intense, iterative process. In the case of queries 5 and 6, this iteration would require all 1,200 sensor outputs for each day within our time frame. While this is technically possible using XQuery, it was felt the complexity of the algorithm required was beyond the scope of our research.

Accesing one transformed file, we were able to identify the window of interest using the preceding-sibling and following-sibling axes alone (see Algorithm 2). By only considering sustained windows (in excess of 15 minutes in the case of queries 5 and 6), we removed noisy status changes from our calculations.

When evaluating query 5, we use the start time and length of the window during which a station was full as the input when calculating the number of spaces available

Algorithm 2. Determining Window

```
for $r in $s/station/result[status="bad sum" or status="error"]
    for $p in $r/preceding-sibling::result[1][status!="bad sum"
            and status!="error"]
        let $end := $p/following-sibling::result[status!="bad
                sum" and status!="error"][1]
```

Algorithm 3. Availability at neighbouring station

```
declare function local:timeRange( $n as xs:integer,
            $now as xs:dateTime,
            $numMinutes as xs:integer,
            $scrape as xs:string ) as xs:integer {
    for $i in 0 to $numMinutes
        let $lTime := $now + xs:dayTimeDuration( concat( "PT",
                $i, "M" ) )
        let $lHour := hours-from-dateTime( $lTime )
        let $lMinute := minutes-from-dateTime( $lTime )

        let $s := collection( $scrape )/stations[time/hour=$lHour
                and time/minute=$lMinute]/station[id=$n]
        let $result := if( $s/error = 0 )
            then $s/spaces
            else ()
        return $result
};
```

at a neighbouring station. The average spaces over all such time windows provides a measure of this neighbouring stations usage.

Some correlation between stations can be seen in the sample output of Table 2 eg. George's Quay, Townsend Street and Custom House. A status of `full` at one station however does not necessarily have any effect on a neighbouring station eg. Merrion Square East and West. This information is of use to users as they return bikes and providers as they plan expansions and improvements.

Table 2. Associated Usage Listing (sample)

Name of Station	Neigbouring station and number of free spaces
George's Quay	Custom House = 0, Townsend Street = 0, Custom House Quay = 5
Grantham Street	Stephen's Green South = 17
Townsend Street	Pearse Street = 10, George's Quay = 6, Custom House = 1
Merrion Square West	Merrion Square East = 20, South Leinster Street = 0

6 Conclusions

In this paper, we presented *Web Sensor Warehouse*, a system for automatic harvesting, processing, enriching and transforming of data. We also provided more accurate information regarding the data generated by sensors outputing to the Web. To do this, we used a popular web service, dublinbikes.ie, one of many bicycle rental and drop off services across Europe. The target user base for the WSW system is both users, requiring accuracy of information, and providers, making strategic decisions impacting the future of the services. Our prototype and experiments demonstrate the usability of the system, with all of our queries expressed in XQuery, a standard XML language.

Our current focus is twofold. Firstly, building an ongoing mining utility to examine the accuracy of data provided by checking how quickly it will be out of date, based on prior patterns and temporal data. Secondly, turning our harvesting component to similar systems across Europe to search for patterns between different locations and to check scalability for our warehouse.

Acknowledgements. The authors would like to thank interns Graham Rhodes and Cian Scolard for their support in helping to develop the data harvesting component.

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Managing On-Demand Business Applications with Hierarchical Service Level Agreements*

Wolfgang Theilmann[1], Ulrich Winkler[2],
Jens Happe[1], and Ildefons Magrans de Abril[2]

SAP Research, SAP AG,
[1] Vincent-Priessnitz-Str. 1, 76131 Karlsruhe, Germany
[2] The Concourse, Queen's Road, Belfast BT3 9DT, United Kingdom
{wolfgang.theilmann,ulrich.winkler,jens.happe}@sap.com
http://www.sap.com/research

Abstract. Business applications are facing an increasing demand for being delivered as on-demand services. Service Level Agreements (SLAs) are a common way for specifying the exact conditions how services are delivered, both for business- and IT-based services. At the same time, the paradigm of service-orientation is expanded to different domains such as business and IT-based services. Services are composed out of other services across domains. Consequently, SLAs have to managed across such service hierarchies.

In this paper, we present the lessons learned from applying a generic, multi-layer SLA management framework to the context of on-demand business applications. We explain the implemented architecture and show how a hierarchy of services can be managed via SLAs. We explain in detail how SLAs can be established for different layers and how SLA terms are translated across the service hierarchy.

Keywords: Service Level Agreement, SaaS, IaaS, SLA hierarchy, SLA translation.

1 Introduction

Traditional business applications have been largely delivered as on-premise deployed solutions. However, the need for increased business agility and decreased cost of ownership results in an increasing demand for business applications being delivered as on-demand services (following paradigms of cloud computing[2] and software as a service). Service customers expect increased agility and reduced effort as they can simply use business aplications without taking care of their management. Service providers consider this model for exploiting economies of scale but also face increased challenges in terms of expected quality and price sensitivity.

* The research leading to these results is partially supported by the European Communitys Seventh Framework Programme (FP7/2001-2013) under grant agreement no.216556.

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 97–106, 2010.
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Service Level Agreements (SLAs) are a common way to formally specify the exact conditions (both functional and nonfunctional) under which services are or are to be delivered. Typical SLAs are specified at the top-level interface between a service provider and a service customer. They can be used to monitor whether the actual service delivery complies with the agreed SLA terms. In case of SLA violations, penalties or compensations can be directly derived.

Recently, the paradigm of services is not just applied at a single layer but is used across the complete business and IT stack including for example business services, application services, middleware services, and infrastructure services[1]. Services can be composed and hence depend on other services from either within the same service provider domain or from external parties. Consequently, the service levels of a composed service form a SLA hierarchy. SLAs in a SLA hierarchy cannot be determined and managed in isolation but only in conjunction with the service levels of depending sub-services.

The research project SLA@SOI [3] develops a generic framework for managing SLAs across complete service stacks including business and IT perspectives. The framework provides generic support for specifying, negotiating and monitoring SLAs as well as a plugin-in architecture with hooks for domain specific extensions. In order to apply the framework in a specific use case, the actual SLA hierarchy needs to be established and the related SLA planning, translation, provisioning and adjustment functionality must be realized.

In this paper, we present the lessons learned from applying the generic SLA management framework [4] to the context of on-demand business applications such as enterprise resource planning applications. We explain the implemented architecture for SLA management and show how SLAs can be specified at different layers of the service hierarchy. We explain in detail how SLA terms can be translated along the service hierarchy.

The remainder of this paper is organized as follows. After setting the stage with foundations and related work in Section 2 we explain the framework's architecture and our use-case specific extensions to that framework in Section 3. Section 4 explains the actual SLA hierarchies and SLA translations for our use case and Section 5 provides a summary and outlook.

2 Foundations

2.1 Service Hierarchies for Business Applications

As founding concept for the remainder of this paper, we introduce a basic reference structure for business applications, which sets the basis for structuring business systems in a service-oriented way.

We also provide a brief sketch of respective aspects relevant for SLA management. Actual applications may of course differ from this structure. Our setting comes with four levels: business solution, application, middleware, and infrastructure.

Business Solution. At the top-most level, customers do not simply expect running software applications but a complete solution to their business needs.

These business solutions go beyond the pure provisioning of software and may include aspects such as human support (for setup and maintenance), compliance (security and archiving) or legal constraints. Any SLA at this level has to be expressed in business terms, i.e. must be understandable for business people in the customer organizations. Lower-level technical details such as CPU specifications etc. should be omitted.

Application. At the second level comes the actual business application, i.e. the software solution that is offered to the customer. SLAs about business applications must be expressed in the language of the business analyst, i.e. the person that eventually uses the software. They cover aspects such as number of orders that can be processed per minute.

Middleware. Business applications rely on common middleware components, such as as application server, messaging system, and database. Middleware componets may be even shared across different applications and customers, such Middleware platforms can be also considered as service offering, where the middleware eventually hosts or is used by different client aaplications. Typical examples for this approach are Google App Engine [5] or Amazon storage services [6]. SLAs about middleware components must be expressed in the language of application (platform) developers and are independant from the actual application. They cover aspects such as application server transaction response time or throughput.

Infrastructure. The lowest layer comprises infrastructure resources (both physical and virtual ones) which are offered as services following the cloud [2] paradigm. SLAs about infrastructure are expressed on basic technical parameters such as the number of CPU cores, memory size, etc.

2.2 Related Work

SLA-Management for on-demand business applications has been partially addressed from different angles. Closest known work can be considered in the areas of IT management and SLA translation. SLAs have been added as a concept to multiple IT management tools and frameworks (Governance Interoperability Framework [7] and CentraSite [8]). However, all of these tools support SLA management basically at a single 'service' layer and do not address multiple layers and their interrelations. SLA translation has been heaviliy researched under different contexts in multiple areas. A thourough overview on SLA translation techniques can be found in [10]. Most prominent in here are so-called Layered Queuing Network (LQN) Models, which support the explicit modelling of a system of layers and subsequent performance analysis. Another prominent approach from [9] uses coloured Petri nets in order to capture dependencies within a chain of services. Our work differs from these appraches as it (1) manages SLAs across multiple layers, (2) integrates the aspects of planning/prediction with actual IT management, (3) considers a larger set of SLA parameters, and (4) is applied to an existing commercial business application.

3 SLA Management Architecture

In the following, we describe an SLA management architecture that can be used for business applications. First, we show the building blocks of the basic framework that we apply. Then, we present the actual architecture.

3.1 Generic SLA Management Framework

The generic SLA Management Framework developed in [3] provides the following core concepts:

A baseline *SLA Model* allows for specifying arbitrary SLAs including a specification of the service and the guarantee terms around states (which must be always assured) and actions (which have to be executed if a specified precondition applies). The SLA model is completely domain independent and comes with a set of predefined metrics and functional expressions but also allows for domain-specific extensions.

The *SLA Manager* component can manage a set of SLAs, SLA templates and instances via a registry. Furthermore, it provides functionality for SLA negotiation and monitoring. It contains abstract planning and provisioning modules which have to be refined for specific domains. SLA Managers can be instantiated for arbitrary domains. Default implementations for business, software, and infrastructure domains are provided by the framework.

The *Service Manager* component is in charge of managing the elements necessary to instantiate a service. Again, it can be instantiated for arbitrary domains. Default implementations are provided for software and infrastructure services.

The *Service Evaluation* component allows for predicting the non-functional behaviour of a fully specified composed service. A composed service is called fully specified when all dependencies to lower-level services have been resolved and their SLAs are known. A default implementation for preformance/reliability prediction of software services is provided by the framework.

Last, the framework comes with a *Business Manager* component that is able to maintain customer and provider relationships and uses business rules to steer the actual SLA management activities like negotiation, planning and adjustment according to business priorities.

Overall, the farmework is very flexible in terms of how many of these components are instatiated in a concrete setup and how they are connected together. The detailed specification can be found in [4].

3.2 SLA Management for Business Applications

Following the reference structure sketched in Section 2 our SLA management solution for business applications needs to support SLAs at 4 layers: business, application, middleware and infrastructure.

The SLA management framework offers a range of possibilities to manage the service stack of our on-demand business applications. At one extreme, we might have a separated SLA manager and a separate service manager for each of these

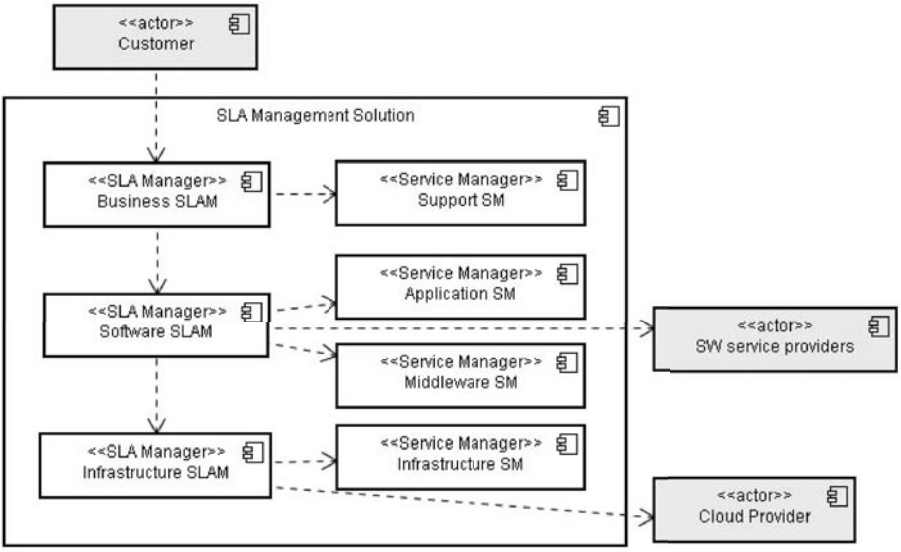

Fig. 1. SLA managament architecture for business applications

layers. At the other extreme, all knowledge could be also brought into one single SLA manager, and one single service manager respectively.

The solution we adopted is shown in Figure 1 and comprises 3 SLA managers (SLAMs) and 4 service managers (SMs). Arrows show general dependency relationships. With this design, we reflect an organization that is divided into 3 main units, the business/sales unit, the software unit and the infrastructure unit. Each unit is enabled to oversee all the SLAs it is currently operating. Furthermore, we have 4 different types of artefacts that contribute to the realized services. Each domain requires different administration knowledge and, therefore, we have encapsulated these types into 4 separate service managers: for support services (including human-resources that need to be planned and managed), for application services, for middleware services, and for infrastructure services. Last, the design also includes the consideration of 2 types of 3rd party service providers: cloud providers and software providers. Cloud providers offer infrastructure services, which can be considered at any time, e.g. in order to handle burst load scenarios. Software service providers can be considered, if the actual business application should be complemented with some additional functionality.

4 SLA Hierarchies

This section provides profound insights of SLAs needed for dealing with the aforementioned four layers of services. In particular, we highlight the translation of SLA terms across these layers.

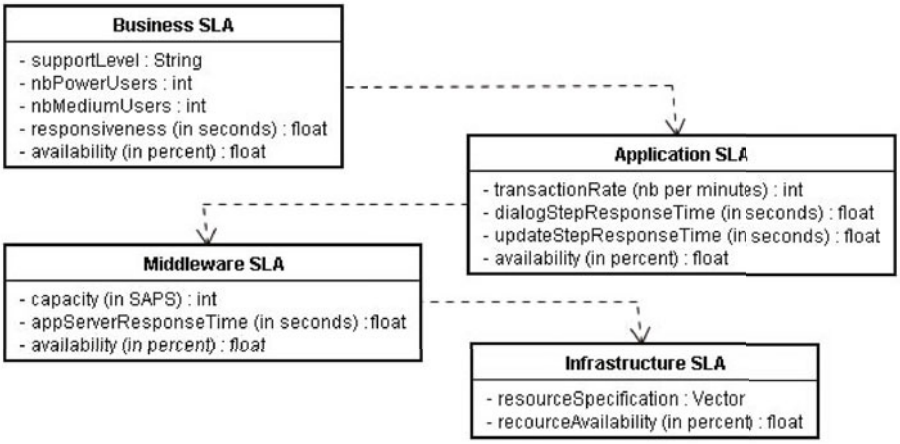

Fig. 2. Simplified sketch of the 4-layer SLA hierarchy

Figure 2 provides an overview of the SLA hierarchy and lists some main SLA terms for each layer. While business and infrastructure SLAs are managed by their repsective SLA manager, application and middleware SLAs are both managed by the Software SLA manager. In the following, we discuss the management/translation of the respective terms.

4.1 SLA Terms and Translation

In [10] we distinguished 4 types of SLA translation between metric and configurables: Metric-to-Metric (e.g. service response time to resource utilization), Metric-to-Configurable (e.g. response time to number of servers), Configurable-to-Metric (server specification to response time) and Configurable-to-Configurable (e.g. service composition to deployment options). All of these are relevant for business applications. We highlight the most important ones in the following.

Solution terms. These are about conditions not directly related to the business application but describing aspects of the overall business solution. An example for such a term is the *supportLevel* which describes the detailed conditions how support around the software artefact is given (e.g. in terms of person days contributed to the setup or in terms of response time obligations for reacting to customer complaints). Solution terms can be managed by a business SLA manager in cooperation with business-level service managers. In our example we have a support service manager, which can decide and manage the capacities of the support organization. Other solution terms may also impact lower level SLA managers (e.g. agreed backup schedules) and require appropriate translation.

Workload. In order to plan the amount of resources needed to run a service it is of great importance to understand and specify the way how customers want to use the service. For example, it makes a huge difference if a customer needs a

service only seldomly and has a single user working with it or if a customer has a complete department relying on the service in their daily work. However, it is typically hard for customers to exactly predict the workload they will put on a service and also rather unlikely that they can express this workload in technical terms. A typical approach is therefore to specify the usage profile in terms of user classes, i.e. specifying the expected number of power users (working more ore less constantly with a service), medium users and low users[11].

This specification for a business SLA now requires translation to the lower-level SLAs:

(1) The first step is to translate the user number and types into expected transaction rates, specified in the application SLA. This can be basically done on the basis how user types are defined for the respective application.

(2) At the next layer, transaction rates must be translated into a load characterization of the middleware. There are many different ways how this could be achieved, such as benchmarking or model-based prediction. For our needs we adopt an application-based benchmark which allows for translating an application transaction mix into an application-independant and infrastructure-independent middleware capacity value, the so-called SAPS measure[11].

(3) Last, middleware workloads could be theoretically also translated into infrastructure specifications (e.g. relying on micro-benchmarks). However, so far there is no generally applicable solution to this available. Instead, common practice is to benchamrk different infrastructure setups against the respective middleware. Again, the same benchmark approach as in step 2 can be used here to specify the capacity of a specific system setup[11]. This allows for direct translation of the specifed workload to the offered system capacity where both are expressed via the same metric (SAPS).

Performance. The performance of a service or system is typically expressed in terms of throughput and response time. In our setup, the workload specification directly specifies also the maximum throughput. If the provider guarantees a certain response time (under the condition that the customer workload stays within the agreed boundaries) he has indirectly also guaranteed the maximum throughput (which is exactly the maximum workload).

The specification of response time now can be done in the following way:

At business-level the specification of responsiveness may be relatively abstract (compared to the more precise and detailed definitions for application and middleware SLAs. It might simply state the general goal that the service shall react within a certain time frame (without specifying the specific application-level operations or interactions).

At application-level this is further refined by stating that a specific classes of interactions (here dialog steps and update steps) shall be executed within a certain time frame. The actual translation towards middleware can be done via model-based or measurement-based approaches or a mixture of these. For example, in [12] we applied a specific queueing model which is calibrated by a set of benchmarking experiments.

At middleware-level the response time relates to generic elements of the middleware elements such as application server response time on http requests or DB response time on SQL operations. Furthermore, response time also relates to middleware configurations such as thread pool or cache size.

At infrastructure-level response time characteristics could be specified for aspects such as cache access time, disk seek operation, or floating point operations. For our case, such characteristics are too fine grained and not practical for actual SLAs. Instead, we translate response time into a configuration including parameters such as CPU speed, memory architecture, and network characteristics.

Availability. The availability of business solutions has to distinguish planned downtimes (e.g. due to maintenance activities) and unplanned downtimes (due to actual failures at either software or hardware level). Both can and should be treated separately. Planned downtimes are typically specified via explicit time intervals within the calendar week (e.g. a maintenance window of 2 hours every Saturday night between 2 and 4 am). Planned downtimes can be systematically translated in a bottom-up fashion as the combination of infrastructure downtimes, middleware downtimes and application downtimes. Unplanned downtimes are specified as a percentage value via a ratio of mean time between failure and mean time to repair. Unplanned downtimes can be systematically translated in a bottom-up fashion via statistical methods[13].

Energy. Energy consumption is playing an increasing role in SLA specifications as customers are keen to report on their environmental compliance. Therefore, energy consumption is not just maintained at infrastructure-level but is also to be aggregated over the different SLA layers which can be done via simple summarizaton. Energy is not in our SLAs as depicted in Fig. 2.

Cost. Costs for services at different layers can be easily aggregated as the sum of costs for lower-level services plus actual costs occuring at the actual level. At business level costs are typically relabelled to a price as here also margins are added to assure the profitability of a service offer. Cost is not in our SLAs as depicted in Fig. 2.

4.2 Integrated Planning

Having understood how the individual SLA terms can be translated, the next challenge is to provide support for their integrated planning. This comes with two main challenges: top-down vs. bottom-up translation and tradeoff decisions between different parameters.

Top-down vs. bottom-up translation. Some parameters can be translated in a top-down manner, i.e. there is a direct way to derive lower-level SLA terms from higher-level ones. For example, a workload definition for a business process can be translated to a workload of it's single steps if the structure of the process is known. Other parameters must be translated in a bottom-up fashion, i.e. a higher-level term can be predicted if the values of lower-level terms are known. For example, response time and availability can be only translated bottom-up as there is a complete design space for them to be explored (knowing response

time of a process does not determine exactly one possible response time of its individual steps).

We implemented an integrated planning algorithm that includes top-down and bottom-up translations. The algorithm works as follows. An incoming SLA request is analysed and the service manager responsible for the given service type is asked for possible service implementations, including their dependencies to lower-level services. Top-down translation is applied for selected SLA terms in order to narrow the search space for SLA templates for lower-level services. Eventually retrieved SLA templates are then included in a bottom-up translation phase in order to predict the exact SLA guarantees that can be achieved. This algorithm is recursively executed for each layer of SLAs, i.e. in our case at four layers.

Trade off decisions. Conceptually, SLA planning is a multi criteria optimization problem where different parameters should be optimized but where optimization of a single parameter might spoil the quality of another parameter. We addressed this issue via an evolutionary multi-objective optimization (MOO) algorithm. This one is based on a cost model (including infrastructure and energy costs) and a performance model (based on queueing networks with finite capacity regions). It computes the Pareto front of optimal tradeoff solutions. Predefined business rules or humans can eventually use this Pareto front in order to select a solution for a specific setup. Figure 3 provides an example for such a Pareto front. It shows the tradeoff between costs and performance (response time) for the TCP benchmark. Further details on this approach can be found in [12].

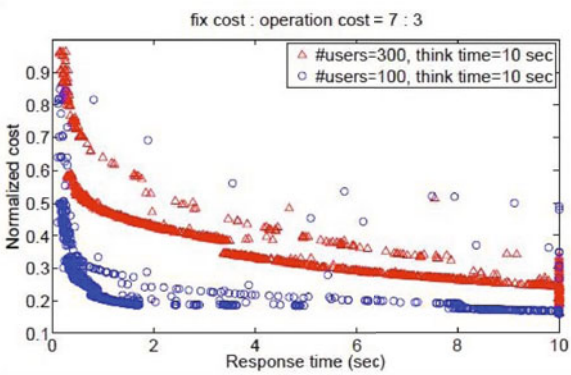

Fig. 3. Pareto front for a cost - performance tradeoff analysis

5 Conclusions

In this paper, we presented the lessons learned from applying a generic, multi-layer SLA management framework to the context of on-demand business applications. We described how a hierarchy of 4 service domains (business, application,

middleware, and infrastructure) can be established. We specified for selected SLA terms their representation at the various layers and their translation across layers. Last, we also sketched a way for integrated planning that combines top-down and bottom-up translation and also allows for determination of tradeoffs.

Future work will focus on adjustment activities that analyse monitored SLA violations and decide on corrective actions to safeguard SLAs. Again, this is a multi-layer challenge as both, SLA violations and adjustment actions can occur at different layers and require proper synchronisation. We plan to explore complex event processing technologies which feed monitored metrics into a reasoning engine which supports temporal logic expressions.

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Controlling Access to RDF Graphs

Giorgos Flouris[1], Irini Fundulaki[1], Maria Michou[1], and Grigoris Antoniou[1,2]

[1] Institute of Computer Science, FORTH, Greece
[2] Computer Science Department, University of Crete, Greece
{fgeo,fundul,michou,antoniou}@ics.forth.gr

Abstract. One of the current barriers towards realizing the huge potential of Future Internet is the protection of sensitive information, i.e., the ability to selectively expose (or hide) information to (from) users depending on their access privileges. Given that RDF has established itself as the de facto standard for data representation over the Web, our work focuses on controlling access to RDF data. We present a *high-level access control specification language* that allows *fine-grained* specification of access control permissions (at triple level) and formally define its semantics. We adopt an *annotation-based* enforcement model, where a user can explicitly associate data items with annotations specifying whether the item is *accessible or not*. In addition, we discuss the implementation of our framework, propose a set of dimensions that should be considered when defining a benchmark to evaluate the different access control enforcement models and present the results of our experiments conducted on different Semantic Web platforms.

1 Introduction

RDF [14] has been established as a data representation standard over the Web and is expected to play a critical role in the realization of the Future Internet. Several commercial and academic efforts, such as the W3C Linking Open Data initiative [22], target the development of RDF repositories, while the number of applications that publish and exchange possibly sensitive RDF data continuously increases in different domains ranging from bioinformatics [21] to e-government applications [5]. Unfortunately, the potential of these efforts and the realization of the Future Internet is undermined by the lack of an effective mechanism for controlling access to such data. In light of the sensitive nature of such information, the issue of *securing* RDF content and *ensuring the selective exposure of information* to different classes of users depending on their access privileges is becoming all the more important. The building blocks of an access control system are the *specification language*, that allows the expression of access control permissions and policies, and the *enforcement mechanism*, responsible for applying the latter to the data, by denying access to non-accessible data.

It is imperative to support access control at a *fine granularity* (i.e., triple level), and not only at a *coarse-grained level* (i.e., repository level), as done by existing RDF repositories such as Sesame [9] and Jena [2]. To enforce access control to

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 107–117, 2010.
Springer-Verlag Berlin Heidelberg 2010

an RDF repository, we advocate a framework which is repository independent, portable across platforms, and in which *fine-grained access control* is enforced by a component built on top of the RDF repository (as in [6]). More specifically, our contributions in this paper are: *(i)* a high level *access control specification language* for RDF graphs focusing on read-only operations; *(ii)* a formal definition of the language's semantics, based on the triple patterns of SPARQL [20]; *(iii)* an *annotation-based enforcement model* where each triple is automatically marked as accessible or inaccessible based on the available annotations and the access control policy; (*iv*) a set of dimensions that should be taken into account when defining a benchmark for the evaluation of enforcement models for access control; and *(v)* an implementation and experimentation of our framework for different platforms.

The rest of the paper is structured as follows: related work is discussed in Section 2; Section 3 provides a brief introduction to RDF and RDFS; Sections 4 and 5 introduce the access control specification language and its formal semantics respectively; we present the enforcement model, implementation and experiments in Section 6, and conclude in Section 7.

2 Related Work

There have been only a few works dealing with the problem of access control for RDF graphs, and most of them lack formal semantics for the proposed framework. Authors in [6] propose a specification language using graph patterns as defined in SPARQL [20]. Unlike our approach, in [6], triples are not annotated with accessibility information, but the enforcement mechanism is implemented through the injection of the permissions in the query, ensuring that only accessible triples will be obtained. Unfortunately, formal semantics are missing, making impossible to check the correctness of the proposed query-enhancement algorithm. In [23], a specification language for defining permissions for update operations on RDF data is presented, but the authors do not discuss the formal semantics of the language and do not provide an enforcement mechanism, a system implementation, or experiments. Approaches that consider RDFS [7] entailment appear in [17], [18]. In [17], the authors discuss how conflicts can be resolved using RDFS subsumption hierarchies, whereas in [18], inferences are computed without revealing information that might have been explicitly unauthorized. In [11], access control requirements for the context of a Semantic Wiki application are presented, but formal semantics are not discussed and no access control enforcement mechanism is provided. Finally, an approach for access control on XML data, which is very similar to ours, appears in [16].

3 Preliminaries

This paper focuses on RDF graphs [14], i.e., data expressed using *RDF triples* of the form (*subject*, *predicate*, *object*). Given two disjoint and infinite sets U, L, denoting the URIs and literals respectively, an *RDF triple* is any element of the set $U \times U \times (U \cup L)$.

subject(s)	*predicate*(p)	*object*(o)
foaf:*Person*	rdfs:subClassOf	*foaf*:*Agent*
foaf:*age*	rdfs:domain	*foaf*:*Agent*
foaf:*age*	rdfs:range	*rdfs*:*Literal*
foaf:*mbox*	rdfs:domain	*foaf*:*Agent*
foaf:*mbox*	rdfs:range	*rdfs*:*Literal*
foaf:*firstName*	rdfs:domain	*foaf*:*Person*
foaf:*firstName*	rdfs:range	*rdfs*:*Literal*
&*a*	rdf:type	*foaf*:*Person*
&*a*	*foaf*:*age*	17
&*a*	*foaf*:*mbox*	<mailto:alice@fun.com>
&*a*	*foaf*:*mbox*	<mailto:alice@work.com>
&*a*	*foaf*:*firstName*	Alice
&*b*	rdf:type	*foaf*:*Person*
&*b*	*foaf*:*mbox*	<mailto:bob@work.com>
&*b*	*foaf*:*firstName*	Bob

Fig. 1. An RDF Graph $\mathcal{G}$

An *RDF graph* $\mathcal{G}$ is a set of RDF triples. RDF triples can express both schema and data information and can be visualized in a graph, whose nodes are the subjects and objects of triples, and arcs the predicates. Note that, for simplicity, we ignore non-universally identified resources (blank nodes). The RDF Schema (RDFS) language [7] provides a built-in vocabulary for asserting user-defined schemas in the RDF data model, as well as semantics for defining various object types (classes, properties, instances) and relationships between such objects.

Example 1. Throughout this paper, we will use, for illustration purposes, an example taken from the FOAF [1] application. More specifically, we will use the RDF graph consisting of the (data and schema) RDF triples shown in Fig. 1.

4 RDF Access Control Framework

At the core of our access control framework lie the notions of *access control permission* and *access control policy*. Intuitively, an access control permission is used to explicitly set certain triple(s) in an RDF graph to be *accessible* (or *inaccessible*). An access control policy includes a set of access control permissions and information that determines whether a triple is accessible when it is not associated with any permission, or when conflicts arise.

In this work, we concentrate on access control permissions for *read-only queries* only; the specification of permissions for update operations (e.g., write, modify) poses extra difficulties which are out of the scope of this paper and will be considered in the future. As already mentioned, we will use fine-grained access control, where the smallest unit of protection is the RDF triple. Thus, given a user, role, or set of users, an access control permission grants or denies to/from the user the ability to read (i.e., access) the protected RDF triple. In practical applications, it often makes sense to impose access control permissions on sets of triples (say, all triples satisfying a certain property), rather than individual ones. For instance, in Example 1, we may want to deny access to the triples referring to the first names of persons who are less than 18 years old. Therefore, we also allow permissions to be provided for sets of triples.

To express such access control permissions we will use the notion of *triple patterns* from the SPARQL Language Specification [20]. We consider $\mathcal{V}$ to be a set of variables (denoted by $?x, ?y, \ldots$), disjoint from U,L. We define a *triple pattern* to be an element of the set $(\mathcal{V} \cup U) \times (\mathcal{V} \cup U) \times (\mathcal{V} \cup U \cup L)$. Triple

patterns are used to denote the triples in an RDF graph that have a specific form, and can be further restricted using constraints to determine the triples that *are in the scope of* a certain access control permission:

Definition 1. *An* access control permission $\mathcal{R}$ *is of the form*

$$\mathcal{R} = \mathsf{include/exclude}\ (x,\ p,\ y)\ \mathsf{where}\ \mathcal{TP}, \mathcal{C}$$

where (i) $(x,\ p,\ y)$ *is a triple pattern* (ii) $\mathcal{TP}$ *is a conjunction of triple patterns and* (iii) $\mathcal{C}$ *is a conjunction of constraints of the form* $u\ op\ c$ *where* $u \in \mathcal{V}$, $op \in \{=, <, >, \neq\}$ *and* $c \in \mathcal{V} \cup U \cup L$.

Intuitively, an access control permission denotes the RDF triple(s) in an RDF graph $\mathcal{G}$ that are *accessible* (*positive* permissions of the form include $(x,\ p,\ y)$ where $\mathcal{TP}, \mathcal{C}$), or *inaccessible* (*negative* permissions of the form exclude $(x,\ p,\ y)$ where $\mathcal{TP}, \mathcal{C}$). Essentially, an access control permission can be thought of as a query whose evaluation over an RDF graph results in a (possibly empty) set of triples (the triples in the scope of the permission); said triples are granted or denied access.

Example 2. Consider the access control permissions shown in Fig. 2. Permission $\mathcal{R}_1$ states that the names (triples of the form $(?x,\ foaf{:}firstName,\ ?y)$) of individuals $?x$ which are under 18 years old (denoted by a triple of the form $(?x,\ foaf{:}age,\ ?z)$ where $?z < 18$) should be inaccessible (exclude). Permission $\mathcal{R}_2$ states that all subsumption relationships should be inaccessible as well. Similarly, permission $\mathcal{R}_3$, states that the names of all persons with an email address should be accessible (include). Fig. 3 shows the accessible and inaccessible triples for the RDF graph shown in Fig. 1, and the related permissions.

Access control permissions explicitly grant or deny access to a certain triple (or set of triples). It would be unrealistic to assume that explicit access rights are set for all triples in an RDF graph, or that permissions are always unambiguous. In our example, the triple $(\&a,\ foaf{:}age,\ \texttt{17})$ does not have any permission set (*missing permissions*), whereas the triple $(\&a,\ foaf{:}firstName,\ \texttt{Alice})$ is marked as both accessible and inaccessible (*ambiguous permissions*). To determine whether such triples should be accessible, we use the notion of *access control policy*:

$\mathcal{R}_1$: exclude $(?x,\ foaf{:}firstName,\ ?y)$ where $(?x,\ foaf{:}age,\ ?z), ?z < 18$
$\mathcal{R}_2$: exclude $(?x,\ \mathsf{rdfs{:}subClassOf},\ ?y)$
$\mathcal{R}_3$: include $(?x,\ foaf{:}firstName,\ ?y)$ where $(?x,\ \mathsf{rdf{:}type},\ foaf{:}Person), (?x,\ foaf{:}mbox,\ ?z)$

Fig. 2. Access Control Permissions for the RDF Graph in Fig. 1

Accessible Triples	$\mathcal{R}$
$(\&a,\ foaf{:}firstName,\ \texttt{Alice})$	$\mathcal{R}_3$
$(\&b,\ foaf{:}firstName,\ \texttt{Bob})$	$\mathcal{R}_3$

Inaccessible Triples	$\mathcal{R}$
$(\&a,\ foaf{:}firstName,\ \texttt{Alice})$	$\mathcal{R}_1$
$(foaf{:}Person,\ \mathsf{rdfs{:}subClassOf},\ foaf{:}Agent)$	$\mathcal{R}_2$

Fig. 3. Accessible and Inaccessible Triples for the RDF Graph in Fig. 1

Definition 2. *An* access control policy *is a tuple of the form* $\mathcal{P} = (\mathbb{P}, \mathbb{N}, ds, cr)$ *where (i)* $\mathbb{P}$ *is a set of positive permissions (ii)* $\mathbb{N}$ *is a set of negative permissions (iii)* $ds \in \{+, -\}$ *is the* default semantics *that indicates whether access to a triple is* granted *(+) or* denied *(−) by default when access control permissions are missing (iv)* $cr \in \{+, -\}$ *is the* conflict resolution *policy and specifies whether access to a triple is* granted *(+) or* denied *(−) when it is in the scope of both positive and negative permissions.*

Example 3. Consider an access control policy $(\mathbb{P}, \mathbb{N}, ds, cr)$ where $\mathbb{P} = \{\mathcal{R}_3\}, \mathbb{N} = \{\mathcal{R}_1, \mathcal{R}_2\}$ (see Fig. 2), with default semantics *deny* ($ds = -$) and conflict resolution policy again *deny* ($cr = -$). In this case, triple (&*a*, *foaf*:*firstName*, `Alice`) that is in the scope of both a positive ($\mathcal{R}_3$) and a negative ($\mathcal{R}_1$) permission (see Fig. 3) is eventually denied access ($cr = -$). Similarly, the triple (&*a*, *foaf*:*age*, `17`), which is not in the scope of any permission, is inaccessible ($ds = -$).

5 Formal Semantics

To formally define the semantics of access control permissions and policies, we will use the notion of *mapping* (as in [19]), which is a function $\mu : \mathcal{V} \mapsto U \cup L$. For ease of presentation, we will denote mappings as sets of pairs, e.g., $\{(?x, \&a), (?y, \texttt{Alice})\}$ is the mapping that maps variables $?x$ and $?y$ to values $\&a$ and `Alice` respectively. We overload the notion of a mapping to apply to triple patterns: in particular, for a triple pattern $tp = (x,\ p,\ y)$, $\mu(tp)$ denotes the triple obtained by replacing the variables in tp with their values according to μ. This way, mappings are used to map triple patterns to triples, by replacing their variables with the corresponding URIs or literals (per μ).

For a triple pattern tp and an RDF graph $\mathcal{G}$, we denote by $\langle\!\langle tp \rangle\!\rangle_{\mathcal{G}}$ the set of mappings obtained by evaluating tp in $\mathcal{G}$, i.e., the mappings that map tp into triples in $\mathcal{G}$. Formally, $\langle\!\langle tp \rangle\!\rangle_{\mathcal{G}} = \{\mu | \mu(tp) \in \mathcal{G}\}$. For example, for the RDF graph $\mathcal{G}$ in Fig. 1 and $tp = (?x,\ foaf{:}firstName,\ ?y)$, two mappings are obtained, namely: $\langle\!\langle tp \rangle\!\rangle_{\mathcal{G}} = \{\{(?x, \&a), (?y, \texttt{Alice})\}, \{(?x, \&b), (?y, \texttt{Bob})\}\}$.

To define the semantics of a conjunction of triple patterns $\mathcal{TP} = (tp_1, \ldots, tp_n)$, we will use the notion of *compatible mappings* introduced in [19]: the mappings $\mu_1, \ldots, \mu_k$ are *compatible*, iff they have the same value for their common variables. Equivalently, $\mu_1, \ldots, \mu_k$ are compatible iff $\bigcup_{i=1,\ldots,k} \mu_i$ is a mapping. The semantics of $\mathcal{TP}$ for an RDF graph $\mathcal{G}$, denoted by $\langle\!\langle \mathcal{TP} \rangle\!\rangle_{\mathcal{G}}$, is defined as:

$$\langle\!\langle \mathcal{TP} \rangle\!\rangle_{\mathcal{G}} = \{ \bigcup_{i=1,\ldots,n} \mu_i \mid \mu_i \in \langle\!\langle tp_i \rangle\!\rangle_{\mathcal{G}} \text{ for } i = 1, \ldots, n, \text{ and } \mu_1, \ldots, \mu_n \textit{ are compatible}\}$$

Intuitively, $\langle\!\langle \mathcal{TP}) \rangle\!\rangle_{\mathcal{G}}$ corresponds to the mappings which map all $tp_i \in \mathcal{TP}$ into triples in $\mathcal{G}$. For instance, for $\mathcal{TP} = ((?x,\ \mathsf{rdf{:}type},\ foaf{:}Person), (?x,\ foaf{:}mbox,\ ?z))$ (see $\mathcal{R}_3$ in Fig. 2) and RDF graph $\mathcal{G}$ in Fig. 1, we get:

$$\begin{aligned}\langle\!\langle \mathcal{TP} \rangle\!\rangle_{\mathcal{G}} = \{\ & \{(?x, \&a), (?z, mailto{:}alice@fun.com)\},\\ & \{(?x, \&a), (?z, mailto{:}alice@work.com)\},\\ & \{(?x, \&b), (?z, mailto{:}bob@work.com)\}\}\end{aligned}$$

For a constraint $\gamma = u\ op\ c$, for $c \in U \cup L$, we say that a mapping μ *satisfies* $u\ op\ c$, denoted by $\mu \vdash (u\ op\ c)$ iff $\mu(u)\ op\ c$ holds; similarly, if $c \in \mathcal{V}$, $\mu \vdash u\ op\ c$ iff $\mu(u)\ op\ \mu(c)$ holds. More generally, we say that a mapping μ satisfies a conjunction of constraints $\mathcal{C}$ (denoted by $\mu \vdash \mathcal{C}$) iff $\mu \vdash \gamma$ for all $\gamma \in \mathcal{C}$. For instance, the mapping $\{(?x, \&a), (?z, \mathtt{17})\}$ satisfies the constraint $?z < 18$.

We can now define the semantics of an access control permission $\mathcal{R} =$ **include/exclude** tp_0 **where** $\mathcal{TP}, \mathcal{C}$ over some graph $\mathcal{G}$, denoted by $\langle\!\langle \mathcal{R} \rangle\!\rangle_{\mathcal{G}}$. The set $\langle\!\langle \mathcal{R} \rangle\!\rangle_{\mathcal{G}}$ corresponds to the mappings that map the triple patterns in $\mathcal{TP}$, as well as tp_0, into triples in $\mathcal{G}$, and satisfy $\mathcal{C}$; formally, for $\mathcal{TP} = \{tp_1 \ldots, tp_n\}$:

$$\langle\!\langle \mathcal{R} \rangle\!\rangle_{\mathcal{G}} = \{\mu \mid \mu \in \langle\!\langle \{tp_0, tp_1, \ldots, tp_n\} \rangle\!\rangle_{\mathcal{G}}, \mu \vdash \mathcal{C}\}$$

The triples that are in the scope of $\mathcal{R}$ are exactly those that tp_0 is mapped to, under some mapping in $\langle\!\langle \mathcal{R} \rangle\!\rangle_{\mathcal{G}}$; such triples are denoted by $[[\mathcal{R}]]_{\mathcal{G}}$. Formally:

$$[[\mathcal{R}]]_{\mathcal{G}} = \{t \ \mid \ \text{there exists } \mu \in \langle\!\langle \mathcal{R} \rangle\!\rangle_{\mathcal{G}} \text{ such that } \mu(tp_0) = t\ \}$$

Back to our example, Fig. 3 shows the triples in the RDF graph $\mathcal{G}$ of Fig. 1 that are in the scope of permissions $\mathcal{R}_i$ from Fig. 2 ($[[R_i]]_{\mathcal{G}}$).

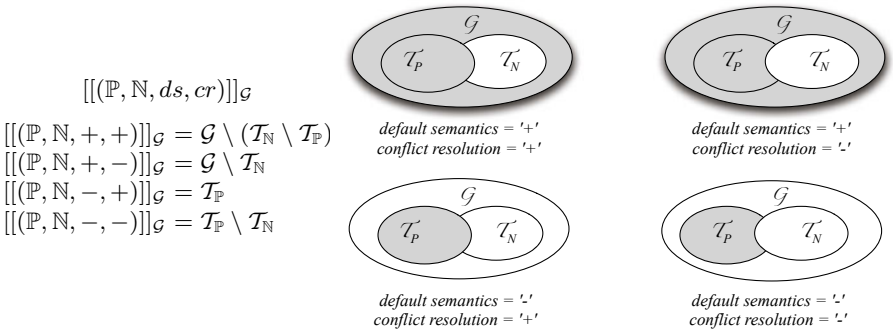


Fig. 4. Access Control Policy Semantics

Now consider some access control policy $\mathcal{P} = (\mathbb{P}, \mathbb{N}, ds, cr)$ and an RDF graph $\mathcal{G}$. The accessible triples of $\mathcal{G}$, given $\mathcal{P}$, are determined by the default semantics (ds) and the conflict resolution policy (cr), in addition to the actual access control permissions, $\mathbb{P}$ and $\mathbb{N}$. Let $\mathcal{T}_{\mathbb{P}} = \bigcup_{\mathcal{R} \in \mathbb{P}} [[\mathcal{R}]]_{\mathcal{G}}$ and $\mathcal{T}_{\mathbb{N}} = \bigcup_{\mathcal{R} \in \mathbb{N}} [[\mathcal{R}]]_{\mathcal{G}}$ be the set of triples in the scope of positive and negative permissions respectively. Triples in $\mathcal{G} \setminus (\mathcal{T}_{\mathbb{P}} \cup \mathcal{T}_{\mathbb{N}})$ (triples that are not in the scope of any permission) or in $\mathcal{T}_{\mathbb{P}} \cap \mathcal{T}_{\mathbb{N}}$ (triples that are in the scope of both positive and negative permissions) may be accessible or not, depending on the values of default semantics (ds) and conflict resolution (cr) respectively. We denote by $[[(\mathbb{P}, \mathbb{N}, ds, cr)]]_{\mathcal{G}}$ the accessible triples for policy $\mathcal{P} = (\mathbb{P}, \mathbb{N}, ds, cr)$. In Fig. 4 we illustrate the policy semantics for the different values of ds and cr (accessible triples are depicted in gray).

6 Implementation and Experiments

In this section we discuss our access control enforcement platform. In particular, we present the architecture of the system, the implementation of the policies' semantics (enforcement mechanism), the dimensions we used to set up our experiments, and the evaluation of our implementation.

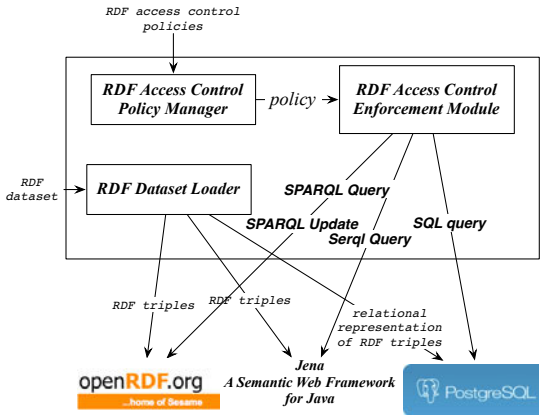


Fig. 5. System Architecture

Architecture: We implemented a main memory platform which serves as an additional access control layer on top of an arbitrary RDF repository. Our goal was for our system to be portable across platforms, so it was designed in a repository-independent way. The system's architecture is shown in Fig. 5. It is comprised of the following modules, all implemented in Java: the RDF Dataset Loader, responsible for loading the RDF triples in the underlying repositories, the RDF Access Control Policy Manager that loads in memory the access control policies and the RDF Access Control Enforcement Module, which translates the access control policies into the appropriate programs that compute the accessible triples of an RDF dataset and annotates accordingly the data in the repositories with accessibility information. Those programs can be SPARQL [20], Serql [8] and Sparul [25] queries (for the corresponding RDF repositories) or SQL queries (when the repository is a relational database).

Access Control Enforcement: To enforce an access control policy, we translate it into a SPARQL/Serql/Sparul/SQL query (depending on the query language supported by the underlying repository) which incorporates the access control policy's default semantics and conflict resolution policy. The triples in the result of the evaluation of this query are exactly the accessible triples in the RDF graph. We use this knowledge to appropriately *annotate* each triple as being accessible or inaccessible. Note that annotations could be (and have been) used for other purposes as well, e.g., to implement trust, uncertainty or provenance information [13,12,15]. Conceptually, annotations can be represented by adding a fourth column to an RDF triple (hence obtaining a quadruple), denoting whether the triple is accessible or not; if several different user roles need to be supported, one column per role should be added. Physically, annotations can be stored using the *named graphs* mechanism of RDF repositories [10], or,

in the case of relational backend, using a large triple table with four columns, following the schema oblivious approach.

Due to lack of space, we will only present the SPARQL/Serql queries used to annotate the RDF triples when the default semantics and conflict resolution policy is "deny" (the rest is similar). Given a policy $\mathcal{P} = (\mathbb{P}, \mathbb{N}, -, -)$, annotations are made using a query of the form $(Q_{\mathbb{P}} \setminus Q_{\mathbb{N}})$ where $Q_{\mathbb{P}}$, $Q_{\mathbb{N}}$ are the queries that compute the set of triples in the scope of permissions in $\mathbb{P}$ and $\mathbb{N}$ respectively (see also Fig. 4). In Fig. 6, we show the form of $Q_{\mathbb{P}}$ and $Q_{\mathbb{N}}$ queries, where tp_0^i is the triple pattern in an access control permission $\mathcal{R}_i$ = include/exclude tp_0^i where $\mathcal{TP}_i, \mathcal{C}_i$ and $expr_i$ is a SPARQL *graph pattern*, i.e., a join of triple patterns and filters that appear in the where clause of $\mathcal{R}_i$. Given that SPARQL does not support the set minus operator ("\" in Fig. 4) between triple sets, we had to implement this operator using main memory set manipulation. In the case of Serql, we use the MINUS operator of the language.

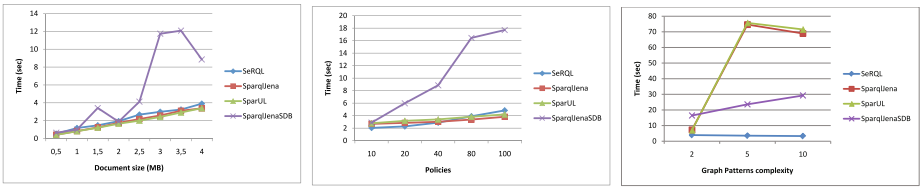
CONSTRUCT $\{tp_0^1$ AND tp_0^2 AND $\ldots tp_0^k\}$
WHERE $expr_1$
UNION
WHERE $expr_2$
...
UNION
WHERE $expr_k$

Fig. 6. Form of Queries $Q_{\mathbb{P}}$ and $Q_{\mathbb{N}}$

Experiments: Our experiments measured the time required to annotate the set of RDF triples, using the above methodology, in state-of-the-art RDF repositories (Sesame [4,9], Jena [2]) or relational backend (Postgres [3]). All our experiments ran on a 2.2GHz Intel Core 2 Duo running Ubuntu v9.10 Linux, with 4GB of physical memory. We used the SP2Bench [24] data generator to obtain the input RDF graphs. We implemented our approach on top of Jena v2.6.2 using the Java engine ARQ v2.8.2, SDB v1.3.1 which links ARQ to an SQL database back-end (Postgresql v8.4), the Java implementation Sesame v2.3.1 and a relational database (Postgresql v8.4). For Jena we tested the SparqlJenaModule and SparqlJenaSDBModule (processing SPARQL queries) as well as the SPARULModule (processing SPARQL/Update language queries) modules. SparqlJenaModule and SPARULModule load the datasets into main memory whereas the SparqlJenaSDBModule stores the datasets to a Postgresql database. For Sesame we used the SeRQLModule, which processes SPARQL [20] and SeRQL [8] queries in memory.

We measured the time required for the annotation as a function of four different parameters: *(i) document size* (*doc*), i.e., the size of the input RDF graph (size ranging between 500KB-4MB with a 500KB increase); *(ii) policy size* (*ps*), i.e., the number of permissions in the access control policy (for sizes of 10, 20, 40, 80 and 100, with an equal share of positive/negative permissions in each case); *(iii) permission size* (*gp*), i.e., the number of triple patterns and constraints in the where clause of each access control permission (values considered: 2, 5, 10); and *(iv) policy parameters*, i.e., the values of the ds, cr parameters of the input policy (all 4 combinations considered).

Evaluation: Fig. 7 shows a subset of the results of our experiments. In each graph, the annotation time is presented as a function of one of the above parameters *(i)-(iv)*, for fixed values for the other parameters (see Fig. 7). Due to lack of space[1] we do not include the results of experiments when varying the policy parameters (default semantics and conflict resolution policy): we report here on the (*deny*, *deny*) case only because it is the most common one. The results show that our approach scales along the considered parameters. All the platforms that we ran our experiments on demonstrated a linear behavior as document, policy sizes and permission complexity increased (except the Jena SPARUL and SPARQL Modules). We do not show here the results of our experiments for the pure relational solution since it proved to be extremely expensive (in some cases causing an increase in annotation time of more than 100%) when compared to the RDF-based solutions. We believe that this is caused by the large number of self-joins (required to implement the WHERE clause of the respective SQL query) on the large triple table.



(a) $gp = 2, ps = 80$ (b) $gp = 2, doc = 4MB$ (c) $ps = 80$, $doc = 4MB$

Fig. 7. Experiments

7 Conclusions

We addressed the problem of selectively exposing information in RDF graphs to different classes of users depending on their access privileges, a problem of great importance for the Future Internet, given the sensitive nature of several datasets. We advocated in favor of a *fine-grained solution*, in which the smallest unit of protection is the RDF triple. In our proposal, triples are specifically annotated as accessible (or inaccessible) using access control permissions, which can be set on sets of triples using *triple patterns* as defined in SPARQL. We then showed how our policies can be translated into SPARQL queries which are subsequently evaluated on different Semantic Web platforms in order to assess the applicability of our approach.

In the future, we will consider extensions of our framework to support RDFS entailment semantics, in effect allowing the determination of the (in)accessibility

[1] The full set of our experimental results can be found at `www.ics.forth.gr/isl/RAC`.

of non-explicit triples. Another possible extension is the incorporation of blank nodes in our model, as well as the consideration of security levels, i.e., non-binary accessibility annotations.

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An Authoring Tool for User Generated Mobile Services

José Danado[1], Marcin Davies[2], Paulo Ricca[1], and Anna Fensel[2]

[1] YDreams – YLabs Research
Edifício YDreams – Madan Parque Sul
P-2825-149 Caparica - Portugal
{jose.danado,paulo.ricca}@ydreams.com
[2] Telecommunications Research Center Vienna (FTW)
Donau-City-Strasse 1
1220 Vienna - Austria
{davies,fensel}@ftw.at

Abstract. Imagine what kind of applications become possible when our mobile devices not only present data but provide valuable information to other users. Users become able to instantaneously create services and to publish content and knowledge on their own mobile device, which can be discovered and accessed remotely by other mobile users in a simple way. To achieve the vision of customizable and context aware user-generated mobile services, we present a mobile authoring tool for end-users to create, customize and deploy mobile services while on-the-go. This tool is designed to allow users with different levels of technical expertise to create mobile services. The paper also gives insight on the performed usability evaluations, namely user interviews and an online survey.

Keywords: Mobile Service Creation, Usability Evaluation, Context Aware Knowledge Acquisition, End-User Development.

1 Introduction

This work contributes to the overall goal of designing a mobile services infrastructure, supported by mobile tools, and an end-user platform to create focused, knowledge-based mobile services, called *microservices*. Microservices are small, sharply focused applications with their own graphical user interface, which allow users to obtain and provide information – such as opinions, recommendations, location or speed – to fellow users. Microservices are created and consumed by end-users solely through their own mobile terminals, encouraging spontaneous and inspired on-the-go creation. Microservices can be shared and downloaded to allow every end-user to become a microservice creator, provider and consumer, in one word, a "prosumer".

As an essential part of the infrastructure, the mobile authoring tool, called Service Creation Kit (SCK), allows mobile users to create new microservices and deploy them into the *m:Ciudad* [6] framework. As more and more microservices become available, the value of the platform increases with the growth of the user base. Users will be able to access services related to their preferences and create microservices on-the-go that

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 118–127, 2010.
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fit their needs. User-friendly mobile service creation and discovery is a major goal of *m:Ciudad*. The presented service authoring tool provides an enhanced user experience during service creation by hiding the complexity of mobile service creation as much as possible.

SCK development and testing was assessed through usability tests, consisting of individual interviews and an online survey. The performed evaluations were used to collect information and provide recommendations on improving the design of the SCK. Based on the presented work, we propose directions for further research related to: the type of users willing to employ user-generated services technology, the scenarios where such technology could be applied, and metaphors that could be used in an authoring tool for mobile generated microservices.

This paper is structured as follows: Section 2 provides an overview on related work. Section 3 introduces two main authoring prototype views evaluated for microservices development. Section 4 presents evaluation results gathered from an online survey and individual interviews. Section 5 describes a first version of the authoring tool and the metaphors used in the SCK interface based on the presented evaluation of the prototypes. Finally, Section 6 concludes the paper and offers an outlook on future work.

2 Related Work

Desktop interfaces have been the subject of successful development in the design of guidelines improving their efficiency and effectiveness. In contrast to the desktop, mobile end-user development is an unknown concept for the majority of mobile users, with unproven and unexplored issues related to characteristics and limitations of current mobile device interfaces and user-specific usage context. Moreover, mobile devices are easily accessible to users making them the preferred device to access services while on the go. However, the extent to which users would want to create mobile services on the mobile device and make them accessible to others is yet to be determined. Thus, this is a challenging field of research.

Häkkilä et al. [3][5] introduced *Context Studio*, a tool that enables the user to define context-action rules in order to activate mobile phone functions when the rule conditions are fulfilled. Paper mockups were used to assess the usability with individual interviews for later development. Afterwards, the application was evaluated through interviews based on Nielsen's heuristics [7]. Such work has inspired our approach that allows users to develop general services, to explore mobile phone sensors, and to create microservices adapted to their needs anytime and anywhere.

User-generated mobile services are active content components or lightweight applications which allow users to obtain and provide information, e.g. opinions, recommendations, location or speed, and functionality to fellow users while "on-the-go" [9]. Such services have been shown to be highly desirable for the end-user on mobile platforms, e.g. for location based services addressing virtual communities and entertainment [4]. Unlike, for example, the Social Semantic Web [1], many aspects of semantic user-generated services are yet to be explored. Microservice creation can also take advantage of semantics to receive adequate suggestions during the service creation process, which is another aspect also considered by *m:Ciudad*.

3 Authoring Tool Prototyping

Service creation on a mobile device is influenced by various factors, such as screen size, input methods and technical experience of the users. This section introduces our prototypes offering two methods of microservice creation to accommodate different levels of user expertise ("beginner" and "advanced").

3.1 Beginner's View – Question-Based Approach

The first view allows the development of a microservice based on templates. In this mode, a user is able to configure information related to the service she is creating, i.e. the name of the Service, the visual theme to be used or the access permissions.

During microservice creation or customization, the user is presented with a set of questions where she is invited to provide the answer that best fits her preferences. An answer can be provided by selecting one of the provided answers or by filling out a text box. Thus, selection of the best option was performed through a carousel.

According to usability heuristics presented by Jakob Nielsen [7], a carousel provides an approach to *match items between system and real world*, i.e. icons can be used to illustrate the option being selected; *give control to the user*, for example, the user is able to provide input to the system and see her option reflected on the system; *provide a consistent approach when used correctly*, i.e. the metaphor is consistent while using the UI control; *prevent users from making mistakes*, i.e. the system only provides adequate option not allowing the user to select options that may corrupt the mobile service; and is *visually appealing*, i.e. the UI control provides a friendly way to present information.

Figures 1.a) and 1.b) show the carousel-like interface of the question-based view to select between different options. The control is adapted for each scenario and is replaced when it makes sense, as seen on a), where the user is able to select the type of microservice to create and, b), the user is able to select the information components to use in the interface.

3.2 Advanced View – Block-Based Approach

The advanced view targets users with some programming skills as well as knowledge about m:Ciudad architectural details. In this mode, the users are able to have a finer control over the development and behavior of the microservice.

Advanced service creation in m:Ciudad is built on top of a XML-based language named U+ Service Description Language (UDL). UDL is used for presentation, logic and profile/metadata of the service. Each tag in UDL is visualized as a block. Blocks are connected in a tree structure and each block can have one or more inputs and a single output. Inputs for a parent block are fed by the outputs of child blocks.

Regarding visualization of such blocks in the device and since screen size is a very limiting factor while designing a service using a block-based language, two different modes were created for UDL visualization, namely a world view and a detail view. The world view mode targets a broader view of the microservice and is intended to get a clearer picture of how blocks are related to each other, while the detail view

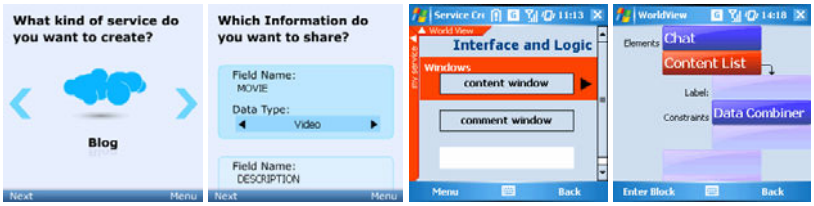

Fig. 1. Beginner view - microservice selection (a), information field and type selection (b), Advanced View – UDL selection (c), world view (d)

focuses on a unique block and its inputs. Using both modes, a user can perform a high-level management of the service creation while still being able to focus on particular behaviors. Figure 1.c) shows how the different components of the UDL can be selected; and the tree of blocks shown in Figure 1.d) depicts their hierarchy.

4 Evaluation

Assessment of SCK interfaces was performed in two parts. First, the authoring views were evaluated through the execution of up to 3 different scenarios for microservice creation. In such part, user interviews took place to assess the effectiveness of the metaphors used.

The second part of the evaluation was an online survey where users could test the previously mentioned service creation view and answer a questionnaire. The survey was supported by an online prototype of the beginner's view and additional videos showing an envisioned usage of the advanced view.

4.1 User Interviews

This test was carried out inside YDreams' Headquarters with 5 collaborators from various educational backgrounds and lasted for about 60 minutes. The collaborators were 2 males and 3 females; 3 had experience in developing applications, i.e. during their career they had the opportunity to develop an application whether for servers, desktops, or mobile devices; One was considered an advanced user, i.e. he was able to install, configure and solve problems in applications; and another collaborator was considered a regular user, i.e. was able to manage regular applications. Ages for the test group ranged from 30 until 36 years old. Lastly to describe the profile of the user's test group, 2 were already aware of m:Ciudad, and the remaining ones were introduced to the project during the interview.

The number of users to assess the interface usability was influenced by Bevan et al. [2] results, where they state that 80% of usability findings are discovered after 5 participants. Additionally, tests were carried out through the usage of a standard "Note Taker's Guide" [8] adapted to the context of the SCK interface. The purpose was to test the viability of the approaches that served as basis for the SCK, as well as the user interface of both views, beginner and advanced. Each user experienced both views through 3 related scenarios. The first scenario was created to assess the usage of a

carousel interface for microservice creation. In this scenario, the user should select a template to use and define the name of the microservice and finally to choose a theme to use.

A second scenario was created to assess microservice customization for novice users. In this scenario, the user is invited to create a new microservice, add visual elements to be shared between users and preview the service. Last scenario was created to evaluate the advanced view and related building blocks. In this scenario, the user was asked to add presentation, logic, content and metadata UDL blocks.

4.1.1 Methods

The method used to evaluate the beginner and advanced views followed the guidelines described below:

1. The trials were executed with only one participant and one tester at a time;
2. At the beginning of the tests, questions about technical expertise were asked in order to understand the technical profile of the user;
3. The user was then presented with a very quick background of the project, namely what could be done with m:Ciudad and what was a microservice;
4. The participant was given the scenarios to accomplish;
5. During execution of each scenario, the tester took notes related to found problems, and additional remarks, such as the performance, easiness or curious thoughts about each view;
6. If the subject was starting to feel frustrated failing to perform a task, the tester would annotate the problem and explain how to surpass it;
7. In the end, the user was asked some global questions about how she felt using both views and about the m:Ciudad project itself.

4.1.2 Results

Overall, the tests showed that users with a technical background can understand the concepts behind the creation of a microservice. The tests with the beginner view showed that changes were required for the interface, mainly through the usage of navigation tips and avoidance of technical jargon to improve usage of the interface. In the advanced view, basic concepts, such as block tree hierarchy or block-based development, were easily perceived, which was surprising since the users had never worked with a similar system, but further tests are required to evaluate the development of m:Ciudad microservices.

The main comments gathered from usability tests are related to the UDL and the mental model behind microservices execution, namely, the design of the interface, data manipulation, and content interaction. It was concluded that both views are still complex to use although the tests suggested that the interface could be improved through introduction of help tips, short descriptive text, icons, colour codes and templates. Another result from the tests was the need for an intermediate view, in order to ease the learning curve between the beginner view and the advanced view (cf. [10]). Such a view would allow users to further customize microservices without getting into technical details. Therefore, the learning steps for end-user development of microservices would cope with all stages of user experience.

4.2 Online Survey

For assessing both views in the authoring tool, an online questionnaire was designed. The online survey was run on the open source software LimeSurvey[1] and was available in English and French to reach a broad audience of users.

The announcements about the questionnaire were circulated on technology-related, as well as general purpose, mailing lists and on social networks (e.g. Facebook, Twitter) inviting interested participants to take part in the study.

The survey contained 38 questions grouped into three main categories:

1. ***Technical background and experiences:*** Gathering information about users' technical background and past experiences with related technologies *(22 questions maximum, depending on the responder's familiarity with different technologies)*;
2. ***Service creation approaches and user motivation***: Gathering feedback on service creation approaches and understanding the user's motivation for creating services *(7 questions)*; this part also included a demo of the question-based SCK and two videos of both the question-based and the block-based service creation approaches.
3. ***Demographic data*** Recording respondents' demographic characteristics such as age and occupation. We were asking these questions in order to determine if various groups have different opinions and attitudes *(9 questions)*.

The time required to take the survey has been estimated at 10 minutes, when careful study of user interface demos and videos is excluded.

First tests provided a positive feedback regarding the usage of developed prototypes, thus envisioning testing the prototypes with a broader audience. In the scope of these prototypes, assistance was required to avoid undeveloped paths within execution. Thus, a web interface was envisioned to provide a safe testbed for a larger audience and a video was also created to show a usage scenario for the second prototype.

Both approaches were included in the questionnaire herein discussed, and the results are presented in the following section.

4.2.1 Results

For the web questionnaire, we received a total number of 52 full responses over a period of 4 months. 74% of the respondents were male and 98% from Europe. 88% had a university degree (bachelor or higher), and 70% were working in ICT or science fields. 44% of the participants were between 20-29, one third between 30-39, and the remainder (24%) between 40-60 years old. 46% of the respondents confirmed to access the internet from their mobiles. Two thirds of the respondents use the internet at least 4 hours per day.

Regarding the users' technical background and experiences, two thirds of the participants would like to customize applications, including a wide range of technical appliances such as television sets, cars, mobiles, GPS devices, etc. 50% of the participants would like to create small programs by themselves. The kind of

[1] LimeSurvey: http://www.limesurvey.org

programs was again quite diverse, but building web sites and automating tasks were recurrent wishes.

Figure 2.a) shows the kind of technologies people had experience with. In general, predominant web technologies (blogs, chats, etc.) were quite familiar. However, mashup editors were widely unknown (4%). We also asked for specific applications, which offer end-user development possibilities. In this category, Facebook apps were used mostly, followed by Microsoft Office Macros, Dashboard Widgets, Yahoo! Mobile and Nokia WidSets - the remainder were used only by a minority (cf. Figure 2.b).

People were also asked to rate tools they know regarding the following aspects: time, complexity, expressivity, openness, fun factor (on a 5 grade scale, 5 best, 1 worst). As Figure 2.b) shows, only Facebook apps and Microsoft Office Macros had a wide reception amongst users. The other tools were only used by a maximum of 4 users, therefore statistical significance could not be reached. Table 1 shows the detailed results of the Facebook and Office ratings (averages).

Fig. 2. a) Users technological experience level, b) End-user development technologies experienced level

Table 1. Facebook and Office ratings

	Microsoft Office Macros	Facebook Applications
Time	**2.71**	**3.08**
Complexity	**2.71**	**2.58**
Expressivity	**3.43**	**3.25**
Openness	**2.57**	**3.08**
Fun Factor	**1.86**	**3.75**

Users were also asked what would be their motivation to create a new microservice and what approaches would best fit their requirements. Thus, after showing people a typical scenario for service creation (as a small cartoon), participants could try an interactive flash demo and look at a video of the beginner's view service creation approach. As an alternative to the beginner's approach, also the advanced view approach was shown in a video. After showing the videos and demo, people could leave their feedback in a free-form text box. We summarize their opinions in quantitative numbers with regard to different aspects in Table 2 ((+) indicates positive comments, and (-) indicates concerns).

Table 2. User's opinions summary

	General positive feedback (+)	Easiness to use for a typical user (+)	Restrictiveness /complexity (-)	Technical concerns, bugs (-)
Beginner	8	13	6/3	12
Advanced	7	1	0/13	8

In terms of easiness, the question-based approach was received much better ("It seems to be also suitable for beginners to create services. The user is well guided by questions") than the block-based one ("for freaks/nerds mainly"). Some users, however, mentioned that the question-based approach might still be too restricting ("I could imagine creating a service this way, but it would probably be too simple to do anything I really wanted.").

The advanced view was mainly regarded as a professional tool for creating more complex services, and one which might fit better on a desktop than on a mobile. The purposes of creating services were mostly to exchange information (71%), stay in contact (60%), and just for fun (60%). People would mostly create them while waiting (73%), commuting (61%) or at home (60%). 44% would also create services at work, i.e. professionally. Free-of-charge provision/usage was the most frequent answer (73% and 87% respectively). About a quarter of people could imagine either: making small payments or subscribing to services.

5 Deriving an Intermediate View

In the previous section, two views were used to assess usability of the m:Ciudad SCK interfaces towards creation of a mobile microservice. The evaluation results, seen in the previous section, were used to develop new interfaces for the m:Ciudad SCK. Therefore, three views were developed to ease the learning curve while using m:Ciudad SCK: beginner, intermediate and advanced.

In the beginner' view, the user is able to select or fill simple options required to create a new microservice from a template. When compared with the first prototype and regarding the assessment from user interviews, the same metaphors were used in the first version of the m:Ciudad SCK. Nevertheless, additional arrows were added, at the bottom, to support navigation between all steps of the operation as well as a progress status to provide feedback on the microservice creation completion level. Figure 3.a) shows such an approach.

In Figure 3.b), an image of the newly added intermediate view is presented. It allows users to easily connect high-level blocks representing both: the visual appearance, logic and content of a common functionality. In this view, the user only has to drag the blocks into a blank canvas and all the functionality is generated automatically. Examples of high-level blocks may be a "Multimedia Gallery", a "Journal", a "Map" or a "Shoutbox". Then, the user can use blocks individually or she may specify relations between them in order to create a coherent and complex service.

The advanced view was designed to allow visualization and development of a microservice within the scope of the m:Ciudad framework. For such view, a "Lego"

metaphor was used to improve connectivity between UDL blocks. Thus, the user should fill those inputs shown as a "hole" with a Lego pattern and a filled input is displayed as a Lego piece which filled the "hole". Consequently, a user can experience creation of a microservice as a construction, with inputs that can only fit related outputs. Lego supplies the metaphor for construction, where a list of possible blocks shown to continue the operation is suggested by the m:Ciudad SCK.

Each input is accompanied by a label describing the input and an additional label inside defining the accepted type of Blocks. The World View supports a broader view over the service structure, while still allowing the user to edit the service. Figure 3.c) is a perspective of the advanced view and the *Lego* metaphor used during microservice creation.

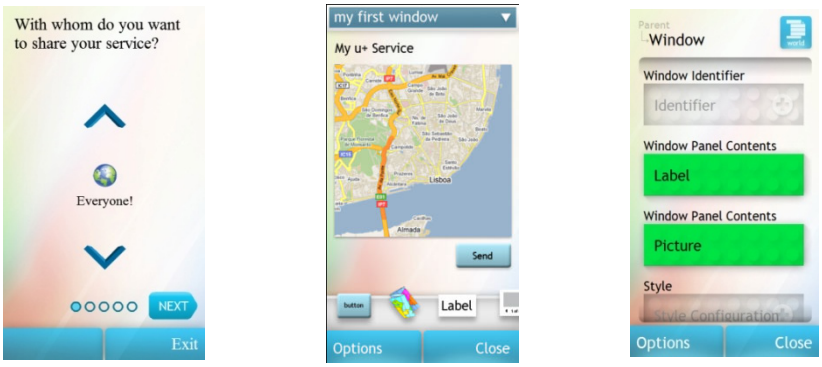

Fig. 3. a) Beginner view, b) intermediate view and c) advanced view

6 Summary and Conclusions

Mobile service creation is still a challenge to be addressed in the future. One of the main advantages of an authoring tool for a mobile device is the ability to use or deploy a service without requiring a desktop machine. Our authoring tool allows users to develop microservices in-situ, when it makes sense.

Therefore, the m:Ciudad SCK is able to thrive in a sense of opportunity and increase mobile services adoption. This sense of opportunity can be used in boosting mobile services usage. In order to achieve such results, development of the m:Ciudad SCK included a study of possible interfaces for mobile service creation, usability tests, an online survey, and development of a prototype including achieved results.

Development of two prototype views for service creation was carried out, targeting users with different technological expertise, namely beginners view and advanced view. Evaluation results demonstrated the user acceptance of the approach, and served as a foundation for deriving a new combined service creation view for an intermediate user. The beginner's view uses a carousel to explore options for mobile service customization in a more controlled environment. The intermediate view explores microservice creation through connection of high-level functionalities.

Finally, the advanced view focus on technical users and gets inspiration from an XML-based language, named UDL through the exploitation of the block concept. Therefore, it requires previous knowledge of the intricacies of m:Ciudad microservices.

Future work targets improvement in the interface, service creation, and further user tests to evaluate the effectiveness of the changes performed in the interface, as well as in each UI control. Moreover, future work includes testing the SCK using suggestions from the microservice creation acquired from ontology-based knowledge gathered in the m:Ciudad infrastructure.

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A Provenance-Based Compliance Framework

Rocío Aldeco-Pérez and Luc Moreau

School of Electronics and Computer Science, University of Southampton,
Southampton SO17 1BJ, UK
{raap06r,l.moreau}@ecs.soton.ac.uk

Abstract. Given the significant amount of personal information available on the Web, verifying its correct use emerges as an important issue. When personal information is published, it should be later used under a set of usage policies. If these policies are not followed, sensitive data could be exposed and used against its owner. Under these circumstances, processing transparency is desirable since it allows users to decide whether information is used appropriately. It has been argued that data provenance can be used as the mechanism to underpin such a transparency. Thereby, if provenance of data is available, *processing becomes transparent* since the provenance of data can be analysed against usage policies to decide whether processing was performed in compliance with such policies. The aim of this paper is to present a Provenance-based Compliance Framework that uses provenance to verify the compliance of processing to predefined information usage policies. It consists of a provenance-based view of past processing of information, a representation of processing policies and a comparison stage in which the past processing is analysed against the processing policies. This paper also presents an implementation using a very common on-line activity: on-line shopping[1].

1 Introduction

Due to the increasing number of Internet services that manage sensitive personal information, the proper use of this information becomes a determinant issue for users who want to access these services with a guarantee that their personal information is not being misuse. Some research [14,4,8] is focused on developing better techniques to avoid information misuse by restricting access to information. However, access restriction alone cannot properly solve this problem on the Web, where information is widely and public available. When information becomes accessible, verifying its correct use after it was processed is also important. In order to solve this problem, users and organisations should be able to define policies under which personal information can be processed. In this context, information accountability, a property according to which users can analyse the way in which their information was used, is desirable. Weitzner *et al.* [16] and the W3C Provenance Incubator group [15] have argued that provenance, which consists of causal dependencies between data and events explaining what contributed to a result in a specific state [11], could be used to support information accountability to

[1] This research was partially supported by the Programme Alβan, the European Union Programme of High Level Scholarships for Latin America, (E06D103956MX) and by the Mexican Council CONACyT (182546).

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 128–137, 2010.
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help users to answer questions related to the processing of information [16]. If provenance of data is available, processing becomes transparent since the provenance of data can be analysed against usage policies to decide whether processing was performed in compliance with such policies [1]. Information related to a specific processing can be obtained from provenance information by means of a provenance query [10], the result of which can be analysed to decide if the processing was performed in accordance with a set of usage policies.

In order to support this vision, we have created a provenance-based Compliance Framework that consists of a view of past information processing, a representation of the policies that processing should follow and a comparison stage in which the past processing is analysed against the processing policies. By using this framework, it is possible to decide if an application processed information in compliance to the predefined information usage policies. The framework components exploit provenance for representing both past execution and the rules to comply with. They are platform-independent and reusable, as they can be applied to different systems to verify different policies. They are also represented as an OPM [12] specialisation. In that way, any system using OPM can make use of this framework. At the same time, our framework could be used to create automatic auditing tools for verifying diverse policies over data processing.

The aim of this paper is to present this Compliance Framework; specifically, the contributions of this paper are: *(i)* The Compliance Framework components, which comprises the Processing View and the Usage Policies Definition, *(ii)* The Compliance Framework Analysis Stage, in which its components are compared to check the correct processing of information, and *(iii)* an implementation prototype.

The remainder of this paper is structured as follows. In §2, an application example is presented. In §3, the Compliance Framework's components are presented and explained. In §4, the framework's Analysis Stage and the algorithms used to verify some requirements are explained. In §5, an implementation of the previously explained example is presented. Finally, §6 discusses some related work and §7 offers some concluding remarks.

2 Application Example

Consider the following scenario: Alice is trying to get pregnant, and so has decided to take a fertility treatment (clomid). She decided to buy her treatment using the web page of a pharmacy. In order to get her treatment, she needs to provide her *name*, *address*, *date of birth*, *gender* and *social security number*. At the same time, but unrelated to her attempt to get pregnant, she applies for a job in the same pharmacy - and she is rejected. She suspects that the pharmacy may have checked its records related to her name and realised that she has plans to have a family, and as a result marked her as a high risk employee for expensive maternity costs. If this is true, the company obviously misused Alice's personal information. When she sent her personal information to the pharmacy, she did that with the purpose of getting her treatment. From the point of view of the company, the purpose is *on-line sales*. The on-line sales purpose could include verifying the existence and the sales of the medicine (manage stock), charging the amount to her card and sending the medicine to her home. The company can create a record of the monthly sales to manage medicines' stock. This record includes the *medicine's name* and the *quantity sold*. Nevertheless, such a record cannot contain the

name of the people that bought that item, as the purpose of that record is not to identify each person. By exposing the way in which the data sent by Alice was used, we can verify if the pharmacy correctly used her information and, if not, make it accountable. This example is represented in Figure 1 and analysed in the next sections to explain the framework.

3 Compliance Framework

The Compliance Framework consists of a past processing view, which is called *Processing View*, a policies representation, which is called *Usage Policies Definition*, and an analysis stage. The Processing View and the Usage Policies Definition are represented by provenance graphs that are directed acyclic graphs whose vertices represent data and edges relationships between such data. Data includes *purposes* (p_i), which are the intentions for which a set of data is to be collected, *tasks* (t_i), which are the processes performed over data, *data* that is to be collected (D_{Ci}) and processed (D_{Ii}), and *results* (r_i), which are the outputs of a task.

3.1 Processing View

The Processing View (PV) (Figure 1(a)) represents a provenance graph captured at execution time. We assume the entities involved in the processing of private information are capturing the corresponding provenance information following the approach presented in [1]. Also, to trust in the outcomes of the analysis stage, we assume that this view is secured in the way described in [2]. This view represents a process in which an application requests a set of data from a user, making explicit the purpose for which such a set **is acquired** (*collection purpose*). After checking the application purpose, the user send the requested multiset of data instances (*collected data*). The goal of the application is to achieve the collection purpose. For that reason, a *task* **is initiated by** the given purpose (*processing purpose*). Such a task **uses** a multiset of data instances that is a **subset of** the collected data (*used data*). Later, the task is executed with the used data as input and **generates** *results*. Such results could be used as collected data in the execution of another task. The *Processing View Graph* G_V can be defined as:

Definition 1 (Processing View Graph). *Let us consider a set of purposes P_V, a set of tasks T_V, a multiset of collected data instances D_{C_V}, a multiset of used data instances D_{U_V}, a multiset of results R_V and a set of relationship's names Rel_V =* {wasInitiatedBy, used, contained, wasGeneratedBy, wasAcquiredFor}.

A Processing View Graph $G_V = (V_V, E_V, Rel_V)$ is a directed acyclic graph, where $V_V \subseteq P_V \cup T_V \cup D_{C_V} \cup D_{U_V} \cup R_V$, $E_V \subseteq (T_V \times P_V \times Rel_V) \cup (T_V \times D_{U_V} \times Rel_V) \cup (D_{U_V} \times D_{C_V} \times Rel_V) \cup (R_V \times T_V \times Rel_V) \cup (D_{C_V} \times P_V \times Rel_V) \cup (D_{U_V} \times R_V \times Rel_V)$.

3.2 Usage Policies Definition

The Usage Policies Definition (UPD) (Figure 1(b)) is a representation of the processing policies that are verified with the framework. These policies should be followed by applications while users' personal information is being processed. Each policy represents the way in which a set of data can be used, i.e. which tasks can use a certain multiset of

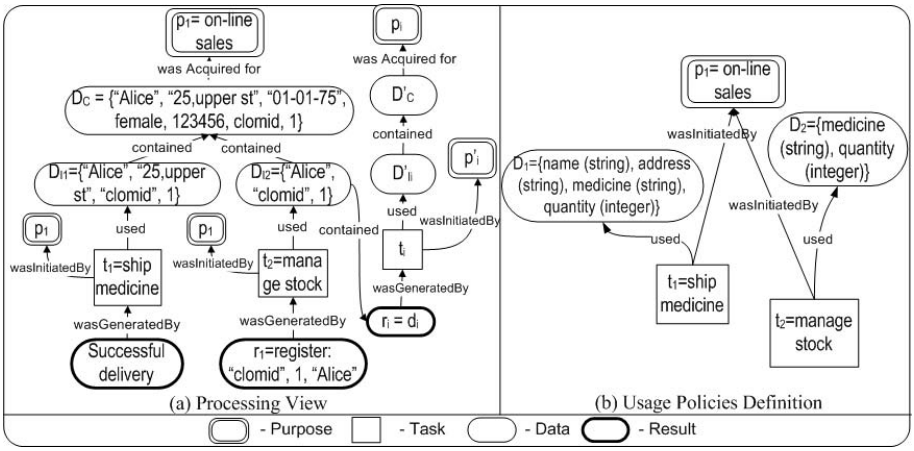

Fig. 1. Compliance Framework Components

data to accomplish which purpose. This definition contains a set of purposes from which users' data is collected, the set of tasks that **are initiated by** a specific purpose and a multiset of data types that a specific task **uses** in its execution. Note that this component contains more than one purpose and each purpose has more than one task. However, one task could be related to more than one purpose. The *Usage Policies Definition* G_R can be defined as:

Definition 2 (Usage Policies Definition Graph). *Let us consider a set of purposes* P_R*, a set of tasks* T_R*, a multiset of data types* D_R*, and a set of relationship's names* $Rel_R = \{\mathsf{wasInitiatedBy}, \mathsf{used}\}$.

A Usage Policies Definition Graph $G_R = (V_R, E_R, Rel_R)$ *is a directed acyclic graph, where* $V_R \subseteq P_R \cup T_R \cup D_R$*,* $E_R \subseteq (T_R \times P_R \times Rel_R) \cup (T_R \times D_R \times Rel_R)$.

4 Analysis Stage

In this stage, by using the already described components, it is possible to verify several information usage requirements, such as *(A) Purpose Compliance:* Processing of data is compatible with the purpose for which it was captured and *(B) Minimun Information Set:* Only information to be processed was captured. Due to space restrictions, we focus on *(A)*. The verification of these requirements is performed by comparing the past processing, described in the Processing View, against the expected processing, represented by the Usage Policies Definition. To do this, information related to the requirement is extracted from the Processing View in form of a subgraph. Then, the data and the relationships of such a subgraph are compared with the content of the Usage Policies Definition. The Purpose Compliance requirement states that the data used when performing a task should be data to only accomplish the initially stated purpose. The verification of this requirement includes to check if the correct type of data was used, if data was used to accomplish the stated valid purposes and, if data is reused, the processing is also made according to the stated purposes. From which, we derive the subrequirements *(1)*

Used Data Compliance, *(2) Purposes Validation* and *(3) Reusing Results*, respectively. Next, the extracted subgraphs related to each of these subrequirements are formally defined, the comparison process is explained presenting its corresponding algorithms and an example.

(1) Used Data Compliance. Here, we verify that the used data has the correct data type related to the task that used it and the purpose that initiated such a task. We also verify that the purpose and the task are in the corresponding Usage Policies Definition. To this end, we extract from a Processing View Graph the provenance of a result. Specifically, we focus on the task that generated such a result, the data instances that such a task used and the purpose that initiated this task. This information is expressed as a subgraph of A_1, which is called *Used Data Compliance*.

Definition 3 (Used Data Compliance Subgraph (A_1)). *Given a Processing View Graph G_V, we can extract a Used Data Compliance Subgraph A_1 such that,*

$$V_{A_1} = \{p_i, D_{U_i}, t_i\}$$
$$E_{A_1} = \{(t_i, p_i, \mathsf{wasInitiatedBy}), (t_i, D_{U_i}, \mathsf{used})\}$$
$$\textit{where } p_i \in P_V, t_i \in T_V, D_{U_i} \in D_V \in R_V \textit{ and } i = \textit{number of tasks}.$$

Algorithm 1. Data Processed According to a Valid Purpose

```
input: G_R = {V_R, E_R, Rel_R}, A_1 = {V_{A_1}, E_{A_1}, Rel_{A_1}}, p_i ∈ V_{A_1}, t_i ∈ V_{A_1}, D_{U_i} ⊆
V_{A_1}, D_j ⊆ V_R
if p_i ∈ V_R then
    if t_i ∈ V_R then
        for all x ∈ Rel_{A_1} and y ∈ Rel_R do
            if x = y then return -1                    ▷ Label not matched
        for all x ∈ D_{U_i} do
            if type(x) ⊈ D_j then return -2            ▷ Type not matched
    else return -3                                     ▷ Not registered task
else return -4                                         ▷ Not registered purpose
return 1                                               ▷ Compliance
```

An example of this subgraph is presented in Figure 2(a). The UPD shows that the task "manage stock" should use a string (medicine's name) and an integer (medicine's quantity). However, as the PV depicts, this task used an extra data item: name. Therefore, the processing presented in Figure 2(a) does not satisfy the Used Data Compliance Subrequirement. The verification process of the Used Data Compliance subrequirement is presented in Algorithm 1[2].

(2) Purposes Validation. Here, we verify if the data was processed with a purpose related to the one it was collected for. To this end, we extract from a Processing View Graph the provenance of a set of tasks related to one set of collected data. Specifically,

[2] The type of each instance is included in the provenance information and obtained using the accessor $type$.

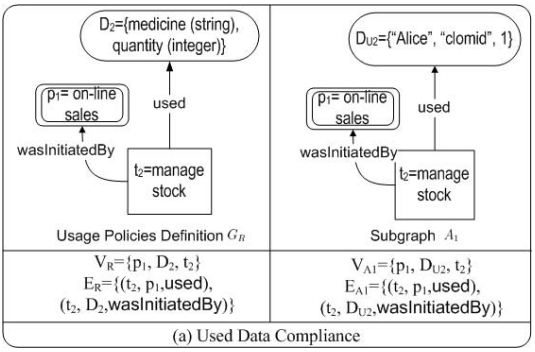

Fig. 2. Extracted Subgraphs

we focus on the purposes from which such tasks were initiated and the purposes from which the collected data was acquired. This is expressed in Figure 2(b) as the Purposes Validation subgraph, which is defined below.

Definition 4 (Purposes Validation Subgraph (A_2)). *Given a Processing View Graph G_V, we can extract a Purposes Validation Subgraph A_2 such that,*

$$V_{A_2} = \{p_i, D_C, D_{U_i}, t_i, p'_i\}$$
$$E_{A_2} = \{(t_i, p'_i, \mathsf{wasInitiatedBy}), (t_i, D_{U_i}, \mathsf{used}), (D_{U_i}, D_C, \mathsf{contained}), (D_C, p_i, \mathsf{wasAcquiredFor})\}$$
$$\textit{where } p_i, p'_i \in P_V, t_i \in T_V, D_{U_i} \in D_V, D_C \in D_V \textit{ and } i = \textit{number of purposes}$$

The Purposes Validation Subgraph is used to create a set of *Processing Purposes* (P'), which contains all p'_i, and a set of *Collection Purposes* (P''), which contains all p_i.

Algorithm 2. Processing Purposes vs Collection Purposes

input: $G_R = \{V_R, E_R, Rel_R\}, A_2 = \{V_{A_2}, E_{A_2}, Rel_{A_2}\}, P' = \{p'_i\}, P'' = \{p_i\}$
if $P' \subseteq P''$ **then**
  if $P' \subseteq V_R \wedge P'' \subseteq V_R$ **then return** 1 ▷ Compliance
  else return -1 ▷ Not registered purpose
else return -2 ▷ Not compatible purpose

Later, we verify if the Processing Purposes are contained in the Collection Purposes, i.e. data was processed according to the purposes it was collected for. It is also verified if both sets contain valid purposes, i.e. they are contained in the UPD. The example presented in Figure 2(b) is in compliance with this subrequirement. The explained comparison process is presented in Algorithm 2.

(3) Reusing Data. The previous requirements are used to verify the purposes related to data that was collected and processed by the same entity. However, an entity may use data generated by tasks executed by other entities. This data is referred as "reused

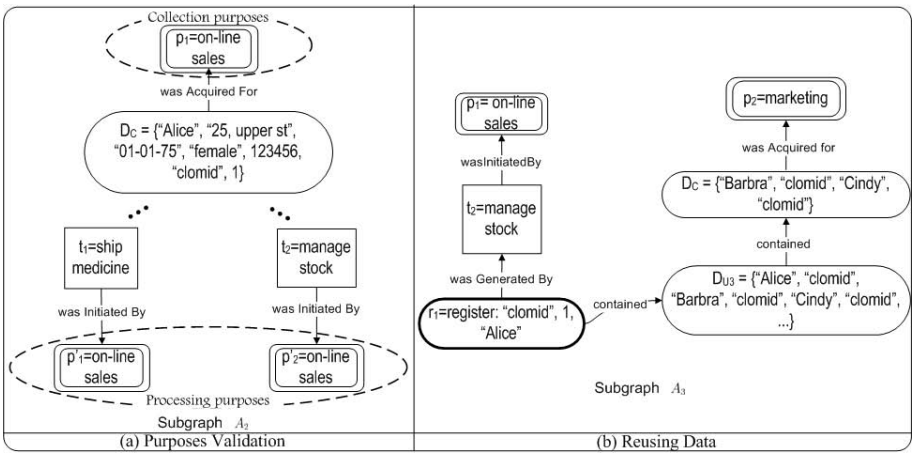

Fig. 3. Extracted Subgraphs

data" and it has its own processing purpose related to the task that produced it. To be in compliance to Requirement A, such a processing purpose should be related to the purposes for which data is reused. To this end, we extract from a Processing View Graph the provenance of the reused results that are the intersection of a collected data multiset related to one entity and a used data multiset related to one task executed by the same entity. Specifically, we focus on the task that generated such a reused data and the purpose from which such a task was initiated. This is expressed by the Reusing Data Subgraph of G_V, which is defined below.

Definition 5 (Reusing Data Subgraph (A_3)). *Given a Processing View Graph G_V, we can extract a Reusing Data Subgraph A_3 such that,*

$$V_{A_3} = \{R_j, t_j, p'_j\}$$
$$E_{A_3} = \{(R_j, t_j, \mathsf{wasInitiatedBy}), (t_j, p'_j, \mathsf{wasGeneratedBy})\}$$
$$\textit{where } p_j \in P_V, t_j \in T_V, R_j \in R_V \textit{ and } j = \textit{number of reused data}$$

Algorithm 3. Reusing Data

input: $G_R = \{V_R, E_R, Rel_R\}, A_3 = \{V_{A_3}, E_{A_3}, Rel_{A_3}\}, P' \subseteq G_R, P'' \subseteq G_R, p \in V_{A_3}$
if $p \in P'$ and $p \in P''$ **then return** 1 ▷ Compatible purpose
else return -1 ▷ Not compatible purpose

In the example presented in Figure 3(a), the reusing of the result of the task "manage stock" is presented. Some data items of this result are used in a new task with purpose "marketing". However, the result was initiated by the purpose "on-line sale". Therefore, reusing this result is not in compliance with the initial purpose. The verification process of this subrequirement is presented in Algorithm 3.

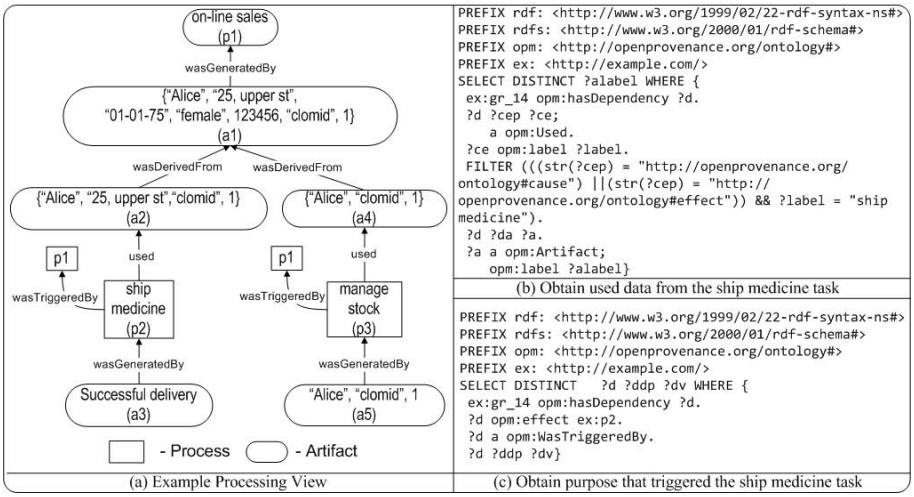

Fig. 4. Implementation Example

5 Implementation

To show how the Provenance-based Compliance Framework is used, we implement the previously presented example. Due to the lack of space, we focus on the queries used to obtain the information we need to verify the subrequirement "Used Data Compliance". To this end, the processing view graph, which is presented in Figure 4(a), is represented as an RDF OPM Graph [12] and the queries are implemented using SPARQL. Initially, we verify the *ship medicine* task. To this end, we extract the used data (D_U) using the SPARQL query presented in Figure 4(b). We also need to extract the purpose (p) that initiated the *ship medicine* task by executing the query presented in 4(c). The task name (t) is extracted by using a similar query, which is not shown for the given space restrictions. Then, the obtained information (D_U, p, t) along with the corresponding Usage Policies Definition, are used as the input of Algorithm 1. After executing this algorithm, the result is that the task *ship medicine* is in compliance with the given requirement, so we can say that the processing of Alice's data by this task was in compliance. The task *manage stock* can be analysed in a similar way. If we extract the necessary elements related to this task and apply Algorithm 1, the result will be no compliance. The reason is that a piece of data that is not part of the policies has been used: *name*. An inventory of a pharmacy does not need the name of the customers, just the item that was sold and the quantity of it. Then, we can say that the processing of Alice's information by this task was not in compliance, and therefore, such an inventory could be used against her interest. In this case, to be marked as a high risk employee.

6 Related Work

Recently, researchers have realised that provenance can be used to verify the compliance of different policies related to the use of personal information [6,5,13] and others use different technologies to support compliance [3,9,7].

Hanson *et al.*[6] present a data-purpose algebra to annotate data with usage restrictions. With these restrictions, requirements similar to the ones presented here can be checked. In this approach, each data is annotated after execution, contrary to our approach, where provenance information created at execution time is used. In terms of implementation, they just present a prototype to verify requirements compliance. Gil *et al.* [5] argue that computational workflow systems can be used to ensure and enforce the appropriate use of sensitive personal data. Then, compliance, transparency and accountability can be supported by these systems. To this end, they define a set of requirements that are similar to the ones we present here. Later, using workflow system to support process transparency, the defined requirements can be verified. They also propose to use this technology to enforce policies and negotiate them. In our work, we propose that given the openness of the Web to support compliance is more flexible than enforcement. Finally, they do not offer a practical solution as the one presented in this paper. Ringelstein *et al.* [13] use a modified version of OPM to express processing execution and an extended version of XACML to represent permission and restriction policies. Then, by modelling conditions, enforcement is supported. As we previously mention, we support accountability, not enforcement. We also create OPM-based rules, which can be easily checked against any OPM-based view without any previous transformations.

In the business processes context, Ly *et al.*[9] present a similar work to check the compliance of rules related to the quality of a product. They requirements are different as they verify the executed order of processes. They use process-aware information systems and process models to model the rules. Awad *et al.* [3] check compliance of control flow and data flow in business process models by using BPMN to express process models and BPMN-Q to represent policies. They design a set of queries and present how violations occur by comparing predefined patterns with the queries. These two approaches do not support the verification of use of data items and are not based in an open model, such as OPM. Kang *et al.* [7] also present a similar approach that helps users to conform with existing policies in a social networks context. This is achieved by making them aware of data usage restrictions defined by the users. One drawback of this work is that they just define a small set of policies, contrary to our framework, in which any usage policy can be defined. In that way, our framework can be applied to a diversity of contexts.

There is also work on authorisation and enforcement of rules in databases [14,4,8]. However, given the openness of information on the Web and the possibility of inferring information using previously published information, authorisation and enforcement are very challenging to implement.

7 Conclusions

In this paper, we present the provenance-based Compliance Framework by explaining its components and how this framework can be applied to an application example over an on-line shopping scenario. With this example, we explain how our framework helps us verify one (of more) requirement related to the processing of information. Our work demonstrates that by using the Provenance-based Compliance Framework, individuals or institutions, which used information in a different manner from the stated, can be held

accountable for misuse. Our framework comprises one past processing view, one novel policies definition, which are platform independent and reusable, and one comparison stage, which is easy to implement in an automatic way.

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Network Virtualization - Opportunities and Challenges for Operators

Jorge Carapinha[1], Peter Feil[2], Paul Weissmann[2], Saemundur E. Thorsteinsson[3], Çağrı Etemoğlu[4], Ólafur Ingþórsson[3], Selami Çiftçi[4], and Márcio Melo[1]

[1] Portugal Telecom Inovação
[2] Deutsche Telekom AG
[3] Síminn hf.
[4] Türk Telekom A.Ş.
jorgec@ptinovacao.pt, peter.feil@telekom.de, paul.weissmann@telekom.de, saemi@siminn.is, cagri.etemoglu@turktelekom.com.tr, olafuring@siminn.is, selami.ciftci@turktelekom.com.tr, marcio-d-melo@ptinovacao.pt

Abstract. In the last few years, the concept of network virtualization has gained a lot of attention both from industry and research projects. This paper evaluates the potential of network virtualization from an operator's perspective, with the short-term goal of optimizing service delivery and rollout, and on a longer term as an enabler of technology integration and migration. Based on possible scenarios for implementing and using network virtualization, new business roles and models are examined. Open issues and topics for further evaluation are identified. In summary, the objective is to identify the challenges but also new opportunities for telecom operators raised by network virtualization.

Keywords: Network Virtualization, VN, VPN.

1 Introduction

Initially proposed in the context of Future Internet initiatives as a flexible platform for research and experimentation, network virtualization has gained an increasing prominence as a potential key component of next-generation Internet architectures by facilitating technological diversity and enabling evolution, thus overcoming the so-called Internet "ossification" problem.

Lately, the deployment of network virtualization has been increasingly advocated in short-to-medium term scenarios, e.g. [2], [3], [4], [5]. The capability to establish multiple isolated network domains over a common infrastructure opens up a range of new possibilities for operators, in several contexts and business scenarios [1].

Virtualization is a potential enabler of profound changes, both in the IT and network domains, and is expected to bridge the gap between these two worlds, traditionally living quite apart. In particular, the combination of cloud computing and network virtualization is likely to open up an immense field of opportunities for operators. Virtualization of computational and storage resources is already commonplace in operational environments, but bringing virtualization to networks has proven to be a lot more challenging.

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 138–147, 2010.
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In spite of the significant potential of the concept, it is clear that deployment of network virtualization requires overcoming major obstacles. Although some of the problems have been addressed in literature, the proposed approaches are usually oriented to research environments, largely overlooking key constraints and requirements of commercial operator networks.

Network virtualization has followed the usual development cycle, starting with research and testbed experimentation [6], [7]. However, evaluation of the concept with a view to deployment in carrier-grade commercial operator environments is still largely unaccomplished.

The network virtualization concept is not entirely new – to some extent, network-based Virtual Private Networks (VPNs), which are essentially separate networks sharing a common infrastructure, can also be seen as a materialization of this idea. However, VPNs cannot be decoupled from the underlying infrastructure and should be seen more as a service, rather than a real network. VPN customers have access to something they perceive as network cloud interconnecting their private network domains, but without any control or even visibility to the protocols running inside the VPN cloud.

Contrary to a VPN, a virtual network (VN) accomplishes a full separation from the underlying infrastructure. VNs are by definition technology agnostic, in the sense that VNs sharing a common physical infrastructure may internally run different protocols, decoupled from the infrastructure layers below. The physical infrastructure may be stretched across multiple administrative domains, i.e. may be owned by multiple infrastructure providers. The foundation for the VN is the physical infrastructure, consisting of links (e.g. fibers, copper lines, wireless connections) and nodes (e.g. routers, switches, servers and respective interfaces). The physical resources are virtualized through partitioning or slicing, thereby making the resources available to a number of isolated VNs.

This paper aims at providing a contribution in this direction by identifying challenges but also new opportunities for operators. The structure of the paper is as follows: after this brief introduction, section 2 presents relevant scenarios, which can benefit from network virtualization. Section 3 identifies potential opportunities to be exploited, as well as open issues and gaps, pointing towards the way forward for operators with the challenges and opportunities network virtualization is expected to bring. Finally, section 4 summarizes the main conclusions.

2 Scenarios for Network Virtualization Uptake

2.1 Network as a Service

The “Network as a Service” concept could be realized in a commercial environment with VN services provided to third parties. This is in line with the business model defined by the 4WARD project [1], characterized by a clear separation of the roles of the infrastructure provider (InP), responsible for operating the physical infrastructure, and the virtual network operator (VNO), responsible for operating the virtual networks. Optionally, an additional player called virtual network provider (VNP) may be involved, particularly in cases where a VN spans multiple infrastructure domains to act as a broker between the InP and the VNO.

This model may be driven by business incentives (new revenues for the InPs, lower CAPEX/OPEX costs for VNOs compared to building a network based on physical resources) or regulatory measures (open access and sharing of the network infrastructure).

Three basic scenarios may be envisaged between the virtual and physical network providers:

a) The VNO is supposed to select and establish a direct business relationship with the InPs that are able to fulfill the requirements, wherever they are located. This is typically the case if the VN is based on a single or few infrastructure domains. On the other hand, it might become complicated to handle if the VN extends across a considerable number of infrastructure domains.
b) To overcome the limitations of the first scenario, the VNP provides an intermediation role between the VNO and the InP. The VNP is responsible for finding the network resources at the best possible condition (price, quality of service, etc.) and offering them essentially as an empty container for the VNO to configure the VN. The main advantage of this scenario is that it enables VNOs to roll out large VNs without the need to establish business relationships with the potentially many InPs involved. The VNOs would have to deal with a single VNP who handles relations to the InPs. However, the fact that all the interactions between the VNO and the InPs have to be realized through the VNP contributes to complicating the whole process.
c) This scenario is similar to the previous one, except that in this case the participation of the VNP is restricted to the VN setup phase. Once the network is established, the VNP drops out of the picture and the VNO has a direct relationship with the InP. Thus, the VNP has essentially a brokerage function, facilitating the collection of the virtual resources.

A specialization of these basic scenarios would be enterprise VPNs:

d) VNs can be seen as an advanced form of enterprise VPN services, by enabling high dynamicity and elasticity. Standardized VNs can be managed and controlled separately from the infrastructure, which in many cases represent a significant advantage over specialized VPNs.

Several gaps and open issues still exist today: Reliability of a VN is ultimately determined by the reliability of the underlying infrastructure. Virtualization introduces an additional level of complexity and represents a potential extra source of failure. Interoperability between different heterogeneous domains is a challenge that requires standardization, especially with interconnection of non-contiguous network domains. Even if appropriate computational resources at the node level can be guaranteed, it is not trivial to guarantee strict isolation of link resources.

2.2 Cloud Computing and CDNs

Both cloud services and content distribution networks (CDNs) represent a remote delivery of computing resources, whether hardware or software resources, most often via the public or managed Internet. The requested data crosses multiple networks before it is delivered to the end user, mostly as a “best-effort” endeavor.

With virtualized networking technology cloud access and content delivery could be improved with end-to-end VNs and customized networking solutions for individual customers, including premium CDNs. A possible scenario is VNs created for each customer (or customer group) accessing the cloud services or content. The underlying VN resources can be based on fixed or mobile wireless networks.

Numerous stakeholders would be affected: Cloud service or CDN providers have high-capacity access connections to one or more network operators and provide the actual data or content; InP, VNP and VNO play the roles described in the previous section; end-users obtain managed access to Cloud services and content through the VNO, simplifying service provisioning and presentation.

The scenario assumes a virtual connection from the service/content provider to the end-user. Numerous tasks and technologies need to be in place for this, including: Networking infrastructure components/nodes with virtualization and load-balancing capabilities, networking protocols – perhaps similar to MPLS VPN – that provide secure data transmission with guaranteed quality of service; intelligent systems capable of performing dynamic seamless handover from one network to another – automatically establishing and tearing down VN connections.

Although the scenario represents an interesting paradigm, there are several outstanding issues: Limitations of supporting distributed cloud service provisioning and the requirement for VNs to span multiple computing or CDN farms; seamless networking handover technologies are still immature or inefficient; potentially complicating service delivery processes or business models.

2.3 Network Partitioning and Dynamic Resource Allocation

The idea is to provide the customers with a better user experience while dealing with commercial Telco services on a VN rather than directly on regular dedicated infrastructure. Services will be provided via dedicated VN slices on either single or multiple physical infrastructure domains. Varying by different commercial services, the VNs may have highly isolated, high performance, high security and privacy or alternating degrees of these features. VNs will be initially provisioned for each service and later adaptively maintained through dynamic resource allocation mechanisms.

Possible commercial services that may run on a VN topology to achieve higher performance and quality with decreased costs include: IPTV running on top of a VN for high quality, uninterrupted audio and video streaming achieved by dedicated bandwidth and isolated network resources; VoIP: using a VN with dedicated resources for VoIP service will ensure the highest possible voice quality; Enterprise services enabling the “virtual enterprise” concept which makes doing business simple by reducing operational costs, improving productivity and supporting new customer-centric services by combining a consistent user experience with “anywhere, anytime” information access.

Within the Telco operator, Services Divisions (SeD) will demand the required VN topology to be prepared from the Infrastructure Division (InD), which decides whether or not it can fulfill the request. If the request can be satisfied, the InD composes the requested VN and hands over full control to the SeD, which constructs the desired service and starts serving its subscribers over the VNs.

Several gaps and open issues remain. Dealing with the increasing number of services as well as subscribers for each service would be challenging. Scalability and dynamic resource allocation must be considered thoroughly when designing and operating the VN and corresponding services. Isolation is required between the VNs running separate Telco services. Depending on the business model and service type, a VN might require complete isolation while another one might share its resources when it is idle. In some business cases, the operator might own the complete physical infrastructure used by the VNs, so additional Telco companies acting as infrastructure providers for a VN might be required (in cases where a VN is spanning more than one operator's infrastructure). In such cases, dealing laterally with multiple Telco companies must be considered as an important issue.

2.4 Experimentation and Technology Migration

Network virtualization can play an important role in running experiments and introducing new networking protocols and solutions on production infrastructure next to production traffic. Currently testing and experiments require separate test networks and environments, which differ in scale and quality from production infrastructure and make technology migration cumbersome.

Running experiments and introducing technologies on production infrastructure confined to closely guarded virtual islands would alleviate the need for a large number of separate and isolated experimentation networks and lessen the impact of large-scale technology rollouts. At the same time this would allow network experiments to run on a scale larger than in confined laboratory and research networks, and close the gap between them and production environments, making parallel and rapid technology migration into providers' backbones and access networks possible.

The business model would be internal to operators, which utilize their existing infrastructure for research and development purposes and technology migration. The infrastructure division would provision VN circuits or slices on their existing infrastructure to internal customers (R&D, product development, and technology) to either run experiments or do staged, small-scale technology introduction. The legacy production network would either run as-is on the current infrastructure or be provisioned in the same way as VNs.

The requirements for separate test networks in business units could be reduced, including lower CAPEX for building and upgrading them and lower OPEX through a reduced need for non-operations units to run their own testing networks. Using VNs for technology migration has several business cases by limiting the scale of the efforts to smaller scopes at first. Thus, the operator would gain a relative market advantage by allowing the introduction of potentially disruptive technologies at a higher pace due to smaller and more targeted efforts. This would allow quicker reactions to innovations and market demands.

On an operational level VNs could lessen or distribute over time the OPEX needed for introducing disruptive technologies; CAPEX could be reduced by alleviating the need for special infrastructure in case of new technologies and conflicting customer demands – legacy networking and the new protocols could run on the same infrastructure for different customers, making investments for parallel solutions for new customers/technologies unnecessary.

However, allowing experimenters and technology introduction access to the virtualized production backbone might pose significant technical and political questions: Isolation for both security and resource allocation needs to be proven in real-world scenarios and tested in edge cases. In addition to the technical isolation requirements, a political consensus is needed between all the corporate divisions to allow the use of VNs on production infrastructure. Furthermore, it needs to be investigated if the VNs for experiments need host virtualization on the network device level as well. Limiting the virtualization to the links or circuits may not be sufficient for all cases, which may need access to (logical) network devices in the path as well.

3 Opportunities and Challenges for Operators

Our aim in this section is to briefly discuss how virtualization could take place in future networks and to give insight how feasible it is to fully implement VNs in reality, by elaborating the current challenges.

3.1 New Business Models and Opportunities

The widespread use of network virtualization both improves existing business and delivery models and introduces new role players/stakeholders to fill the gap between the virtualized network architecture and the existing one. The core of the new business models is in operators' backbone networks and centers on the possible independence from infrastructure constraints and boundaries for offering new network services. New players could emerge as independent providers of virtual networking resources not only to fill some gaps but also to re-shape the existing landscape.

A main attraction of network virtualization as an improvement to the existing business and delivery models is the probable benefit in flexibility currently lacking in the legacy networks. After successful rollout and configuration, provision of networks could be as easy as a simple service request for a new network that spans multiple providers that are invisible to the end-user or service providers. With the flexibility enabled by network virtualization, a wide variety of quality of service (QoS) levels could be guaranteed to the end users by offering different options while connected to the network.

The majority of scenarios use network virtualization as delivery model for services ranging from creating tailored networks for individual customers for cloud services and computing access to offering complete network packages as a commercial service. Virtualized networks on top of the existing infrastructure itself enable Telcos to experiment and integrate new services without affecting their existing business. In this way, Telcos would be able to plan their offerings to their customers and also add new services seamlessly. Network virtualization would offer an interesting way forward for operating and managing shared infrastructure resources by multiple parties on multiple levels. Telco operators could provide all parts of the network, from the physical infrastructure to VN operation, while other players could opt for focusing on the operation of VNs leased from virtual network providers.

Aside from improving existing business and delivery models, a whole new class of operators could emerge that only acquire and resell VN resources from existing Telco

operators and offer these virtualized bundles as VNs directly to customers. The benefit for customers would be getting a large-scale network from a single source, the virtual network providers.

Network virtualization could lead to a break of the existing market monopoly of the current network equipment vendors. Today's offerings of horizontally and vertically integrated solutions preclude market entry of new vendors and technologies that may threaten the business model of existing vendors. Virtualized networks on top of commodity hardware could lead to a comparable reduction in CAPEX and increase in vendor independence in operators' networks and data centers.

3.2 Challenges and Possible Roadblocks

Deployment of network virtualization may be hindered by a number of issues. To properly evaluate how challenges can represent real obstacles and affect the widespread deployment of network virtualization, it will be necessary to analyze the different use cases separately, as they are differently affected by the various challenges.

Carrier-Grade Reliability. VNs must be reliable, at least as reliable as a physical network counterpart. Today most of the available products with network virtualization capabilities are mainly targeted at the high-end segment of the market. On the other hand, very promising, flexible and adaptable technologies such as OpenFlow are perceived as research tools and have not yet reached a point of maturity to enable large scale deployment.

Scalability. The importance of scalability as a network virtualization requirement is particularly relevant in the cases where the number of VNs is expected to grow. For example, it is not clear to what extent traffic can be aggregated while still preserving fine grained resource control, as well as migration flexibility.

Isolation. VNs are by definition based on shared resources and this brings up the isolation problem, especially when the number of VNs sharing the same infrastructure is very high. On the other hand, the strictness of isolation varies according to the specific use case. In a Network as a Service scenario, isolation will be obviously a fundamental requirement; on the other hand, isolation is not as important in the network partitioning use case.

Security. Security issues will certainly represent a potential factor to discourage adopters of network virtualization, at least in an initial phase. Information privacy is crucial and must be guaranteed by whatever mechanisms are available (e.g. data encryption). Also, a well-defined degree of isolation has to be guaranteed to prevent VNs from mutually affecting each other adversely – either purposefully (e.g. DoS attacks) or inadvertently.

Interoperability. Many VNs will span multiple network infrastructure domains. Interoperability is a crucial requirement to enable widespread deployment of network virtualization. Standardization will be required to enable interoperability between VNs, as well as interoperability between virtualized and non-virtualized networks. In addition, interoperability between VNOs and InPs will also be required, through standardized vertical interfaces.

Operational Complexity. Since multiple stakeholders are supposed to be involved in provisioning and controlling of VNs, operational complexity represents a significant challenge. For example, once the VN is established and running, the role played by the VNP seems to be redundant. On the other hand, removing the VNP from the process would make the establishment of multi-domain VNs very complex.

Quality Management and Handling of Failures. VNOs are supposed to provide services to end users and service quality should be monitored and measured against specific SLAs. The extra level of complexity introduced by virtualization complicates this process significantly. In such an environment, the handling of failures is another complex challenge, as the identification of the source of the problem might not be trivial.

Programmability. Programmability is supposed to represent one of the main features offered by network virtualization in research and experimentation environments. This may be feasible using software-based routers that are based on open source, however, it is so far not feasible in commercial equipment.

Accountability. Accountability is required to track and report utilization of physical resources and is essential from the business and regulation perspective. The severity of this challenge would be proportional to the number of end users in the virtualized networks and amplified further with the diversity of quality of service demands.

Monetization. Last but not least, to become a successful technology, network virtualization must bring significant economic advantages to all the stakeholders involved. For example, it is not clear that the role of infrastructure provider will be an attractive proposition to operators.

As mentioned before, each network virtualization challenge should be evaluated in the scope of a specific use case. Table 1 summarizes how important the challenges mentioned in this section are for the scenarios identified in the previous section.

Overcoming all of the challenges at once may not be possible so high priority challenges should be addressed first. Analyzing use scenarios reveals that carrier-grade reliability and security are two of the most crucial challenges in the realization of VNs. VNs' solutions to these challenges have also significant impact on the customers to embrace this new technology.

In addition to this, standardization is the key requirement not only to clarify the unclear points in VN architecture and regulations but also to resolve the interoperability issues between different stakeholders and between VNs and legacy networks. Hence, standardization expedites the process of deployment of VNs and not having mature standards can be seen as a possible roadblock. Without standards, independent hardware and software vendors, who play key roles for VN uptake, can not engage in this process in a unified manner. In this way, the invaluable resources of the stakeholders such as time, money and manpower are not wasted and the challenges and open issues are resolved through the collaboration of the stakeholders. Standardization is still in a relatively early stage, but it should be noted that relevant activities in IRTF [8] and ITU-T [9] have started recently.

Table 1. Network virtualization challenges vs. use cases

	Scenarios			
	Cloud computing/CDN	Network as a Service	Network partitioning	Experimentation and Technology Migration
Carrier-grade reliability	***	***	***	***
Scalability	**	***	*	*
Isolation	***	***	**	**
Interoperability	**	***	**	***
Security	***	***	***	***
Operational complexity	***	***	**	*
Quality management	***	***	**	***
Programmability	*	*	*	***
Accountability	**	***	*	-
Monetization	**	***	**	-

*** Crucial challenge; will surely represent an obstacle against deployment if an appropriate solution cannot be found.
** Major challenge; may represent an obstacle to widespread deployment.
* Minor challenge; should be addressed, but does not represent a major obstacle in this specific use case.
- Not relevant in the scope of this specific use case.

4 Conclusions

Network virtualization was initially proposed as a key enabler of new Internet approaches. However, for network operators, the potential benefits of network virtualization are not limited to the evolution of the Future Internet – actually, they are multifold and can be explored in several time scales: Minimizing the cost of network ownership by sharing infrastructure; generating new revenues by reselling infrastructure to third parties or building new services; enhancing agility by facilitating rapid network deployment; enabling the coexistence of production and testing environments and enabling smooth technology migration. From the operators' point of view, the business incentives to deploy network virtualization are clear.

However, before any technology can be considered ready for deployment in large commercial environments, a number of key requirements will have to be fulfilled, most notably scalability, security and carrier-grade reliability. So far, network virtualization has been successfully demonstrated in small-scale research test beds, but it is clear that there is still a way to go before technology can be considered mature enough for large-scale deployment. There are complex challenges and potential hurdles to be faced. Standardization, still in a relatively incipient stage, will surely play a key role to enable interoperability between different vendors and between different operators.

Last but not least, network virtualization should not be analyzed isolated from other relevant emerging trends, particularly cloud computing. Integration of network- and IT resources will certainly raise an immense field of opportunities for operators and network virtualization will have a role to play in this scenario.

Acknowledgment

This work has been carried out in the framework of the Eurescom[1] P1956 study "Network Virtualisation - Opportunities and Challenges". In particular, the authors are thankful to Mr. Adam Kapovits for the fruitful discussions, as well as for insightful comments and suggestions for improvement.

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Cloud Computing and the Impact on Enterprise IT

Maximilian Ahrens

Zimory GmbH
ahrens@zimory.com

Abstract. This paper examines the impact of the cloud computing paradigm on enterprise IT structures. The first section of the paper defines the term cloud computing with the relevant aspects for this paper. In the second chapter the paper shows the potential of cloud computing for the enterprise IT. In the last section the paper will briefly reflect how the future development of infrastructure technologies will create more potential for enterprise IT cloud cases in the future.

1 Introduction

Cloud computing is one of the most discussed issues in enterprise IT. Despite this discussion the use cases in cloud computing are mostly derived from a non-enterprise environment. As cloud computing was first used to provide IT infrastructure to consumer web applications the original cloud services were developed to meet their requirements. Enterprises have different requirements for their use cases.

Therefore this paper examines the demand of enterprise IT in the cloud for the segment of infrastructure as a service (IaaS) clouds. In the course of this two use cases are introduced and analyzed.

Some of the challenges that where found during the analysis cannot be sufficiently addressed by today's technologies. Two of those issues are described in the last section, describing the demand of new infrastructure technologies to further develop cloud computing for enterprises.

2 Cloud Computing

Cloud Computing is on-demand access to virtualized IT resources that are sourced inside or outside of a data center, scalable, shared by others, simple to use, paid for via subscription or as you go, and transparently accessible over the network. There is no common definition of clouds at the moment, neither from academic nor industry communities, but in general it is described as the provisioning of services at different layers, Infrastructure as a Service, Platform as a Service (PaaS) and Software as a Service (SaaS) (see (Youseff, Butrico, & Da Silva, 2008), (Buyya et al., 2009), (Briscoe & Marinos, 2009)).

The lowest layer of clouds is represented by IaaS, where the object provided is a "server instance" with a network connection or other services like (virtual) storage devices. The customer obtains infrastructure components, which can be used for executing virtual machines.

A.J. Berre et al. (Eds.): FIS 2010, LNCS 6369, pp. 148–155, 2010.
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On a layer above, PaaS offerings basically provide an execution platform or interpreter service for user-defined application codes. PaaS contains pure runtime environments as well as programmable interfaces that allow the usage of pre-defined application services.

The upmost layer in this three layer model is SaaS, which describes the delivery of specific applications comparable to the ASP (Application Service Provisioning) model. These applications can be generic applications (e.g. webmail) as also specific business or research applications (CRM or GRID services).

In addition, Cloud Services can be divided in "internal" and "external". Whereas the external Cloud Services are offered as multi-tenant applications over the internet, internal services only available in the intranet of a specific organization.

The important aspect in all three cases is the service delivery to the customer, which is always performed in a transparent manner independent of the actual network location. The technical key characteristic of all these concepts is the usage of X86 virtualization, which allows the isolation of workloads independent of the single application on standard platforms. In combination with additional non-technical aspects, like on-demand usage and payment, Cloud Computing is currently considered as a new way of large scale computing.

Nevertheless today's Cloud Computing platforms are limited in functionality, often to predefined sets of applications and in terms of access capabilities. This leads to the rise of many small platforms for specific use cases, and following the diversity, faces many different obstacles on the technical and legal and operational level. This creates the risk of a new complexity.

The specific challenge for enterprise IT customers to manage the possibilities of the cloud are analyzed in the remainder.

3 Enterprise IT in the Cloud – The Challenges

Understanding the challenges of enterprise IT in the cloud requires looking at the traditional cloud use cases first. Cloud computing was used primarily for consumer oriented web applications. For these applications cloud computing could not only provide new technological capabilities to scale applications quicker, but it offered also the ability to gradually increase the costs with the addition of users. This is relevant in the typically user driven revenue models of these applications and allowed many services to be commercially successful.

This main use case in cloud computing defined technology and setup of the offerings. When enterprises began to analyze how the cloud can optimize their IT landscapes a couple of challenges arose.

Time To Market. A very important aspect of cloud computing for enterprises is the ability to reduce the time to market. Bringing services to the market or internal users in less time is crucial for business success.

Load optimization. Many enterprises have already an optimized datacenter environment. Compared to public cloud environments where a huge amount of independent customers and applications allow a statistical load balancing and thus load optimization, enterprise environments cannot deliver this. Enterprises have a smaller amount

of applications, each having a larger size. Also many applications are interconnected. Because of this peak loads can occur at the same time.

Legacy application. Enterprises have complex landscapes of existing applications. When planning the migration of services into the cloud these existing applications must be taken into consideration. In particular technical aspects like network requirements, scaling technologies and platform compatibility need to be inspected carefully.

Security Zones. Security is a huge topic in the cloud – as well for consumer oriented applications as also for enterprise IT use cases. The difference between the two use cases is that consumer applications have a single security zone as all customers are treated equally secure. Enterprises instead need to differentiate these security aspects in more complex ways. Internal users, partners, suppliers and customers and visitors are user groups that need access to different levels of information and thus need to be separated in different security zones. This requirement often led to the introduction of more complex network schemes to control access at this level. These cannot easily be transferred to the cloud.

SLAs and Consistency. SLAs in the cloud are different than those of usual enterprise hosting contracts. Cloud SLAs refer to the availability of the entire platform, but not to the availability of a single instance running in the platform. Therefore single instances can fail more often without breaching the SLA commitment. For an enterprise, which is used to single instance SLAs this has a twofold impact. On the one hand side the enterprise must implement failover mechanisms to manage cases where a single instance fails, but the overall platform remains functioning. On the other side a user must manage consistency issues when a single instance fails. Data that is stored within the instance cannot be recovered.

Datacenter Processes. Enterprises have clear rules and processes how the datacenter operation is done, which are required by legal compliance as well as service quality management. These processes need to be adjusted when self-service principles are being introduced. In order to meet quality and compliance goals cloud computing customers need to introduce new processes as well as change existing ones.

Despite these challenges cloud computing is being seen as on of the most important drivers for the enterprise IT in the upcoming years. Similarly to the movement from the mainframe to client server architectures the cloud computing paradigm is seen as a new movement towards centralized IT services. The benefits of cloud computing can mainly be seen in the simplification and thus decrease of time to market and the financial benefits. These will be explained in more detail in the next section using two use cases.

4 Enterprise IT in the Cloud – The Benefits

Enterprises using cloud computing can benefit in different ways from the cloud depending on the requirements of the specific use case. These should be demonstrated by two different use cases in this section.

4.1 Use Case 1: Solution Introduction in Enterprises

Internal communication and effective knowledge management is one of the most challenging tasks of modern knowledge driven enterprises.

Innovative software solutions are recently introduced into enterprise IT landscapes to improve communication and knowledge exchange. An example for these solutions is "enterprise microblogging".

Solutions in these area are currently being adopted by very early adopters, as software products are available only since less than a year. These early adopters usually evaluate the solution by piloting it in defined areas of the company. When introducing these services often the usage grows already during the pilot phase strongly, as the service is getting part of the daily work processes. Also this daily usage changes the requirements regarding the service levels, as downtimes are not tolerable.

Confronted with these challenges cloud computing can improve the introduction of such a service in the following aspects:

1. Time to pilot: in the cloud a customer can create a pilot environment within a couple of hours instead of days. The infrastructure can be provisioned within minutes and the configuration can be conducted also within very short time.
2. Scale: with the user adoption the growing platform demand can be covered by dynamic scaling of the infrastructure.
3. Security / SLAs: in a cloud platform failover mechanisms can be implemented in a simplified manner. This leverages the cloud capabilities to meet the requirements of availability and consistency with the growing usage.
4. Costs: the resource allocation along the real demand not only saves money and thus reduces the total cost of ownership (TCO), but also the cost structure is optimized. Instead of huge amounts of capital expenses, when using the cloud nearly all infrastructure costs are flexible operating expenses. The overall cost improvements are depicted in Figure 1. This figure shows the improvements of an installation where 10 servers have been replaced by instances in a compute cloud.

4.2 Use Case 2: Peak Pooling by Design

Also in the production phase of an application cloud computing can deliver benefits.

In this use case two independent enterprise applications are being grouped together after the analysis of their respective load profiles. As discussed in section 3 the load optimization on a statistical basis is not applicable in the enterprise case. Therefore a load profile analysis can define groups of applications that will not require peak resources at the same time, neither in scheduled load situations nor in unplanned ones. Once these applications are identified, the cloud computing technologies enable the automated scaling of workloads in the peak time and the downscaling after the peak. To gain the commercial benefits the two applications are separated in a base workload part and a peak workload part.

The base workload is being executed on an unshared platform, whereas the peak workload runs in the shared peak platform.

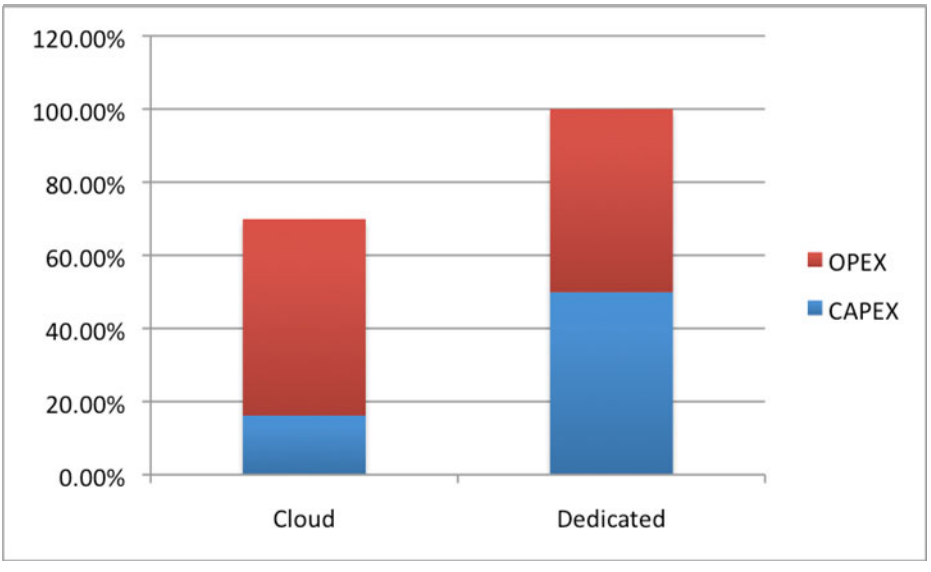

Fig. 1. Cost comparison for a small cloud use case

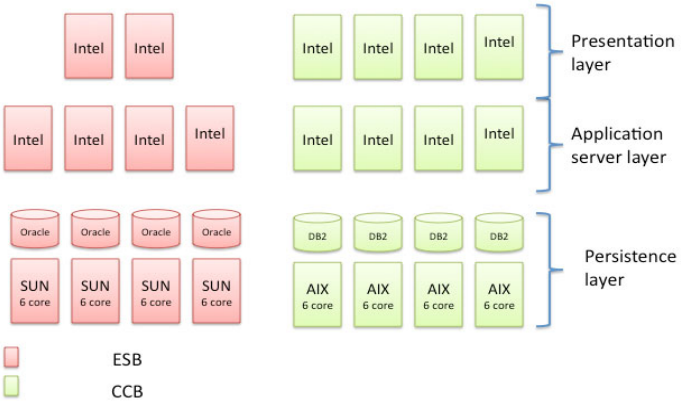

Fig. 2. As-is situation for peak pooling

The overall savings in the presented use case are around 30% of the TCO of the two applications.

To realize these savings several factors are taken into consideration:

1. Reducing hardware cost: This reduction comes from the overall size reduction, but in addition significant savings derive from the usage of standard machines instead of expensive multicore environments.
2. Reducing operational costs: The constant resizing of the platform is happening automatically by the cloud technologies, which decreases the operational costs extremely.

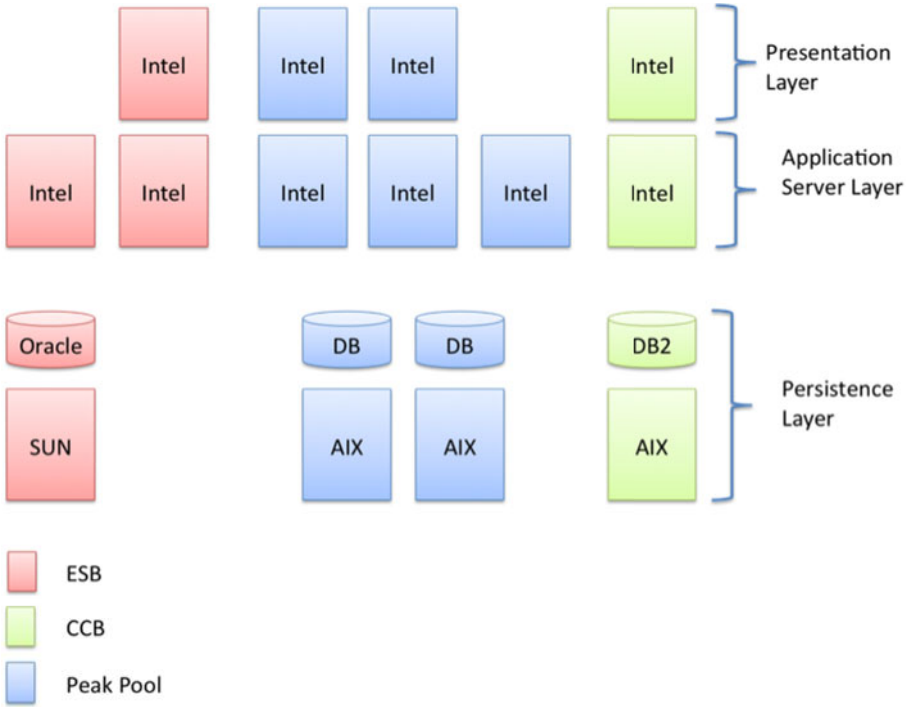

Fig. 3. To-be situation

3. Homogenization of the infrastructure: When peak pooling is implemented, all layers in the pool are running on a single homogenous platform. This simplifies a couple of processes and reduces the per node costs.

This use case demonstrates the potential of cloud computing in enterprises, but also shows the challenges as described in section 3.

- The two applications are legacy applications. The specific challenge with the shown applications is to allow the scalability also in the datalayer.
- The applications are implemented on host based SLAs. Guaranteeing consistency in a cloud environment requires adoption of the applications.
- The data center processes need to be adjusted significantly. In the cloud the application requests and releases itself the required infrastructure resources instead of administrators managing this. The administrators have to manage and maintain the cloud platform instead. Only by implementing these procedural changes the benefits can be realized.

5 Next Generation Infrastructures and the Cloud

Although – as shown in the previous section – cloud computing can deliver huge benefits to the enterprise IT, some of the challenges described in section 3 cannot be sufficiently addressed.

In this section two challenges should be briefly reviewed: SLA / consistency issues and security zones.

5.1 SLA / Consistency

SLAs and consistency are an inherent challenge for cloud computing platforms. On the one hand side creating high SLAs on a per host base does not comply with the business model of cloud computing, on the other hand consistency on a low SLA infrastructure requires constant synchronization. This leads to the demand of having a cloud of multiple locations with a high performance network between these. Only then high volume replications can be performed in short enough time to guarantee consistency across the sites. When consistency is achievable in such a distributed environment also very high SLAs can be guaranteed.

In this scenario a high performance but affordable network with quality of service management functionality is required to be compliant with enterprise demand.

5.2 Security Zones

Network management is one of the most complex tasks in cloud computing. As described in section 3 most cloud computing network structures do not comply with enterprise IT demands. Having a more flexible and secure network structure is a key factor for cloud success in the enterprise IT. When workloads are being resized and moved between locations automatically, also network structures must be able to connect them in an automated and transparent way.

6 Conclusion

Cloud Computing has a huge potential for enterprise IT. Nevertheless to realize the full potential of the cloud, enterprises have to do quite substantial adjustments to their internal structures and applications.

The biggest challenge in realizing the potential of flexibility of the cloud is the network structure. This challenge is caused by bandwidth shortage of many enterprises as well as structural network setups that cannot be adjusted in the dynamic way required by the cloud. New technologies in this area are required to fulfill as well security demands of the enterprise as also the dynamics of the cloud.

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